Chapter 9
NATURAL CONVECTION

Physical Mechanisms of Natural Convection

9-1C Natural convection is the mode of heat transfer that occurs between a solid and a fluid which moves under the influence of natural means. Natural convection differs from forced convection in that fluid motion in natural convection is caused by natural effects such as buoyancy.

9-2C The convection heat transfer coefficient is usually higher in forced convection because of the higher fluid velocities involved.

9-3C The hot boiled egg in a spacecraft will cool faster when the spacecraft is on the ground since there is no gravity in space, and thus there will be no natural convection currents which is due to the buoyancy force.

9-4C The upward force exerted by a fluid on a body completely or partially immersed in it is called the buoyancy or "lifting" force. The buoyancy force is proportional to the density of the medium. Therefore, the buoyancy force is the largest in mercury, followed by in water, air, and the evacuated chamber. Note that in an evacuated chamber there will be no buoyancy force because of absence of any fluid in the medium.

9-5C The buoyancy force is proportional to the density of the medium, and thus is larger in sea water than it is in fresh water. Therefore, the hull of a ship will sink deeper in fresh water because of the smaller buoyancy force acting upwards.

9-6C A spring scale measures the "weight" force acting on it, and the person will weigh less in water because of the upward buoyancy force acting on the person's body.

9-7C The greater the volume expansion coefficient, the greater the change in density with temperature, the greater the buoyancy force, and thus the greater the natural convection currents.

9-8C There cannot be any natural convection heat transfer in a medium that experiences no change in volume with temperature.

9-9C The lines on an interferometer photograph represent isotherms (constant temperature lines) for a gas, which correspond to the lines of constant density. Closely packed lines on a photograph represent a large temperature gradient.

9-10C The Grashof number represents the ratio of the buoyancy force to the viscous force acting on a fluid. The inertial forces in Reynolds number is replaced by the buoyancy forces in Grashof number.

9-11 The volume expansion coefficient is defined as $\beta = \frac{-1}{\rho}\left(\frac{\partial \rho}{\partial T}\right)_P$. For an ideal gas, $P = \rho RT$ or $\rho = \frac{P}{RT}$, and thus $\beta = -\frac{1}{\rho}\left(\frac{\partial (P/RT)}{\partial T}\right)_P = \frac{-1}{\rho}\left(\frac{-P}{RT^2}\right) = \frac{1}{\rho T}\left(\frac{P}{RT}\right) = \frac{1}{\rho T}(\rho) = \frac{1}{T}$

Natural Convection Over Surfaces

9-12C Rayleigh number is the product of the Grashof and Prandtl numbers.

9-13C A vertical cylinder can be treated as a vertical plate when $D \geq \frac{35L}{Gr^{1/4}}$.

9-14C No, a hot surface will cool slower when facing down since the warmer air in this position cannot rise and escape easily.

9-15C The heat flux will be higher at the bottom of the plate since the thickness of the boundary layer which is a measure of thermal resistance is the lowest there.

9-16 A horizontal hot water pipe passes through a large room. The rate of heat loss from the pipe by natural convection and radiation is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** The temperature of the outer surface of the pipe is constant.

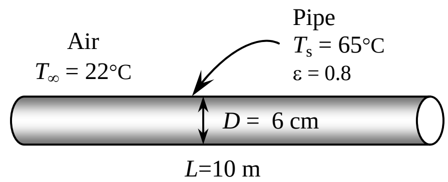
Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (65+22)/2 = 43.5°C$ are (Table A-15)

$$k = 0.02688 \text{ W/m.°C}$$

$$\upsilon = 1.735\times10^{-5} \text{ m}^2\text{/s}$$

$$\text{Pr} = 0.7245$$

$$\beta = \frac{1}{T_f} = \frac{1}{(43.5+273)\text{K}} = 0.00316 \text{ K}^{-1}$$



Analysis (*a*) The characteristic length in this case is the outer diameter of the pipe, $L_c = D = 0.06$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.00316 \text{ K}^{-1})(65-22 \text{ K})(0.06 \text{ m})^3}{(1.735\times10^{-5} \text{ m}^2\text{/s})^2}(0.7245) = 692{,}805$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+(0.559/\text{Pr})^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(692{,}805)^{1/6}}{\left[1+(0.559/0.7245)^{9/16}\right]^{8/27}}\right\}^2 = 13.15$$

$$h = \frac{k}{D}Nu = \frac{0.02688 \text{ W/m.°C}}{0.06 \text{ m}}(13.15) = 5.893 \text{ W/m}^2\text{.°C}$$

$$A_s = \pi DL = \pi(0.06 \text{ m})(10 \text{ m}) = 1.885 \text{ m}^2$$

$$\dot{Q} = hA_s(T_s - T_\infty) = (5.893 \text{ W/m}^2\text{.°C})(1.885 \text{ m}^2)(65-22)\text{°C} = \mathbf{477.6 \text{ W}}$$

(*b*) The radiation heat loss from the pipe is

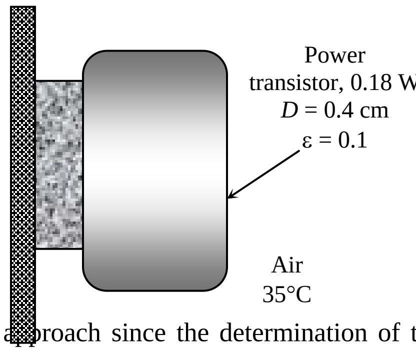
$$\dot{Q}_{rad} = \varepsilon A_s\sigma(T_s{}^4 - T_{surr}{}^4) = (0.8)(1.885 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2\text{.K}^4)\left[(65+273 \text{ K})^4 - (22+273 \text{ K})^4\right] = \mathbf{468.4 \text{ W}}$$

9-17 A power transistor mounted on the wall dissipates 0.18 W. The surface temperature of the transistor is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** Any heat transfer from the base surface is disregarded. **4** The local atmospheric pressure is 1 atm. **5** Air properties are evaluated at 100°C.

Properties The properties of air at 1 atm and the given film temperature of 100°C are (Table A-15)

$$k = 0.03095 \text{ W/m.°C}$$
$$\upsilon = 2.306\times10^{-5} \text{ m}^2\text{/s}$$
$$\text{Pr} = 0.7111$$
$$\beta = \frac{1}{T_f} = \frac{1}{(100+273)\text{ K}} = 0.00268 \text{ K}^{-1}$$



Analysis The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the surface temperature to be 165°C for the evaluation of h. This is the surface temperature that will give a film temperature of 100°C. We will check the accuracy of this guess later and repeat the calculations if necessary.

The transistor loses heat through its cylindrical surface as well as its top surface. For convenience, we take the heat transfer coefficient at the top surface of the transistor to be the same as that of its side surface. (The alternative is to treat the top surface as a vertical plate, but this will double the amount of calculations without providing much improvement in accuracy since the area of the top surface is much smaller and it is circular in shape instead of being rectangular). The characteristic length in this case is the outer diameter of the transistor, $L_c = D = 0.004\text{ m}$. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2}\text{Pr} = \frac{(9.81\text{ m/s}^2)(0.00268\text{ K}^{-1})(165-35\text{ K})(0.004\text{ m})^3}{(2.306\times10^{-5}\text{ m}^2\text{/s})^2}(0.7111) = 292.6$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+(0.559/\text{Pr})^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(292.6)^{1/6}}{\left[1+(0.559/0.7111)^{9/16}\right]^{8/27}}\right\}^2 = 2.039$$

$$h = \frac{k}{D}Nu = \frac{0.03095\text{ W/m.°C}}{0.004\text{ m}}(2.039) = 15.78\text{ W/m}^2\text{.°C}$$

$$A_s = \pi DL + \pi D^2/4 = \pi(0.004\text{ m})(0.0045\text{ m}) + \pi(0.004\text{ m})^2/4 = 0.0000691\text{ m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s^{\,4} - T_{surr}^{\,4})$$
$$0.18\text{ W} = (15.8\text{ W/m}^2\text{.°C})(0.0000691\text{ m}^2)(T_s - 35)\text{ °C}$$
$$+(0.1)(0.0000691\text{ m}^2)(5.67\times10^{-8})\left[(T_s+273)^4 - (25+273\text{ K})^4\right]$$
$$\longrightarrow T_s = 187\text{°C}$$

which is relatively close to the assumed value of 165°C. To improve the accuracy of the result, we repeat the Rayleigh number calculation at new surface temperature of 187°C and determine the surface temperature to be

T_s = **183°C**

Discussion W evaluated the air properties again at 100°C when repeating the calculation at the new surface temperature. It can be shown that the effect of this on the calculated surface temperature is less than 1°C.

9-18 "!PROBLEM 9-18"

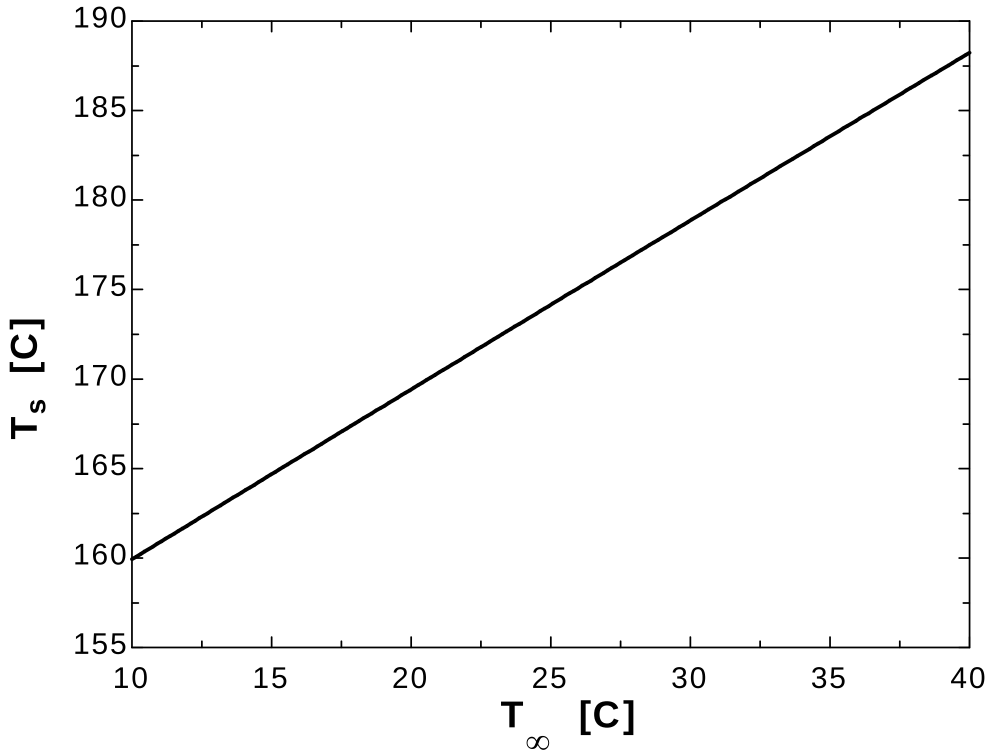
"GIVEN"
Q_dot=0.18 "[W]"

```
"T_infinity=35 [C], parameter to be varied"
L=0.0045 "[m]"
D=0.004 "[m]"
epsilon=0.1
T_surr=T_infinity-10 "[C]"

"PROPERTIES"
Fluid$='air'
k=Conductivity(Fluid$, T=T_film)
Pr=Prandtl(Fluid$, T=T_film)
rho=Density(Fluid$, T=T_film, P=101.3)
mu=Viscosity(Fluid$, T=T_film)
nu=mu/rho
beta=1/(T_film+273)
T_film=1/2*(T_s+T_infinity)
sigma=5.67E-8 "[W/m^2-K^4], Stefan-Boltzmann constant"
g=9.807 "[m/s^2], gravitational acceleration"

"ANALYSIS"
delta=D
Ra=(g*beta*(T_s-T_infinity)*delta^3)/nu^2*Pr
Nusselt=(0.6+(0.387*Ra^(1/6))/(1+(0.559/Pr)^(9/16))^(8/27))^2
h=k/delta*Nusselt
A=pi*D*L+pi*D^2/4
Q_dot=h*A*(T_s-T_infinity)+epsilon*A*sigma*((T_s+273)^4-(T_surr+273)^4)
```

T_∞ [C]	T_s [C]
10	159.9
12	161.8
14	163.7
16	165.6
18	167.5
20	169.4
22	171.3
24	173.2
26	175.1
28	177
30	178.9
32	180.7
34	182.6
36	184.5
38	186.4
40	188.2


190
185
180
175
170
165
160
155
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15
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25
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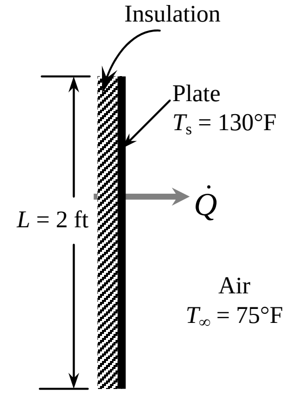
35
40
Ts [C]
T∞ [C]

9-19E A hot plate with an insulated back is considered. The rate of heat loss by natural convection is to be determined for different orientations.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.



Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (130+75)/2 = 102.5$°F are (Table A-15)

$$k = 0.01535 \text{ Btu/h.ft.°F}$$
$$\upsilon = 0.1823\times10^{-3} \text{ ft}^2\text{/s}$$
$$\text{Pr} = 0.7256$$
$$\beta = \frac{1}{T_f} = \frac{1}{(102.5+460)\text{R}} = 0.001778 \text{ R}^{-1}$$

Analysis (*a*) When the plate is vertical, the characteristic length is the height of the plate. $L_c = L = 2$ ft. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\text{Pr} = \frac{(32.2 \text{ ft/s}^2)(0.001778 \text{ R}^{-1})(130-75 \text{ R})(2 \text{ ft})^3}{(0.1823\times10^{-3} \text{ ft}^2\text{/s})^2}(0.7256) = 5.503\times10^8$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(5.503\times10^8)^{1/6}}{\left[1+\left(\frac{0.492}{0.7256}\right)^{9/16}\right]^{8/27}}\right\}^2 = 102.6$$

$$h = \frac{k}{L}Nu = \frac{0.01535 \text{ Btu/h.ft.°F}}{2 \text{ ft}}(102.6) = 0.7869 \text{ Btu/h.ft}^2\text{.°F}$$
$$A_s = L^2 = (2 \text{ ft})^2 = 4 \text{ ft}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (0.7869 \text{ Btu/h.ft}^2\text{.°F})(4 \text{ ft}^2)(130-75)\text{°C} = \mathbf{173.1 Btu/h}$$

(*b*) When the plate is horizontal with hot surface facing up, the characteristic length is determined from

$$L_s = \frac{A_s}{P} = \frac{L^2}{4L} = \frac{L}{4} = \frac{2 \text{ ft}}{4} = 0.5 \text{ ft}.$$

Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\upsilon^2}\text{Pr} = \frac{(32.2 \text{ ft/s}^2)(0.001778 \text{ R}^{-1})(130-75 \text{ R})(0.5 \text{ ft})^3}{(0.1823\times10^{-3} \text{ ft}^2\text{/s})^2}(0.7256) = 8.598\times10^6$$

$$Nu = 0.54Ra^{1/4} = 0.54(8.598\times10^6)^{1/4} = 29.24$$

$$h = \frac{k}{L_c}Nu = \frac{0.01535 \text{ Btu/h.ft.°F}}{0.5 \text{ ft}}(29.24) = 0.8975 \text{ Btu/h.ft}^2\text{.°F}$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (0.8975 \text{ Btu/h.ft}^2\text{.°F})(4 \text{ ft}^2)(130-75)\text{°C} = \mathbf{197.4 \text{ Btu/h}}$$

(*c*) When the plate is horizontal with hot surface facing down, the characteristic length is again $\delta = 0.5$ ft and the Rayleigh number is $Ra = 8.598\times10^6$. Then,

$$Nu = 0.27Ra^{1/4^2} = 0.27(8.598\times10^6)^{1/4} = 14.62$$

$$h = \frac{k}{L_c}Nu = \frac{0.01535 \text{ Btu/h.ft.°F}}{0.5 \text{ ft}}(14.62) = 0.4487 \text{ Btu/h.ft}^2\text{.°F}$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (0.4487\,\text{Btu/h.ft}^2.°\text{F})(4\,\text{ft}^2)(130 - 75)°\text{C} = \mathbf{98.7\,Btu/h}$$

9-20E "!PROBLEM 9-20E"

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"GIVEN"
L=2 "[ft]"
T_infinity=75 "[F]"
"T_s=130 [F], parameter to be varied"

"PROPERTIES"
Fluid$='air'
k=Conductivity(Fluid$, T=T_film)
Pr=Prandtl(Fluid$, T=T_film)
rho=Density(Fluid$, T=T_film, P=14.7)
mu=Viscosity(Fluid$, T=T_film)*Convert(lbm/ft-h, lbm/ft-s)
nu=mu/rho
beta=1/(T_film+460)
T_film=1/2*(T_s+T_infinity)
g=32.2 "[ft/s^2], gravitational acceleration"

"ANALYSIS"
"(a), plate is vertical"
delta_a=L
Ra_a=(g*beta*(T_s-T_infinity)*delta_a^3)/nu^2*Pr
Nusselt_a=0.59*Ra_a^0.25
h_a=k/delta_a*Nusselt_a
A=L^2
Q_dot_a=h_a*A*(T_s-T_infinity)
"(b), plate is horizontal with hot surface facing up"
delta_b=A/p
p=4*L
Ra_b=(g*beta*(T_s-T_infinity)*delta_b^3)/nu^2*Pr
Nusselt_b=0.54*Ra_b^0.25
h_b=k/delta_b*Nusselt_b
Q_dot_b=h_b*A*(T_s-T_infinity)
"(c), plate is horizontal with hot surface facing down"
delta_c=delta_b
Ra_c=Ra_b
Nusselt_c=0.27*Ra_c^0.25
h_c=k/delta_c*Nusselt_c
Q_dot_c=h_c*A*(T_s-T_infinity)
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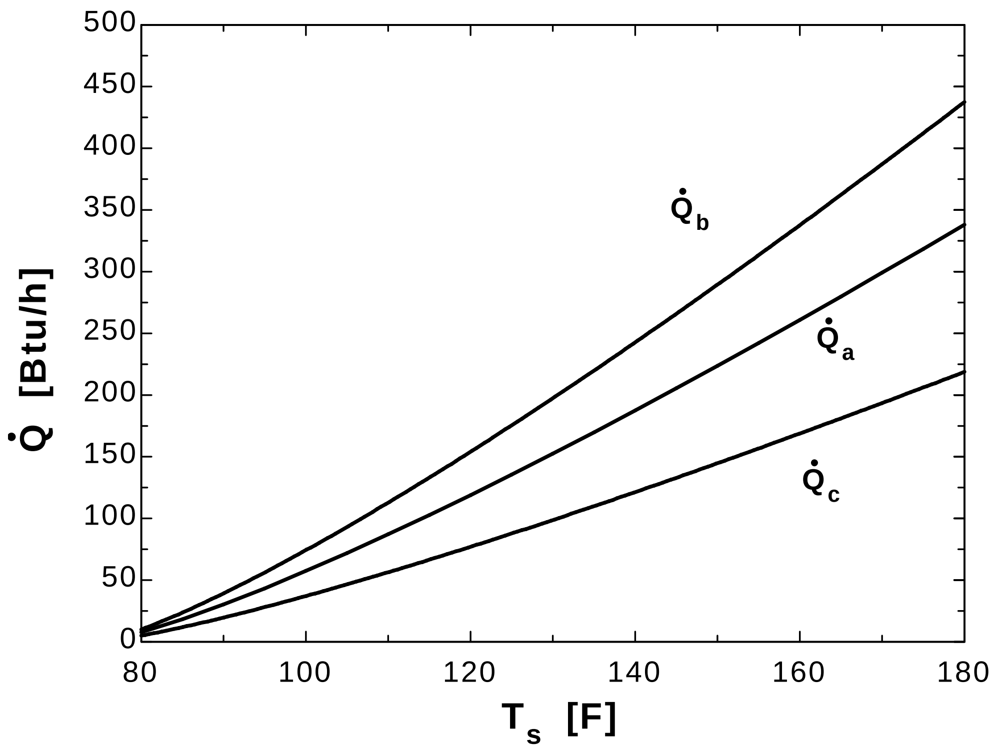
T_s [F]	Q_a [Btu/h]	Q_b [Btu/h]	Q_c [Btu/h]
80	7.714	9.985	4.993
85	18.32	23.72	11.86
90	30.38	39.32	19.66
95	43.47	56.26	28.13
100	57.37	74.26	37.13
105	71.97	93.15	46.58
110	87.15	112.8	56.4
115	102.8	133.1	66.56
120	119	154	77.02
125	135.6	175.5	87.75
130	152.5	197.4	98.72
135	169.9	219.9	109.9
140	187.5	242.7	121.3
145	205.4	265.9	132.9
150	223.7	289.5	144.7
155	242.1	313.4	156.7
160	260.9	337.7	168.8
165	279.9	362.2	181.1
170	299.1	387.1	193.5
175	318.5	412.2	206.1
180	338.1	437.6	218.8

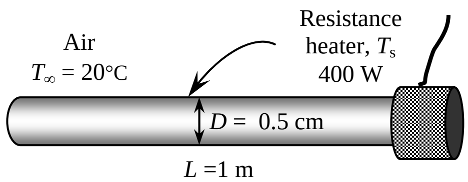


9-21 A cylindrical resistance heater is placed horizontally in a fluid. The outer surface temperature of the resistance wire is to be determined for two different fluids.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** Any heat transfer by radiation is ignored. **5** Properties are evaluated at 500°C for air and 40°C for water.

Properties The properties of air at 1 atm and 500°C are (Table A-15)

$$k = 0.05572\ \text{W/m.°C}$$
$$\upsilon = 7.804\times10^{-5}\ \text{m}^2\text{/s}$$
$$\text{Pr} = 0.6986$$
$$\beta = \frac{1}{T_f} = \frac{1}{(500+273)\text{K}} = 0.001294\ \text{K}^{-1}$$



The properties of water at 40°C are

$$k = 0.631\ \text{W/m.°C}$$
$$\upsilon = \mu/\rho = 0.6582\times10^{-6}\ \text{m}^2\text{/s}$$
$$\text{Pr} = 4.32$$
$$\beta = 0.000377\ \text{K}^{-1}$$

Analysis (*a*) The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the surface temperature to be 1200°C for the calculation of *h*. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the outer diameter of the wire, $L_c = D = 0.005\ \text{m}$. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2}\text{Pr} = \frac{(9.81\ \text{m/s}^2)(0.001294\ \text{K}^{-1})(1200-20)\text{°C}(0.005\ \text{m})^3}{(7.804\times10^{-5}\ \text{m}^2\text{/s})^2}(0.6986) = 214.7$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/\text{Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(214.7)^{1/6}}{\left[1+\left(0.559/0.6986\right)^{9/16}\right]^{8/27}}\right\}^2 = 1.919$$

$$h = \frac{k}{D}Nu = \frac{0.05572\ \text{W/m.°C}}{0.005\ \text{m}}(1.919) = 21.38\ \text{W/m}^2\text{.°C}$$

$$A_s = \pi DL = \pi(0.005\ \text{m})(1\ \text{m}) = 0.01571\ \text{m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty)$$
$$400\ \text{W} = (21.38\ \text{W/m}^2\text{.°C})(0.01571\ \text{m}^2)(T_s - 20)\text{°C}$$
$$T_s = \mathbf{1211°C}$$

which is sufficiently close to the assumed value of 1200°C used in the evaluation of *h*, and thus it is not necessary to repeat calculations.

(*b*) For the case of water, we "guess" the surface temperature to be 40°C. The characteristic length in this case is the outer diameter of the wire, $L_c = D = 0.005\ \text{m}$. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2}\text{Pr} = \frac{(9.81\ \text{m/s}^2)(0.000377\ \text{K}^{-1})(40-20\ \text{K})(0.005\ \text{m})^3}{(0.6582\times10^{-6}\ \text{m}^2\text{/s})^2}(4.32) = 92{,}197$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/\text{Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(92{,}197)^{1/6}}{\left[1+\left(0.559/4.32\right)^{9/16}\right]^{8/27}}\right\}^2 = 8.986$$

$$h = \frac{k}{D} Nu = \frac{0.631\,\text{W/m.°C}}{0.005\,\text{m}}(8.986) = 1134\ \text{W/m}^2\text{.°C}$$

and

$$\dot{Q} = hA_s(T_s - T_\infty)$$

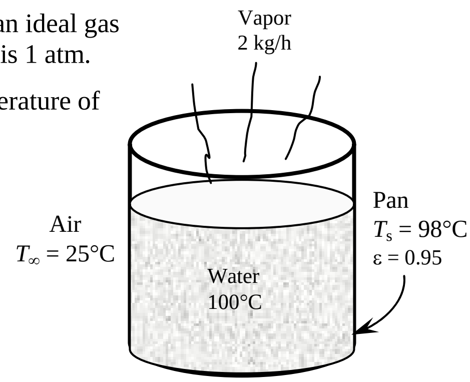
$$400\ \text{W} = (1134\ \text{W/m}^2\text{.°C})(0.01571\,\text{m}^2)(T_s - 20)\text{°C}$$

$$T_s = \mathbf{42.5°C}$$

which is sufficiently close to the assumed value of 40°C in the evaluation of the properties and *h*. The film temperature in this case is $(T_s+T_\infty)/2$ = (42.5+20)/2 =31.3°C, which is close to the value of 40°C used in the evaluation of the properties.

9-22 Water is boiling in a pan that is placed on top of a stove. The rate of heat loss from the cylindrical side surface of the pan by natural convection and radiation and the ratio of heat lost from the side surfaces of the pan to that by the evaporation of water are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.



Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (98+25)/2 = 61.5$°C are (Table A-15)

$$k = 0.02819\ \text{W/m.°C}$$
$$\upsilon = 1.910\times10^{-5}\ \text{m}^2/\text{s}$$
$$\text{Pr} = 0.7198$$
$$\beta = \frac{1}{T_f} = \frac{1}{(61.5+273)\text{K}} = 0.00299\ \text{K}^{-1}$$

Analysis (*a*) The characteristic length in this case is the height of the pan, $L_c = L = 0.12$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\text{Pr} = \frac{(9.81\ \text{m/s}^2)(0.00299\ \text{K}^{-1})(98-25\ \text{K})(0.12\ \text{m})^3}{(1.910\times10^{-5}\ \text{m}^2/\text{s})^2}(0.7198) = 7.299\times10^6$$

We can treat this vertical cylinder as a vertical plate since

$$\frac{35L}{Gr^{1/4}} = \frac{35(0.12)}{(7.299\times10^6\ /\ 0.7198)^{1/4}} = 0.07443 < 0.25 \quad \text{and thus } D \geq \frac{35L}{Gr^{1/4}}$$

Therefore,

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(7.299\times10^6)^{1/6}}{\left[1+\left(\frac{0.492}{0.7198}\right)^{9/16}\right]^{8/27}}\right\}^2 = 28.60$$

$$h = \frac{k}{L}Nu = \frac{0.02819\ \text{W/m.°C}}{0.12\ \text{m}}(28.60) = 6.720\ \text{W/m}^2.\text{°C}$$
$$A_s = \pi DL = \pi(0.25\ \text{m})(0.12\ \text{m}) = 0.09425\ \text{m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (6.720\ \text{W/m}^2.\text{°C})(0.09425\ \text{m}^2)(98-25)\text{°C} = \mathbf{46.2\ W}$$

(*b*) The radiation heat loss from the pan is

$$\dot{Q}_{rad} = \varepsilon A_s\sigma(T_s{}^4 - T_{surr}{}^4)$$
$$= (0.95)(0.09425\ \text{m}^2)(5.67\times10^{-8}\ \text{W/m}^2.\text{K}^4)\left[(98+273\ \text{K})^4 - (25+273\ \text{K})^4\right] = \mathbf{56.1\ W}$$

(*c*) The heat loss by the evaporation of water is

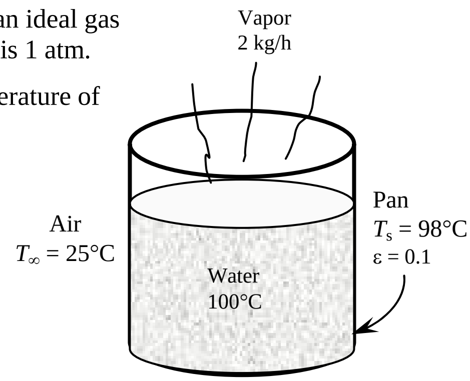
$$\dot{Q} = \dot{m}h_{fg} = (2/3600\ \text{kg/s})(2257\ \text{kJ/kg}) = 1.254\ \text{kW} = 1254\ \text{W}$$

Then the ratio of the heat lost from the side surfaces of the pan to that by the evaporation of water then becomes

$$f = \frac{46.2+56.1}{1254} = 0.082 = \mathbf{8.2\%}$$

9-23 Water is boiling in a pan that is placed on top of a stove. The rate of heat loss from the cylindrical side surface of the pan by natural convection and radiation and the ratio of heat lost from the side surfaces of the pan to that by the evaporation of water are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.



Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (98+25)/2 = 61.5$°C are (Table A-15)

$$k = 0.02819 \text{ W/m.°C}$$
$$\upsilon = 1.910\times10^{-5} \text{ m}^2\text{/s}$$
$$\text{Pr} = 0.7198$$
$$\beta = \frac{1}{T_f} = \frac{1}{(61.5+273)\text{K}} = 0.00299 \text{ K}^{-1}$$

Analysis (*a*) The characteristic length in this case is the height of the pan, $L_c = L = 0.12$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.00299 \text{ K}^{-1})(98-25 \text{ K})(0.12 \text{ m})^3}{(1.910\times10^{-5} \text{ m}^2\text{/s})^2}(0.7198) = 7.299\times10^6$$

We can treat this vertical cylinder as a vertical plate since

$$\frac{35L}{Gr^{1/4}} = \frac{35(0.12)}{(7.299\times10^6 / 0.7198)^{1/4}} = 0.07443 < 0.25 \quad \text{and thus } D \geq \frac{35L}{Gr^{1/4}}$$

Therefore,

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(7.299\times10^6)^{1/6}}{\left[1+\left(\frac{0.492}{0.7198}\right)^{9/16}\right]^{8/27}}\right\}^2 = 28.60$$

$$h = \frac{k}{L}Nu = \frac{0.02819 \text{ W/m.°C}}{0.12 \text{ m}}(28.60) = 6.720 \text{ W/m}^2\text{.°C}$$
$$A_s = \pi DL = \pi(0.25 \text{ m})(0.12 \text{ m}) = 0.09425 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (6.720 \text{ W/m}^2\text{.°C})(0.09425 \text{ m}^2)(98-25)\text{°C} = \mathbf{46.2 \text{ W}}$$

(b) The radiation heat loss from the pan is

$$\dot{Q}_{rad} = \varepsilon A_s\sigma(T_s{}^4 - T_{surr}{}^4)$$
$$= (0.10)(0.09425 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2\text{.K}^4)\left[(98+273 \text{ K})^4 - (25+273 \text{ K})^4\right] = \mathbf{5.9 \text{ W}}$$

(c) The heat loss by the evaporation of water is

$$\dot{Q} = \dot{m}h_{fg} = (2/3600 \text{ kg/s})(2257 \text{ kJ/kg}) = 1.254 \text{ kW} = 1254 \text{ W}$$

Then the ratio of the heat lost from the side surfaces of the pan to that by the evaporation of water then becomes

$$f = \frac{46.2+5.9}{1254} = 0.042 = \mathbf{4.2\%}$$

9-24 Some cans move slowly in a hot water container made of sheet metal. The rate of heat loss from the four side surfaces of the container and the annual cost of those heat losses are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **3** Heat loss from the top surface is disregarded.

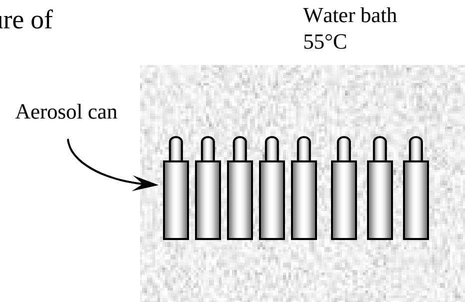
Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (55+20)/2 = 37.5°C$ are (Table A-15)

$$k = 0.02644\ \text{W/m.°C}$$

$$\upsilon = 1.678\times10^{-5}\ \text{m}^2/\text{s}$$

$$\text{Pr} = 0.7261$$

$$\beta = \frac{1}{T_f} = \frac{1}{(37.5+273)\text{K}} = 0.003221\ \text{K}^{-1}$$



Analysis The characteristic length in this case is the height of the bath, $L_c = L = 0.5$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\text{Pr} = \frac{(9.81\ \text{m/s}^2)(0.003221\ \text{K}^{-1})(55-20\ \text{K})(0.5\ \text{m})^3}{(1.678\times10^{-5}\ \text{m}^2/\text{s})^2}(0.7261) = 3.565\times10^8$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(3.565\times10^8)^{1/6}}{\left[1+\left(\frac{0.492}{0.7261}\right)^{9/16}\right]^{8/27}}\right\}^2 = 89.84$$

$$h = \frac{k}{L}Nu = \frac{0.02644\ \text{W/m.°C}}{0.5\ \text{m}}(89.84) = 4.75\ \text{W/m}^2\text{.°C}$$

$$A_s = 2[(0.5\ \text{m})(1\ \text{m}) + (0.5\ \text{m})(3.5\ \text{m})] = 4.5\ \text{m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (4.75\ \text{W/m}^2\text{.°C})(4.5\ \text{m}^2)(55-20)\text{°C} = 748.1\ \text{W}$$

The radiation heat loss is

$$\begin{aligned}\dot{Q}_{rad} &= \varepsilon A_s\sigma(T_s{}^4 - T_{surr}{}^4) \\ &= (0.7)(4.5\ \text{m}^2)(5.67\times10^{-8}\ \text{W/m}^2\text{.K}^4)\left[(55+273\ \text{K})^4 - (20+273\ \text{K})^4\right] = 750.9\ \text{W}\end{aligned}$$

Then the total rate of heat loss becomes

$$\dot{Q}_{total} = \dot{Q}_{\substack{natural\\convection}} + \dot{Q}_{rad} = 748.1 + 750.9 = \mathbf{1499\ W}$$

The amount and cost of the heat loss during one year is

$$Q_{total} = \dot{Q}_{total}\Delta t = (1.499\ \text{kW})(8760\ \text{h}) = 13{,}131\ \text{kWh}$$

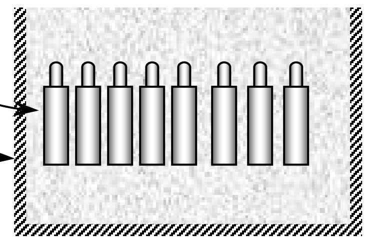
$$\text{Cost} = (13{,}131\ \text{kWh})(\$0.085/\text{kWh}) = \mathbf{\$1116}$$

9-25 Some cans move slowly in a hot water container made of sheet metal. It is proposed to insulate the side and bottom surfaces of the container for \$350. The simple payback period of the insulation to pay for itself from the energy it saves is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **3** Heat loss from the top surface is disregarded.

Properties Insulation will drop the outer surface temperature to a value close to the ambient temperature. The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature, which is unknown. We assume the surface temperature to be 26°C. The properties of air at the anticipated film temperature of (26+20)/2=23°C are (Table A-15)

$$k = 0.02536 \text{ W/m.°C}$$
$$\upsilon = 1.543\times10^{-5} \text{ m}^2/\text{s}$$
$$\Pr = 0.7301$$
$$\beta = \frac{1}{T_f} = \frac{1}{(23+273)\text{K}} = 0.00338 \text{ K}^{-1}$$



Analysis We start the solution process by "guessing" the outer surface temperature to be 26°C. We will check the accuracy of this guess later and repeat the calculations if necessary with a better guess based on the results obtained. The characteristic length in this case is the height of the tank, $L_c = L = 0.5$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\Pr = \frac{(9.81 \text{ m/s}^2)(0.00338 \text{ K}^{-1})(26-20 \text{ K})(0.5 \text{ m})^3}{(1.543\times10^{-5} \text{ m}^2/\text{s})^2}(0.7301) = 7.622\times10^7$$

$$\text{Nu} = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(7.622\times10^7)^{1/6}}{\left[1+\left(\frac{0.492}{0.7301}\right)^{9/16}\right]^{8/27}}\right\}^2 = 56.53$$

$$h = \frac{k}{L}Nu = \frac{0.02536 \text{ W/m.°C}}{0.5 \text{ m}}(56.53) = 2.868 \text{ W/m}^2.\text{°C}$$
$$A_s = 2\left[(0.5 \text{ m})(1.10 \text{ m}) + (0.5 \text{ m})(3.60 \text{ m})\right] = 4.7 \text{ m}^2$$

Then the total rate of heat loss from the outer surface of the insulated tank by convection and radiation becomes

$$\begin{aligned}\dot{Q} = \dot{Q}_{conv} + \dot{Q}_{rad} &= hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s^4 - T_{surr}^4)\\ &= (2.868 \text{ W/m}^2.\text{°C})(4.7 \text{ m}^2)(26-20)\text{°C}\\ &\quad + (0.1)(4.7 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2.\text{K}^4)[(26+273 \text{ K})^4 - (20+273 \text{ K})^4]\\ &= 97.5 \text{ W}\end{aligned}$$

In steady operation, the heat lost by the side surfaces of the tank must be equal to the heat lost from the exposed surface of the insulation by convection and radiation, which must be equal to the heat conducted through the insulation. The second conditions requires the surface temperature to be

$$\dot{Q} = \dot{Q}_{insulation} = kA_s\frac{T_{\text{tank}} - T_s}{L} \quad \rightarrow \quad 97.5 \text{ W} = (0.035 \text{ W/m.°C})(4.7 \text{ m}^2)\frac{(55-T_s)\text{°C}}{0.05 \text{ m}}$$

It gives T_s = 25.38°C, which is very close to the assumed temperature, 26°C. Therefore, there is no need to repeat the calculations.

The total amount of heat loss and its cost during one year are

$$Q_{total} = \dot{Q}_{total}\Delta t = (97.5 \text{ W})(8760 \text{ h}) = 853.7 \text{ kWh}$$

$$\text{Cost} = (853.7\text{ kWh})(\$0.085/\text{kWh}) = \$72.6$$

Then money saved during a one-year period due to insulation becomes

$$\text{Money saved} = \text{Cost}_{\substack{without\\insulation}} - \text{Cost}_{\substack{with\\insulation}} = \$1116 - \$72.6 = \$1043$$

where $1116 is obtained from the solution of Problem 9-24.

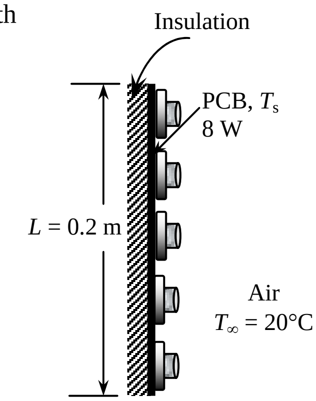
The insulation will pay for itself in

$$\text{Payback period} = \frac{\text{Cost}}{\text{Money saved}} = \frac{\$350}{\$1043/\,yr} = \mathbf{0.3354\ yr = 122\ days}$$

Discussion We would definitely recommend the installation of insulation in this case.

9-26 A printed circuit board (PCB) is placed in a room. The average temperature of the hot surface of the board is to be determined for different orientations.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **3** The heat loss from the back surface of the board is negligible.



Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2 = (45+20)/2 = 32.5$°C are (Table A-15)

$$k = 0.02607 \text{ W/m.°C}$$
$$\upsilon = 1.631\times10^{-5} \text{ m}^2/\text{s}$$
$$\Pr = 0.7275$$
$$\beta = \frac{1}{T_f} = \frac{1}{(32.5+273)\text{K}} = 0.003273 \text{ K}^{-1}$$

Analysis The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown

(*a*) **Vertical PCB** . We start the solution process by "guessing" the surface temperature to be 45°C for the evaluation of the properties and *h*. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the height of the PCB, $L_c = L = 0.2$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\Pr = \frac{(9.81 \text{ m/s}^2)(0.003273 \text{ K}^{-1})(45-20 \text{ K})(0.2 \text{ m})^3}{(1.631\times10^{-5} \text{ m}^2/\text{s})^2}(0.7275) = 1.756\times10^7$$

$$\text{Nu} = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(1.756\times10^7)^{1/6}}{\left[1+\left(\frac{0.492}{0.7275}\right)^{9/16}\right]^{8/27}}\right\}^2 = 36.78$$

$$h = \frac{k}{L}Nu = \frac{0.02607 \text{ W/m.°C}}{0.2 \text{ m}}(36.78) = 4.794 \text{ W/m}^2.\text{°C}$$
$$A_s = (0.15 \text{ m})(0.2 \text{ m}) = 0.03 \text{ m}^2$$

Heat loss by both natural convection and radiation heat can be expressed as

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s{}^4 - T_{surr}{}^4)$$
$$8 \text{ W} = (4.794 \text{ W/m}^2.\text{°C})(0.03 \text{ m}^2)(T_s - 20)\text{°C} + (0.8)(0.03 \text{ m}^2)(5.67\times10^{-8})\left[(T_s+273)^4 - (20+273 \text{ K})^4\right]$$

Its solution is

$$T_s = \mathbf{46.6°\,C}$$

which is sufficiently close to the assumed value of 45°C for the evaluation of the properties and *h*.

(*b*) **Horizontal, hot surface facing up** Again we assume the surface temperature to be 45°C and use the properties evaluated above. The characteristic length in this case is

$$L_c = \frac{A_s}{p} = \frac{(0.20 \text{ m})(0.15 \text{ m})}{2(0.2 \text{ m} + 0.15 \text{ m})} = 0.0429 \text{ m}.$$

Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\upsilon^2}\Pr = \frac{(9.81 \text{ m/s}^2)(0.003273 \text{ K}^{-1})(45-20 \text{ K})(0.0429 \text{ m})^3}{(1.631\times10^{-5} \text{ m}^2/\text{s})^2}(0.7275) = 1.728\times10^5$$

$$Nu = 0.54Ra^{1/4} = 0.54(1.728\times10^5)^{1/4} = 11.01$$

$$h = \frac{k}{L_c}Nu = \frac{0.02607 \text{ W/m.°C}}{0.0429 \text{ m}}(11.01) = 6.696 \text{ W/m}^2.\text{°C}$$

Heat loss by both natural convection and radiation heat can be expressed as

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s{}^4 - T_{surr}{}^4)$$

$$8\text{ W} = (6.696\text{ W/m}^2.°\text{C})(0.03\text{ m}^2)(T_s - 20)°\text{C} + (0.8)(0.03\text{ m}^2)(5.67\times10^{-8})[(T_s + 273)^4 - (20 + 273\text{ K})^4]$$

Its solution is

$$T_s = \mathbf{42.6°C}$$

which is sufficiently close to the assumed value of 45°C in the evaluation of the properties and *h*.

(c) **Horizontal, hot surface facing down** This time we expect the surface temperature to be higher, and assume the surface temperature to be 50 °C. We will check this assumption after obtaining result and repeat calculations with a better assumption, if necessary. The properties of air at the film temperature of

$T_f = \dfrac{T_s + T_\infty}{2} = \dfrac{50+20}{2} = 35°\text{C}$ are (Table A-15)

$$k = 0.02625\text{ W/m.°C}$$

$$\upsilon = 1.655\times10^{-5}\text{ m}^2\text{/s}$$

$$\text{Pr} = 0.7268$$

$$\beta = \frac{1}{T_f} = \frac{1}{(35+273)\text{K}} = 0.003247\text{ K}^{-1}$$

The characteristic length in this case is, from part (b), L_c = 0.0429 m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\upsilon^2}\text{Pr} = \frac{(9.81\text{ m/s}^2)(0.003247\text{ K}^{-1})(50-20\text{ K})(0.0429\text{ m})^3}{(1.655\times10^{-5}\text{ m}^2\text{/s})^2}(0.7268) = 166{,}379$$

$$Nu = 0.27Ra^{1/4} = 0.27(166{,}379)^{1/4} = 5.453$$

$$h = \frac{k}{L_c}Nu = \frac{0.02625\text{ W/m.°C}}{0.0429\text{ m}}(5.453) = 3.340\text{ W/m}^2.°\text{C}$$

Considering both natural convection and radiation heat loses

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s{}^4 - T_{surr}{}^4)$$

$$8\text{ W} = (3.340\text{ W/m}^2.°\text{C})(0.03\text{ m}^2)(T_s - 20)°\text{C} + (0.8)(0.03\text{ m}^2)(5.67\times10^{-8})[(T_s + 273)^4 - (20 + 273\text{ K})^4]$$

Its solution is

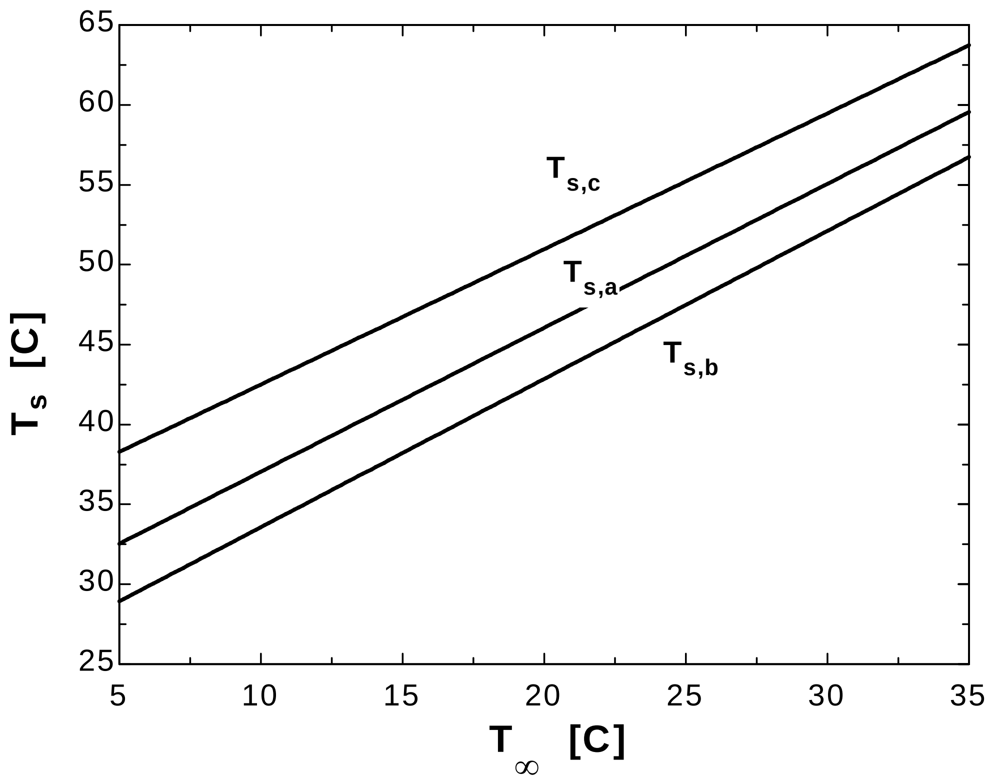
$$T_s = \mathbf{50.7°C}$$

which is very close to the assumed value. Therefore, there is no need to repeat calculations.

9-27 "!PROBLEM 9-27"

```
"GIVEN"
L=0.2 "[m]"
w=0.15 "[m]"
"T_infinity=20 [C], parameter to be varied"
Q_dot=8 "[W]"
epsilon=0.8 "parameter to be varied"
T_surr=T_infinity

"PROPERTIES"
Fluid$='air'
k=Conductivity(Fluid$, T=T_film)
Pr=Prandtl(Fluid$, T=T_film)
rho=Density(Fluid$, T=T_film, P=101.3)
mu=Viscosity(Fluid$, T=T_film)
nu=mu/rho
beta=1/(T_film+273)
T_film=1/2*(T_s_a+T_infinity)
sigma=5.67E-8 "[W/m^2-K^4], Stefan-Boltzmann constant"
g=9.807 "[m/s^2], gravitational acceleration"

"ANALYSIS"
"(a), plate is vertical"
delta_a=L
Ra_a=(g*beta*(T_s_a-T_infinity)*delta_a^3)/nu^2*Pr
Nusselt_a=0.59*Ra_a^0.25
h_a=k/delta_a*Nusselt_a
A=w*L
Q_dot=h_a*A*(T_s_a-T_infinity)+epsilon*A*sigma*((T_s_a+273)^4-(T_surr+273)^4)
"(b), plate is horizontal with hot surface facing up"
delta_b=A/p
p=2*(w+L)
Ra_b=(g*beta*(T_s_b-T_infinity)*delta_b^3)/nu^2*Pr
Nusselt_b=0.54*Ra_b^0.25
h_b=k/delta_b*Nusselt_b
Q_dot=h_b*A*(T_s_b-T_infinity)+epsilon*A*sigma*((T_s_b+273)^4-(T_surr+273)^4)
"(c), plate is horizontal with hot surface facing down"
delta_c=delta_b
Ra_c=Ra_b
Nusselt_c=0.27*Ra_c^0.25
h_c=k/delta_c*Nusselt_c
Q_dot=h_c*A*(T_s_c-T_infinity)+epsilon*A*sigma*((T_s_c+273)^4-(T_surr+273)^4)
```

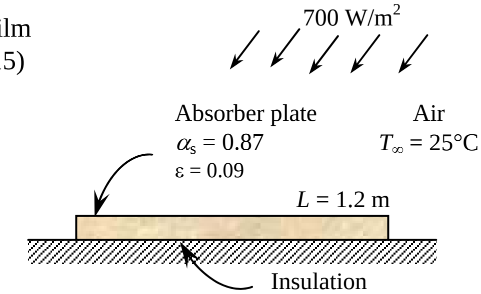
T_∞ [F]	$T_{s,a}$ [C]	$T_{s,b}$ [C]	$T_{s,c}$ [C]
5	32.54	28.93	38.29
7	34.34	30.79	39.97
9	36.14	32.65	41.66
11	37.95	34.51	43.35
13	39.75	36.36	45.04
15	41.55	38.22	46.73
17	43.35	40.07	48.42
19	45.15	41.92	50.12
21	46.95	43.78	51.81
23	48.75	45.63	53.51
25	50.55	47.48	55.21
27	52.35	49.33	56.91
29	54.16	51.19	58.62
31	55.96	53.04	60.32
33	57.76	54.89	62.03
35	59.56	56.74	63.74



9-28 Absorber plates whose back side is heavily insulated is placed horizontally outdoors. Solar radiation is incident on the plate. The equilibrium temperature of the plate is to be determined for two cases.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2 = (115+25)/2 = 70°C$ are (Table A-15)



$$k = 0.02881\ \text{W/m.°C}$$
$$\upsilon = 1.995\times10^{-5}\ \text{m}^2/\text{s}$$
$$\text{Pr} = 0.7177$$
$$\beta = \frac{1}{T_f} = \frac{1}{(70+273)\text{K}} = 0.002915\ \text{K}^{-1}$$

Analysis The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the surface temperature to be 115°C for the evaluation of the properties and *h*. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is $L_c = \frac{A_s}{p} = \frac{(1.2\ \text{m})(0.8\ \text{m})}{2(1.2\ \text{m}+0.8\ \text{m})} = 0.24\ \text{m}$. Then,

$$Ra = \frac{g\beta(T_s-T_\infty)L_c^3}{\upsilon^2}\text{Pr} = \frac{(9.81\ \text{m/s}^2)(0.002915\ \text{K}^{-1})(115-25\ \text{K})(0.24\ \text{m})^3}{(1.995\times10^{-5}\ \text{m}^2/\text{s})^2}(0.7177) = 6.414\times10^7$$

$$Nu = 0.54Ra^{1/4} = 0.54(6.414\times10^7)^{1/4} = 48.33$$

$$h = \frac{k}{L_c}Nu = \frac{0.02881\ \text{W/m.°C}}{0.24\ \text{m}}(48.33) = 5.801\ \text{W/m}^2.\text{°C}$$

$$A_s = (0.8\ \text{m})(1.2\ \text{m}) = 0.96\ \text{m}^2$$

In steady operation, the heat gain by the plate by absorption of solar radiation must be equal to the heat loss by natural convection and radiation. Therefore,

$$\dot{Q} = \alpha\dot{q}A_s = (0.87)(700\ \text{W/m}^2)(0.96\ \text{m}^2) = 584.6\ \text{W}$$

$$\dot{Q} = hA_s(T_s-T_\infty) + \varepsilon A_s\sigma(T_s^{\ 4}-T_{sky}^{\ 4})$$

$$584.6\ \text{W} = (5.801\ \text{W/m}^2.\text{°C})(0.96\ \text{m}^2)(T_s-25)\text{°C} + (0.09)(0.96\ \text{m}^2)(5.67\times10^{-8})[(T_s+273)^4-(10+273\ \text{K})^4]$$

Its solution is $T_s = \mathbf{115.6°C}$

which is identical to the assumed value. Therefore there is no need to repeat calculations.

If the absorber plate is made of ordinary aluminum which has a solar absorptivity of 0.28 and an emissivity of 0.07, the rate of solar gain becomes

$$\dot{Q} = \alpha\dot{q}A_s = (0.28)(700\ \text{W/m}^2)(0.96\ \text{m}^2) = 188.2\ \text{W}$$

Again noting that in steady operation the heat gain by the plate by absorption of solar radiation must be equal to the heat loss by natural convection and radiation, and using the convection coefficient determined above for convenience,

$$\dot{Q} = hA_s(T_s-T_\infty) + \varepsilon A_s\sigma(T_s^{\ 4}-T_{sky}^{\ 4})$$

$$188.2\ \text{W} = (5.801\ \text{W/m}^2.\text{°C})(0.96\ \text{m}^2)(T_s-25)\text{°C} + (0.07)(0.96\ \text{m}^2)(5.67\times10^{-8})[(T_s+273)^4-(10+273\ \text{K})^4]$$

Its solution is T_s = **55.2°C**

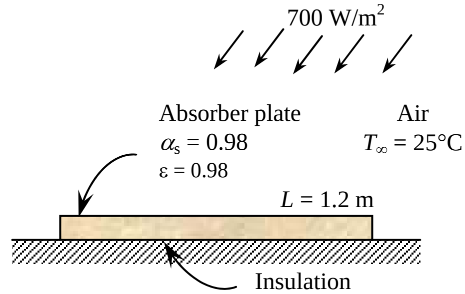
Repeating the calculations at the new film temperature of 40°C, we obtain

h = 4.524 W/m².°C and T_s = **62.8°C**

9-29 An absorber plate whose back side is heavily insulated is placed horizontally outdoors. Solar radiation is incident on the plate. The equilibrium temperature of the plate is to be determined for two cases.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2 = (70+25)/2 = 47.5°C$ are (Table A-15)



$$k = 0.02717 \text{ W/m.°C}$$
$$\upsilon = 1.774\times10^{-5} \text{ m}^2\text{/s}$$
$$\text{Pr} = 0.7235$$
$$\beta = \frac{1}{T_f} = \frac{1}{(47.5+273)\text{K}} = 0.00312 \text{ K}^{-1}$$

Analysis The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the surface temperature to be 70°C for the evaluation of the properties and h. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is $L_c = \dfrac{A_s}{p} = \dfrac{(1.2\text{ m})(0.8\text{ m})}{2(1.2\text{ m}+0.8\text{ m})} = 0.24\text{ m}$. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\upsilon^2}\text{Pr} = \frac{(9.81\text{ m/s}^2)(0.00312\text{ K}^{-1})(70-25\text{ K})(0.24\text{ m})^3}{(1.774\times10^{-5}\text{ m}^2\text{/s})^2}(0.7235) = 4.379\times10^7$$

$$Nu = 0.54Ra^{1/4} = 0.54(4.379\times10^7)^{1/4} = 43.93$$

$$h = \frac{k}{L_c}Nu = \frac{0.02717\text{ W/m.°C}}{0.24\text{ m}}(43.93) = 4.973\text{ W/m}^2\text{.°C}$$

$$A_s = (0.8\text{ m})(1.2\text{ m}) = 0.96\text{ m}^2$$

In steady operation, the heat gain by the plate by absorption of solar radiation must be equal to the heat loss by natural convection and radiation. Therefore,

$$\dot{Q} = \alpha\dot{q}A_s = (0.98)(700\text{ W/m}^2)(0.96\text{ m}^2) = 658.6\text{ W}$$

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s{}^4 - T_{surr}{}^4)$$

$$658.6\text{ W} = (4.973\text{ W/m}^2\text{.°C})(0.96\text{ m}^2)(T_s - 25)\text{°C} + (0.98)(0.96\text{ m}^2)(5.67\times10^{-8})[(T_s+273)^4 - (10+273\text{ K})^4]$$

Its solution is $\quad T_s = \mathbf{73.5°C}$

which is close to the assumed value. Therefore there is no need to repeat calculations.

For a white painted absorber plate, the solar absorptivity is 0.26 and the emissivity is 0.90. Then the rate of solar gain becomes

$$\dot{Q} = \alpha\dot{q}A_s = (0.26)(700\text{ W/m}^2)(0.96\text{ m}^2) = 174.7\text{ W}$$

Again noting that in steady operation the heat gain by the plate by absorption of solar radiation must be equal to the heat loss by natural convection and radiation, and using the convection coefficient determined above for convenience (actually, we should calculate the new h using data at a lower temperature, and iterating if necessary for better accuracy),

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s{}^4 - T_{surr}{}^4)$$

$$174.7\text{ W} = (4.973\text{ W/m}^2\text{.°C})(0.96\text{ m}^2)(T_s - 25)\text{°C} + (0.90)(0.96\text{ m}^2)(5.67\times10^{-8})[(T_s+273)^4 - (10+273\text{ K})^4]$$

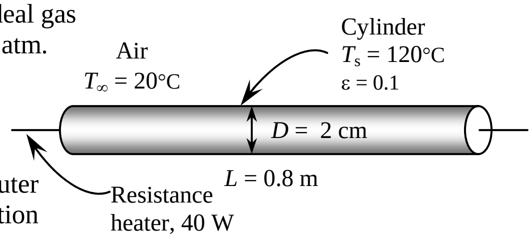
Its solution is $\quad T_s = \mathbf{35.0°C}$

Discussion If we recalculated the h using air properties at 30°C, we would obtain

h = 3.47 W/m^2.°C and T_s = 36.6°C.

9-30 A resistance heater is placed along the centerline of a horizontal cylinder whose two circular side surfaces are well insulated. The natural convection heat transfer coefficient and whether the radiation effect is negligible are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.



Analysis The heat transfer surface area of the cylinder is

$$A = \pi DL = \pi(0.02\text{ m})(0.8\text{ m}) = 0.05027\text{ m}^2$$

Noting that in steady operation the heat dissipated from the outer surface must equal to the electric power consumed, and radiation is negligible, the convection heat transfer is determined to be

$$\dot{Q} = hA_s(T_s - T_\infty) \rightarrow \quad h = \frac{\dot{Q}}{A_s(T_s - T_\infty)} = \frac{40\text{ W}}{(0.05027\text{ m}^2)(120-20)°\text{C}} = \mathbf{7.96\ W/m^2.°C}$$

The radiation heat loss from the cylinder is

$$\begin{aligned}\dot{Q}_{rad} &= \varepsilon A_s \sigma (T_s{}^4 - T_{surr}{}^4) \\ &= (0.1)(0.05027\text{ m}^2)(5.67\times10^{-8}\text{ W/m}^2.\text{K}^4)[(120+273\text{ K})^4 - (20+273\text{ K})^4] = 4.7\text{ W}\end{aligned}$$

Therefore, the fraction of heat loss by radiation is

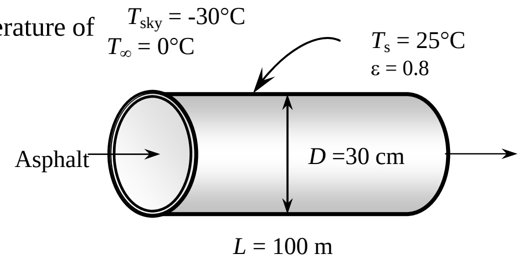
$$\text{Radiation fraction} = \frac{\dot{Q}_{radiation}}{\dot{Q}_{total}} = \frac{4.7\text{ W}}{40\text{ W}} = 0.1175 = 11.8\%$$

which is greater than 5%. Therefore, the radiation effect is still more than acceptable, and corrections must be made for the radiation effect.

9-31 A thick fluid flows through a pipe in calm ambient air. The pipe is heated electrically. The power rating of the electric resistance heater and the cost of electricity during a 10-h period are to be determined.√

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (25+0)/2 = 12.5°C$ are (Table A-15)



$$k = 0.02458 \text{ W/m.°C}$$
$$\upsilon = 1.448\times10^{-5} \text{ m}^2/\text{s}$$
$$\text{Pr} = 0.7330$$
$$\beta = \frac{1}{T_f} = \frac{1}{(12.5+273)\text{K}} = 0.003503 \text{ K}^{-1}$$

Analysis The characteristic length in this case is the outer diameter of the pipe, $L_c = D = 0.3$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003503 \text{ K}^{-1})(25-0 \text{ K})(0.3 \text{ m})^3}{(1.448\times10^{-5} \text{ m}^2/\text{s})^2}(0.7330) = 8.106\times10^7$$

$$Nu = \left\{0.6 + \frac{0.387 Ra^{1/6}}{\left[1+\left(0.559/\text{Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(8.106\times10^7)^{1/6}}{\left[1+\left(0.559/0.7330\right)^{9/16}\right]^{8/27}}\right\}^2 = 53.29$$

$$h = \frac{k}{L_c}Nu = \frac{0.02458 \text{ W/m.°C}}{0.3 \text{ m}}(53.29) = 4.366 \text{ W/m}^2\text{.°C}$$

$$A_s = \pi DL = \pi(0.3 \text{ m})(100 \text{ m}) = 94.25 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (4.366 \text{ W/m}^2\text{.°C})(94.25 \text{ m}^2)(25-0)\text{°C} = 10{,}287 \text{ W}$$

The radiation heat loss from the cylinder is

$$\dot{Q}_{rad} = \varepsilon A_s\sigma(T_s^{\,4} - T_{surr}^{\,4})$$
$$= (0.8)(94.25 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2\text{.K}^4)[(25+273 \text{ K})^4 - (-30+273 \text{ K})^4] = 18{,}808 \text{ W}$$

Then,

$$\dot{Q}_{total} = \dot{Q}_{\substack{natural\\convection}} + \dot{Q}_{radiation} = 10{,}287 + 18{,}808 = 29{,}094 \text{ W} = \mathbf{29.1\ kW}$$

The total amount and cost of heat loss during a 10 hour period is

$$Q = \dot{Q}\Delta t = (29.1 \text{ kW})(10 \text{ h}) = 290.9 \text{ kWh}$$

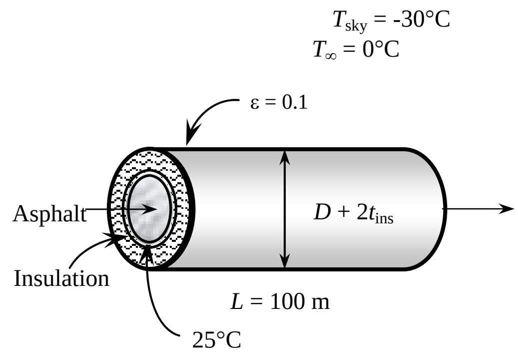
$$\text{Cost} = (290.9 \text{ kWh})(\$0.09/\text{kWh}) = \mathbf{\$26.18}$$

9-32 A fluid flows through a pipe in calm ambient air. The pipe is heated electrically. The thickness of the insulation needed to reduce the losses by 85% and the money saved during 10-h are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.

Properties Insulation will drop the outer surface temperature to a value close to the ambient temperature, and possible below it because of the very low sky temperature for radiation heat loss. For convenience, we use the properties of air at 1 atm and 5°C (the anticipated film temperature) (Table A-15),

$$k = 0.02401\ \text{W/m.°C}$$
$$\upsilon = 1.382\times10^{-5}\ \text{m}^2\text{/s}$$
$$\text{Pr} = 0.7350$$
$$\beta = \frac{1}{T_f} = \frac{1}{(5+273)\text{K}} = 0.003597\ \text{K}^{-1}$$



Analysis The rate of heat loss in the previous problem was obtained to be 29,094 W. Noting that insulation will cut down the heat losses by 85%, the rate of heat loss will be

$$\dot{Q} = (1-0.85)\dot{Q}_{\text{no insulation}} = 0.15\times 29{,}094\ \text{W} = 4364\ \text{W}$$

The amount of energy and money insulation will save during a 10-h period is simply determined from

$$Q_{saved,total} = \dot{Q}_{saved}\Delta t = (0.85\times 29.094\ \text{kW})(10\ \text{h}) = 247.3\ \text{kWh}$$

$$\text{Money saved} = \text{(Energy saved)(Unit cost of energy)} = (247.3\ \text{kWh})(\$0.09/\text{kWh}) = \mathbf{\$22.26}$$

The characteristic length in this case is the outer diameter of the insulated pipe, $L_c = D + 2t_{insul} = 0.3 + 2t_{insul}$ where t_{insul} is the thickness of insulation in m. Then the problem can be formulated for T_s and t_{insul} as follows:

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\upsilon^2}\text{Pr} = \frac{(9.81\ \text{m/s}^2)(0.003597\ \text{K}^{-1})[(T_s-273)\text{K}](0.3+2t_{insul})^3}{(1.382\times10^{-5}\ \text{m}^2\text{/s})^2}(0.7350)$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/\text{Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/0.7350\right)^{9/16}\right]^{8/27}}\right\}^2$$

$$h = \frac{k}{L_c}Nu = \frac{0.02401\ \text{W/m.°C}}{L_c}Nu$$

$$A_s = \pi D_0 L = \pi(0.3+2t_{insul})(100\ \text{m})$$

The total rate of heat loss from the outer surface of the insulated pipe by convection and radiation becomes

$$\dot{Q} = \dot{Q}_{conv} + \dot{Q}_{rad} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s^{\,4} - T_{surr}^{\,4})$$
$$4364 = hA_s(T_s - 273) + (0.1)A_s(5.67\times10^{-8}\ \text{W/m}^2.\text{K}^4)[T_s^4 - (-30+273\ \text{K})^4]$$

In steady operation, the heat lost by the side surfaces of the pipe must be equal to the heat lost from the exposed surface of the insulation by convection and radiation, which must be equal to the heat conducted through the insulation. Therefore,

$$\dot{Q} = \dot{Q}_{insulation} = \frac{2\pi kL(T_{\text{tank}} - T_s)}{\ln(D_o/D)} \rightarrow 4364\ \text{W} = \frac{2\pi(0.035\ \text{W/m.°C})(100\ \text{m})(298-T_s)\text{K}}{\ln[(0.3+2t_{insul})/0.3]}$$

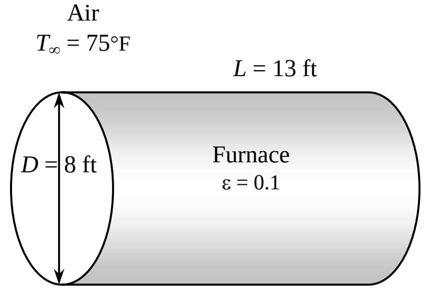
The solution of all of the equations above simultaneously using an equation solver gives T_s = 281.5 K = 8.5°C and t_{insul} = **0.013 m = 1.3 cm**.

Note that the film temperature is (8.5+0)/2 = 4.25°C which is very close to the assumed value of 5°C. Therefore, there is no need to repeat the calculations using properties at this new film temperature.

9-33E An industrial furnace that resembles a horizontal cylindrical enclosure whose end surfaces are well insulated. The highest allowable surface temperature of the furnace and the annual cost of this loss to the plant are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2=(140+75)/2=107.5°F$ are (Table A-15)



$$k = 0.01546 \text{ Btu/h.ft.°F}$$
$$\upsilon = 0.1851\times10^{-3} \text{ ft}^2\text{/s}$$
$$\text{Pr} = 0.7249$$
$$\beta = \frac{1}{T_f} = \frac{1}{(107.5+460)\text{R}} = 0.001762 \text{ R}^{-1}$$

Analysis The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the surface temperature to be 140°F for the evaluation of the properties and h. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the outer diameter of the furnace, $L_c = D = 8$ ft. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2}\text{Pr} = \frac{(32.2 \text{ ft/s}^2)(0.001762 \text{ R}^{-1})(140-75 \text{ R})(8 \text{ ft})^3}{(0.1851\times10^{-3} \text{ ft}^2\text{/s})^2}(0.7249) = 3.996\times10^{10}$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/\text{Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(3.996\times10^{10})^{1/6}}{\left[1+\left(0.559/0.7249\right)^{9/16}\right]^{8/27}}\right\}^2 = 376.9$$

$$h = \frac{k}{D}Nu = \frac{0.01546 \text{ Btu/h.ft.°F}}{8 \text{ ft}}(376.9) = 0.7287 \text{ Btu/h.ft}^2\text{.°F}$$

$$A_s = \pi DL = \pi(8 \text{ ft})(13 \text{ ft}) = 326.7 \text{ ft}^2$$

The total rate of heat generated in the furnace is

$$\dot{Q}_{gen} = (0.82)(48 \text{ therms/h})(100{,}000 \text{ Btu/therm}) = 3.936\times10^6 \text{ Btu/h}$$

Noting that 1% of the heat generated can be dissipated by natural convection and radiation ,

$$\dot{Q} = (0.01)(3.936\times10^6 \text{ Btu/h}) = 39{,}360 \text{ Btu/h}$$

The total rate of heat loss from the furnace by natural convection and radiation can be expressed as

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s^{\,4} - T_{surr}^{\,4})$$

$$39{,}360 \text{ Btu/h} = (0.7287 \text{ Btu/h.ft}^2\text{.°F})(326.7 \text{ ft}^2)[T_s - (75+460 \text{ R})]$$
$$+ (0.85)(326.7 \text{ m}^2)(0.1714\times10^{-8} \text{ Btu/h.ft}^2\text{.R}^4)[T_s^{\,4} - (75+460 \text{ R})^4]$$

Its solution is

$$T_s = 601.8 \text{ R} = \mathbf{141.8°F}$$

which is very close to the assumed value. Therefore, there is no need to repeat calculations.

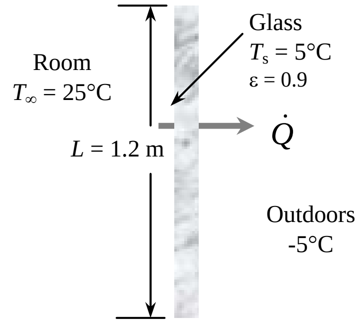
The total amount of heat loss and its cost during a-2800 hour period is

$$Q_{total} = \dot{Q}_{total}\Delta t = (39{,}360 \text{ Btu/h})(2800 \text{ h}) = 1.102\times10^8 \text{ Btu}$$

$$\text{Cost} = (1.102\times10^8/100{,}000 \text{ therm})(\$0.65/\text{therm}) = \mathbf{\$716.4}$$

9-34 A glass window is considered. The convection heat transfer coefficient on the inner side of the window, the rate of total heat transfer through the window, and the combined natural convection and radiation heat transfer coefficient on the outer surface of the window are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.



Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (5+25)/2 = 15°C$ are (Table A-15)

$$k = 0.02476 \text{ W/m.°C}$$
$$\upsilon = 1.471\times10^{-5} \text{ m}^2\text{/s}$$
$$\text{Pr} = 0.7323$$
$$\beta = \frac{1}{T_f} = \frac{1}{(15+273)\text{K}} = 0.003472 \text{ K}^{-1}$$

Analysis (*a*) The characteristic length in this case is the height of the window, $L_c = L = 1.2$ m. Then,

$$Ra = \frac{g\beta(T_\infty - T_s)L_c^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003472 \text{ K}^{-1})(25-5\text{ K})(1.2 \text{ m})^3}{(1.471\times10^{-5} \text{ m}^2\text{/s})^2}(0.7323) = 3.986\times10^9$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(3.986\times10^9)^{1/6}}{\left[1+\left(\frac{0.492}{0.7323}\right)^{9/16}\right]^{8/27}}\right\}^2 = 189.7$$

$$h = \frac{k}{L}Nu = \frac{0.02476 \text{ W/m.°C}}{1.2 \text{ m}}(189.7) = \mathbf{3.915\ W/m^2.°C}$$
$$A_s = (1.2 \text{ m})(2 \text{ m}) = 2.4 \text{ m}^2$$

(*b*) The sum of the natural convection and radiation heat transfer from the room to the window is

$$\dot{Q}_{\text{convection}} = hA_s(T_\infty - T_s) = (3.915 \text{ W/m}^2.°\text{C})(2.4 \text{ m}^2)(25-5)°\text{C} = 187.9 \text{ W}$$
$$\dot{Q}_{\text{radiation}} = \varepsilon A_s\sigma(T_{surr}{}^4 - T_s{}^4)$$
$$= (0.9)(2.4 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2.\text{K}^4)[(25+273\text{ K})^4 - (5+273\text{ K})^4] = 234.3 \text{ W}$$
$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{convection}} + \dot{Q}_{\text{radiation}} = 187.9 + 234.3 = \mathbf{422.2\ W}$$

(*c*) The outer surface temperature of the window can be determined from

$$\dot{Q}_{\text{total}} = \frac{kA_s}{t}(T_{s,i} - T_{s,o}) \longrightarrow T_{s,o} = T_{s,i} - \frac{\dot{Q}_{\text{total}}t}{kA_s} = 5°\text{C} - \frac{(346 \text{ W})(0.006 \text{ m})}{(0.78 \text{ W/m.°C})(2.4 \text{ m}^2)} = 3.65°\text{C}$$

Then the combined natural convection and radiation heat transfer coefficient on the outer window surface becomes

$$\dot{Q}_{\text{total}} = h_{\text{combined}}A_s(T_{s,o} - T_{\infty,o})$$

or $$h_{\text{combined}} = \frac{\dot{Q}_{\text{total}}}{A_s(T_{s,o} - T_{\infty,o})} = \frac{346 \text{ W}}{(2.4 \text{ m}^2)[3.65-(-5)]°\text{C}} = \mathbf{20.35\ W/m^2.°C}$$

Note that $\Delta T = \dot{Q}R$ and thus the thermal resistance R of a layer is proportional to the temperature drop across that layer. Therefore, the fraction of thermal resistance of the glass is equal to the ratio of the temperature drop across the glass to the overall temperature difference,

$$\frac{R_{\text{glass}}}{R_{\text{total}}} = \frac{\Delta T_{\text{glass}}}{\Delta TR_{\text{total}}} = \frac{5-3.65}{25-(-5)} = 0.045 \quad (\text{or } 4.5\%)$$

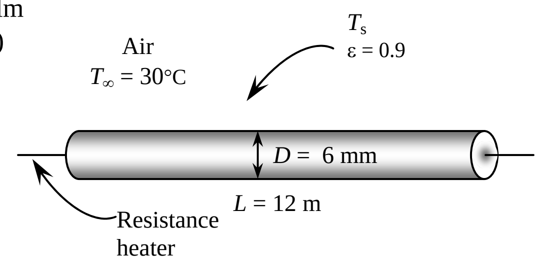
which is low. Thus it is reasonable to neglect the thermal resistance of the glass.

9-35 An insulated electric wire is exposed to calm air. The temperature at the interface of the wire and the plastic insulation is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2 = (50+30)/2 = 40°C$ are (Table A-15)

$$k = 0.02662 \text{ W/m.°C}$$
$$\upsilon = 1.702\times10^{-5} \text{ m}^2/\text{s}$$
$$\text{Pr} = 0.7255$$
$$\beta = \frac{1}{T_f} = \frac{1}{(40+273)\text{K}} = 0.003195 \text{ K}^{-1}$$



Analysis The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the surface temperature to be 50°C for the evaluation of the properties and h. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the outer diameter of the insulated wire $L_c = D = 0.006$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003195 \text{ K}^{-1})(50-30 \text{ K})(0.006 \text{ m})^3}{(1.702\times10^{-5} \text{ m}^2/\text{s})^2}(0.7255) = 339.3$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/\text{Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(339.3)^{1/6}}{\left[1+\left(0.559/0.7255\right)^{9/16}\right]^{8/27}}\right\}^2 = 2.101$$

$$h = \frac{k}{D}Nu = \frac{0.02662 \text{ W/m.°C}}{0.006 \text{ m}}(2.101) = 9.327 \text{ W/m}^2.°\text{C}$$

$$A_s = \pi DL = \pi(0.006 \text{ m})(12 \text{ m}) = 0.2262 \text{ m}^2$$

The rate of heat generation, and thus the rate of heat transfer is

$$\dot{Q} = VI = (8 \text{ V})(10 \text{ A}) = 80 \text{ W}$$

Considering both natural convection and radiation, the total rate of heat loss can be expressed as

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s^4 - T_{surr}^4)$$

$$80 \text{ W} = (9.327 \text{ W/m}^2.°\text{C})(0.226 \text{ m}^2)(T_s - 30)°\text{C}$$
$$+(0.9)(0.2262 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2.\text{K}^4)[(T_s+273)^4 - (30+273 \text{ K})^4]$$

Its solution is

$$T_s = 52.6°\text{C}$$

which is close to the assumed value of 50°C. Then the temperature at the interface of the wire and the plastic cover in steady operation becomes

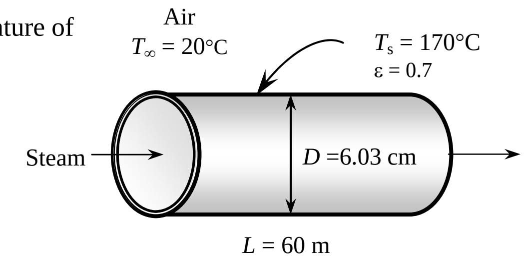
$$\dot{Q} = \frac{2\pi kL}{\ln(D_2/D_1)}(T_i - T_s) \longrightarrow T_i = T_s + \frac{\dot{Q}\ln(D_2/D_1)}{2\pi kL} = 52.6°\text{C} + \frac{(80 \text{ W})\ln(6/3)}{2\pi(0.15 \text{ W/m.°C})(12 \text{ m})} = \mathbf{57.5°C}$$

9-36 A steam pipe extended from one end of a plant to the other with no insulation on it. The rate of heat loss from the steam pipe and the annual cost of those heat losses are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2$ = (170+20)/2 = 95°C are (Table A-15)

$$k = 0.0306 \text{ W/m.°C}$$
$$\upsilon = 2.252\times10^{-5} \text{ m}^2\text{/s}$$
$$\text{Pr} = 0.7121$$
$$\beta = \frac{1}{T_f} = \frac{1}{(95+273)\text{K}} = 0.002717 \text{ K}^{-1}$$



Analysis The characteristic length in this case is the outer diameter of the pipe, $L_c = D = 0.0603\text{ m}$. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2}\text{Pr} = \frac{(9.81\text{ m/s}^2)(0.002717\text{ K}^{-1})(170-20\text{ K})(0.0603\text{ m})^3}{(2.252\times10^{-5}\text{ m}^2\text{/s})^2}(0.7121) = 1.231\times10^6$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/\text{Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(1.231\times10^6)^{1/6}}{\left[1+\left(0.559/0.7121\right)^{9/16}\right]^{8/27}}\right\}^2 = 15.42$$

$$h = \frac{k}{D}Nu = \frac{0.0306\text{ W/m.°C}}{0.0603\text{ m}}(15.42) = 7.823\text{ W/m}^2\text{.°C}$$

$$A_s = \pi DL = \pi(0.0603\text{ m})(60\text{ m}) = 11.37\text{ m}^2$$

Then the total rate of heat transfer by natural convection and radiation becomes

$$\begin{aligned}\dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s^4 - T_{surr}^4)\\ &= (7.823\text{ W/m}^2\text{.°C})(11.37\text{ m}^2)(170-20)\text{°C}\\ &\quad + (0.7)(11.37\text{ m}^2)(5.67\times10^{-8}\text{ W/m}^2\text{.K}^4)[(170+273\text{ K})^4 - (20+273\text{ K})^4]\\ &= 27{,}388\,W = \mathbf{27.4\ kW}\end{aligned}$$

The total amount of gas consumption and its cost during a one-year period is

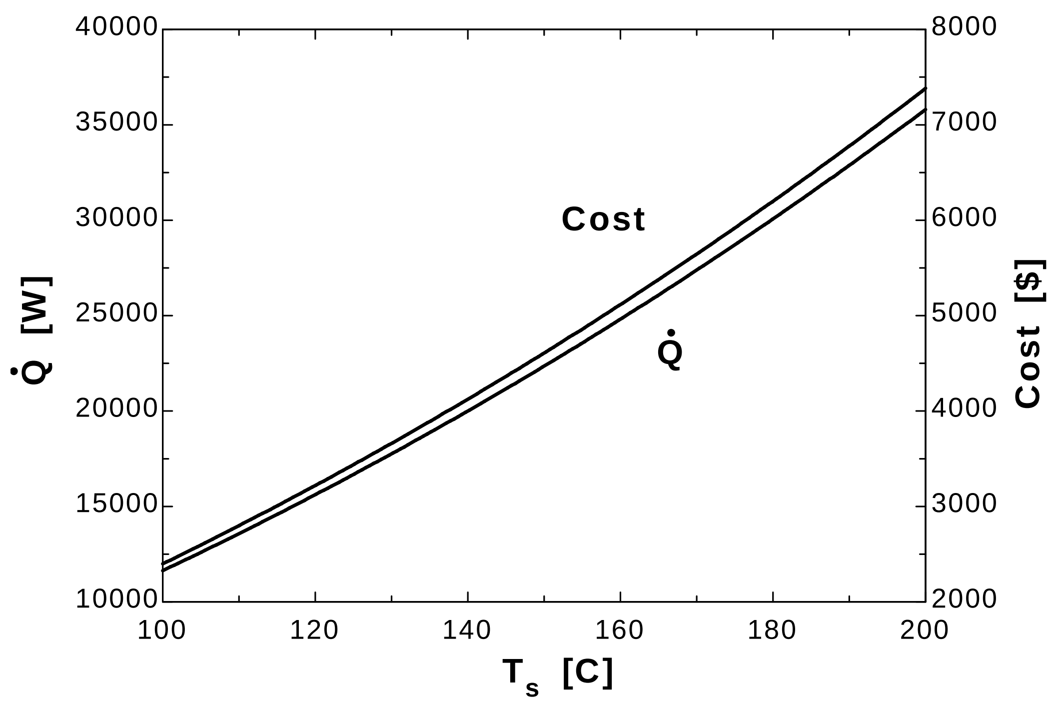
$$Q_{gas} = \frac{\dot{Q}\Delta t}{\eta} = \frac{27.388\text{ kJ/s}}{0.78}\left(\frac{1\text{ therm}}{105{,}500\text{ kJ}}\right)(8760\text{ h/yr}\times3600\text{ s/h}) = 10{,}496\text{ therms/yr}$$

$$\text{Cost} = (10{,}496\text{ therms/yr})(\$0.538/\text{therm}) = \mathbf{\$5647/yr}$$

9-37 "!PROBLEM 9-37"

```
"GIVEN"
L=60 "[m]"
D=0.0603 "[m]"
T_s=170 "[C], parameter to be varied"
T_infinity=20 "[C]"
epsilon=0.7
T_surr=T_infinity
eta_furnace=0.78
UnitCost=0.538 "[$/therm]"
time=24*365 "[h]"

"PROPERTIES"
Fluid$='air'
k=Conductivity(Fluid$, T=T_film)
Pr=Prandtl(Fluid$, T=T_film)
rho=Density(Fluid$, T=T_film, P=101.3)
mu=Viscosity(Fluid$, T=T_film)
nu=mu/rho
beta=1/(T_film+273)
T_film=1/2*(T_s+T_infinity)
sigma=5.67E-8 "[W/m^2-K^4], Stefan-Boltzmann constant"
g=9.807 "[m/s^2], gravitational acceleration"

"ANALYSIS"
delta=D
Ra=(g*beta*(T_s-T_infinity)*delta^3)/nu^2*Pr
Nusselt=(0.6+(0.387*Ra^(1/6))/(1+(0.559/Pr)^(9/16))^(8/27))^2
h=k/delta*Nusselt
A=pi*D*L
Q_dot=h*A*(T_s-T_infinity)+epsilon*A*sigma*((T_s+273)^4-(T_surr+273)^4)
Q_gas=(Q_dot*time)/eta_furnace*Convert(h, s)*Convert(J, kJ)*Convert(kJ, therm)
Cost=Q_gas*UnitCost
```

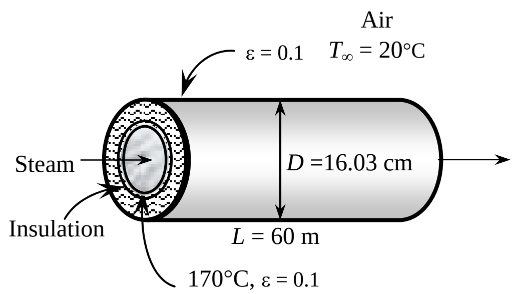
T_s [C]	Q [W]	Cost [$]
100	11636	2399
105	12594	2597
110	13577	2799
115	14585	3007
120	15618	3220
125	16676	3438
130	17760	3661
135	18869	3890
140	20004	4124
145	21166	4364
150	22355	4609
155	23570	4859
160	24814	5116
165	26085	5378
170	27385	5646
175	28713	5920
180	30071	6200
185	31459	6486
190	32877	6778
195	34327	7077
200	35807	7382



9-38 A steam pipe extended from one end of a plant to the other. It is proposed to insulate the steam pipe for \$750. The simple payback period of the insulation to pay for itself from the energy it saves are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2$ = (35+20)/2 = 27.5°C are (Table A-15)



$$k = 0.0257 \text{ W/m.°C}$$
$$\upsilon = 1.584\times10^{-5} \text{ m}^2\text{/s}$$
$$\Pr = 0.7289$$
$$\beta = \frac{1}{T_f} = \frac{1}{(27.5+273)\text{K}} = 0.003328 \text{ K}^{-1}$$

Analysis Insulation will drop the outer surface temperature to a value close to the ambient temperature. The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the outer surface temperature to be 35°C for the evaluation of the properties and h. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the outer diameter of the insulated pipe, $L_c = D = 0.1603 \text{ m}$. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2}\Pr = \frac{(9.81 \text{ m/s}^2)(0.003328 \text{ K}^{-1})(35-20 \text{ K})(0.1603 \text{ m})^3}{(1.584\times10^{-5} \text{ m}^2\text{/s})^2}(0.7289) = 5.856\times10^6$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/\Pr\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(5.856\times10^6)^{1/6}}{\left[1+\left(0.559/0.7289\right)^{9/16}\right]^{8/27}}\right\}^2 = 24.23$$

$$h = \frac{k}{D}Nu = \frac{0.0257 \text{ W/m.°C}}{0.1603 \text{ m}}(24.23) = 3.884 \text{ W/m}^2\text{.°C}$$

$$A_s = \pi DL = \pi(0.1603 \text{ m})(60 \text{ m}) = 30.22 \text{ m}^2$$

Then the total rate of heat loss from the outer surface of the insulated pipe by convection and radiation becomes

$$\begin{aligned}\dot{Q} = \dot{Q}_{conv} + \dot{Q}_{rad} &= hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s{}^4 - T_{surr}{}^4)\\ &= (3.884 \text{ W/m}^2\text{.°C})(30.22 \text{ m}^2)(35-20)\text{°C}\\ &+ (0.1)(30.22 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2\text{.K}^4)[(35+273 \text{ K})^4 - (20+273 \text{ K})^4]\\ &= 2039 \text{ W}\end{aligned}$$

In steady operation, the heat lost from the exposed surface of the insulation by convection and radiation must be equal to the heat conducted through the insulation. This requirement gives the surface temperature to be

$$\dot{Q} = \dot{Q}_{\text{insulation}} = \frac{T_{s.i} - T_s}{R_{ins}} = \frac{T_{s.i} - T_s}{\dfrac{\ln(D_2/D_1)}{2\pi kL}} \rightarrow 2039 \text{ W} = \frac{(170 - T_s)\text{°C}}{\dfrac{\ln(16.03/6.03)}{2\pi(0.038 \text{ W/m.°C})(60 \text{ m})}}$$

It gives 30.8°C for the surface temperature, which is somewhat different than the assumed value of 35°C. Repeating the calculations with other surface temperatures gives

$$T_s = 34.3\text{°C} \quad \text{and } \dot{Q} = 1988 \text{ W}$$

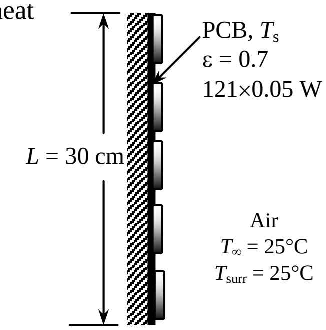
Heat loss and its cost without insulation was determined in the Prob. 9-36 to be 27.388 kW and \$5647. Then the reduction in the heat losses becomes

$$\dot{Q}_{\text{saved}} = 27.388 - 1.988 \cong 25.40 \text{ kW} \quad \text{or} \quad 25.388/27.40 = 0.927 \quad (92.7\%)$$

Therefore, the money saved by insulation will be 0.921×(\$5647/yr) = **\$5237/yr** which will pay for the cost of \$750 in \$750/(\$5237/yr)=0.1432 year = **52.3 days**.

9-39 A circuit board containing square chips is mounted on a vertical wall in a room. The surface temperature of the chips is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** The heat transfer from the back side of the circuit board is negligible.



Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2$ = (35+25)/2 = 30°C are (Table A-15)

$$k = 0.02588 \text{ W/m.°C}$$
$$\upsilon = 1.608\times10^{-5} \text{ m}^2\text{/s}$$
$$\text{Pr} = 0.7282$$
$$\beta = \frac{1}{T_f} = \frac{1}{(30+273)\text{K}} = 0.0033 \text{ K}^{-1}$$

Analysis The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the surface temperature to be 35°C for the evaluation of the properties and h. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the height of the board, $L_c = L = 0.3$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.0033 \text{ K}^{-1})(35-25 \text{ K})(0.3 \text{ m})^3}{(1.608\times10^{-5} \text{ m}^2\text{/s})^2}(0.7282) = 2.463\times10^7$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(2.463\times10^7)^{1/6}}{\left[1+\left(\frac{0.492}{0.7282}\right)^{9/16}\right]^{8/27}}\right\}^2 = 40.57$$

$$h = \frac{k}{L}Nu = \frac{0.02588 \text{ W/m.°C}}{0.3 \text{ m}}(40.57) = 3.50 \text{ W/m}^2\text{.°C}$$
$$A_s = (0.3 \text{ m})^2 = 0.09 \text{ m}^2$$

Considering both natural convection and radiation, the total rate of heat loss can be expressed as

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s^{\,4} - T_{surr}^{\,4})$$

$$(121\times0.05) \text{ W} = (3.50 \text{ W/m}^2\text{.°C})(0.09 \text{ m}^2)(T_s - 25)\text{°C}$$
$$+(0.7)(0.09 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2\text{.K}^4)[(T_s + 273 \text{ K})^4 - (25+273 \text{ K})^4]$$

Its solution is

T_s = **33.5°C**

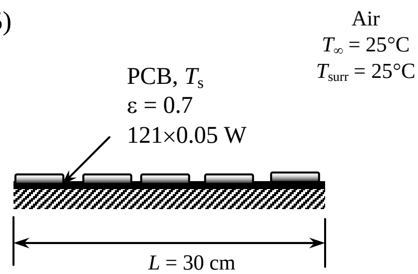
which is sufficiently close to the assumed value in the evaluation of properties and h. Therefore, there is no need to repeat calculations by reevaluating the properties and h at the new film temperature.

9-40 A circuit board containing square chips is positioned horizontally in a room. The surface temperature of the chips is to be determined for two orientations.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** The heat transfer from the back side of the circuit board is negligible.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2 = (35+25)/2 = 30°C$ are (Table A-15)

$$k = 0.02588 \text{ W/m.°C}$$
$$\upsilon = 1.608\times10^{-5} \text{ m}^2\text{/s}$$
$$\Pr = 0.7282$$
$$\beta = \frac{1}{T_f} = \frac{1}{(30+273)\text{K}} = 0.0033 \text{ K}^{-1}$$



Analysis The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the surface temperature to be 35°C for the evaluation of the properties and h. The characteristic length for both cases is determined from

$$L_c = \frac{A_s}{p} = \frac{(0.3\text{ m})^2}{2[(0.3\text{ m})+(0.3\text{ m})]} = 0.075\text{ m}.$$

Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\upsilon^2}\Pr = \frac{(9.81\text{ m/s}^2)(0.00333\text{ K}^{-1})(35-25\text{ K})(0.075\text{ m})^3}{(1.608\times10^{-5}\text{ m}^2\text{/s})^2}(0.7282) = 3.848\times10^5$$

(a) Chips (hot surface) facing up:

$$Nu = 0.54Ra^{1/4} = 0.54(3.848\times10^5)^{1/4} = 13.45$$
$$h = \frac{k}{L_c}Nu = \frac{0.02588\text{ W/m.°C}}{0.075\text{ m}}(13.45) = 4.641\text{ W/m}^2\text{.°C}$$
$$A_s = (0.3\text{ m})^2 = 0.09\text{ m}^2$$

Considering both natural convection and radiation, the total rate of heat loss can be expressed as

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s^4 - T_{surr}^4)$$
$$(121\times0.05)\text{ W} = (4.641\text{ W/m}^2\text{.°C})(0.09\text{ m}^2)(T_s - 25)\text{°C}$$
$$+(0.7)(0.09\text{ m}^2)(5.67\times10^{-8}\text{ W/m}^2\text{.K}^4)[(T_s+273\text{ K})^4 - (25+273\text{ K})^4]$$

Its solution is T_s = **32.5°C**
which is sufficiently close to the assumed value. Therefore, there is no need to repeat calculations.
(b) Chips (hot surface) facing up:

$$Nu = 0.27Ra^{1/4} = 0.27(3.848\times10^5)^{1/4} = 6.725$$
$$h = \frac{k}{L_c}Nu = \frac{0.02588\text{ W/m.°C}}{0.075\text{ m}}(6.725) = 2.321\text{ W/m}^2\text{.°C}$$

Considering both natural convection and radiation, the total rate of heat loss can be expressed as

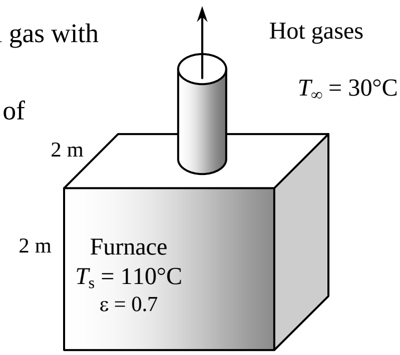
$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s^4 - T_{surr}^4)$$
$$(121\times0.05)\text{ W} = (2.321\text{ W/m}^2\text{.°C})(0.09\text{ m}^2)(T_s - 25)\text{°C}$$
$$+(0.7)(0.09\text{ m}^2)(5.67\times10^{-8}\text{ W/m}^2\text{.K}^4)[(T_s+273\text{ K})^4 - (25+273\text{ K})^4]$$

Its solution is T_s = **35.0°C**
which is identical to the assumed value in the evaluation of properties and h. Therefore, there is no need to repeat calculations.

9-41 It is proposed that the side surfaces of a cubic industrial furnace be insulated for \$550 in order to reduce the heat loss by 90 percent. The thickness of the insulation and the payback period of the insulation to pay for itself from the energy it saves are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (110+30)/2 = 70°C$ are (Table A-15)



$$k = 0.02881\,\text{W/m.°C}$$
$$\upsilon = 1.995\times10^{-5}\ \text{m}^2/\text{s}$$
$$\text{Pr} = 0.7177$$
$$\beta = \frac{1}{T_f} = \frac{1}{(70+273)\text{K}} = 0.002915\,\text{K}^{-1}$$

Analysis The characteristic length in this case is the height of the furnace, $L_c = L = 2$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\text{Pr} = \frac{(9.81\,\text{m/s}^2)(0.002915\,\text{K}^{-1})(110-30\,\text{K})(2\,\text{m})^3}{(1.995\times10^{-5}\ \text{m}^2/\text{s})^2}(0.7177) = 3.301\times10^{10}$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(3.301\times10^{10})^{1/6}}{\left[1+\left(\frac{0.492}{0.7177}\right)^{9/16}\right]^{8/27}}\right\}^2 = 369.2$$

$$h = \frac{k}{L_c}Nu = \frac{0.02881\,\text{W/m.°C}}{2\,\text{m}}(369.2) = 5.318\ \text{W/m}^2.\text{°C}$$
$$A_s = 4(2\,\text{m})^2 = 16\,\text{m}^2$$

Then the heat loss by combined natural convection and radiation becomes

$$\begin{aligned}\dot{Q} &= hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s{}^4 - T_{surr}{}^4)\\ &= (5.318\,\text{W/m}^2.\text{°C})(16\,\text{m}^2)(110-30)\text{°C}\\ &\quad + (0.7)(16\,\text{m}^2)(5.67\times10^{-8}\ \text{W/m}^2.\text{K}^4)[(110+273\,\text{K})^4 - (30+273\,\text{K})^4]\\ &= 15{,}119\ \text{W}\end{aligned}$$

Noting that insulation will reduce the heat losses by 90%, the rate of heat loss after insulation will be

$$\dot{Q}_{\text{saved}} = 0.9\dot{Q}_{\text{no insulation}} = 0.9\times15{,}119\ \text{W} = 13{,}607\ \text{W}$$
$$\dot{Q}_{\text{loss}} = (1-0.9)\dot{Q}_{\text{no insulation}} = 0.1\times15{,}119\ \text{W} = 1512\ \text{W}$$

The furnace operates continuously and thus 8760 h. Then the amount of energy and money the insulation will save becomes

$$\text{Energy saved} = \dot{Q}_{saved}\Delta t = \frac{13.607\ \text{kJ/s}}{0.78}\left(\frac{1\,\text{therm}}{105{,}500\ \text{kJ}}\right)(8760\times3600\ \text{s/yr}) = 5215\ \text{therms/yr}$$

$$\text{Money saved} = (\text{Energy saved})(\text{Unit cost of energy}) = (5215\ \text{therms})(\$0.55/\text{therm}) = \$2868$$

Therefore, the money saved by insulation will pay for the cost of \$550 in

550/(\$2868/yr)=0.1918 yr = **70 days**.

Insulation will lower the outer surface temperature, the Rayleigh and Nusselt numbers, and thus the convection heat transfer coefficient. For the evaluation of the heat transfer coefficient, we assume the surface temperature in this case to be 50°C. The properties of air at the film temperature of $(T_s+T_\infty)/2$ = (50+30)/2 = 40°C are (Table A-15)

$$k = 0.02662 \text{ W/m.°C}$$

$$\upsilon = 1.702\times10^{-5} \text{ m}^2\text{/s}$$

$$\Pr = 0.7255$$

$$\beta = \frac{1}{T_f} = \frac{1}{(40+273)\text{K}} = 0.003195 \text{ K}^{-1}$$

Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\Pr = \frac{(9.81 \text{ m/s}^2)(0.003195 \text{ K}^{-1})(50-30 \text{ K})(2 \text{ m})^3}{(1.702\times10^{-5} \text{ m}^2\text{/s})^2}(0.7255) = 1.256\times10^{10}$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(1.256\times10^{10})^{1/6}}{\left[1+\left(\frac{0.492}{0.7255}\right)^{9/16}\right]^{8/27}}\right\}^2 = 272.0$$

$$h = \frac{k}{L}Nu = \frac{0.02662 \text{ W/m.°C}}{2 \text{ m}}(272.0) = 3.620 \text{ W/m}^2\text{.°C}$$

$$A_s = 4\times(2 \text{ m})(2+2t_{insul}) \text{ m}$$

The total rate of heat loss from the outer surface of the insulated furnace by convection and radiation becomes

$$\dot{Q} = \dot{Q}_{conv} + \dot{Q}_{rad} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s{}^4 - T_{surr}{}^4)$$

$$1512 \text{ W} = (3.620 \text{ W/m}^2\text{.°C})A(T_s - 30)\text{°C} + (0.7)A(5.67\times10^{-8} \text{ W/m}^2\text{.K}^4)[(T_s + 273 \text{ K})^4 - (30+273 \text{ K})^4]$$

I
n steady operation, the heat lost by the side surfaces of the pipe must be equal to the heat lost from the exposed surface of the insulation by convection and radiation, which must be equal to the heat conducted through the insulation. Therefore,

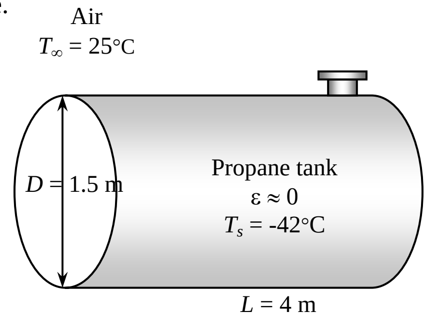
$$\dot{Q} = \dot{Q}_{insulation} = kA_s\frac{(T_{\text{furnace}} - T_s)}{t_{ins}} \rightarrow 1512 \text{ W} = (0.038 \text{ W/m.°C})A_s\frac{(110 - T_s)\text{°C}}{t_{insul}}$$

Solving the two equations above by trial-and error (or better yet, an equation solver) gives

T_s = 48.4°C and t_{insul} = 0.0254 m = **2.54 cm**

9-42 A cylindrical propane tank is exposed to calm ambient air. The propane is slowly vaporized due to a crack developed at the top of the tank. The time it will take for the tank to empty is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** Radiation heat transfer is negligible.



Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2$ = (-42+25)/2 = -8.5°C are (Table A-15)

$$k = 0.02299 \text{ W/m.°C}$$
$$\upsilon = 1.265\times10^{-5} \text{ m}^2\text{/s}$$
$$\text{Pr} = 0.7383$$
$$\beta = \frac{1}{T_f} = \frac{1}{(-8.5+273)\text{K}} = 0.003781 \text{ K}^{-1}$$

Analysis The tank gains heat through its cylindrical surface as well as its circular end surfaces. For convenience, we take the heat transfer coefficient at the end surfaces of the tank to be the same as that of its side surface. (The alternative is to treat the end surfaces as a vertical plate, but this will double the amount of calculations without providing much improvement in accuracy since the area of the end surfaces is much smaller and it is circular in shape rather than being rectangular). The characteristic length in this case is the outer diameter of the tank, $L_c = D = 1.5$ m. Then,

$$Ra = \frac{g\beta(T_\infty - T_s)D^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003781 \text{ K}^{-1})[(25-(-42)\text{ K}](1.5 \text{ m})^3}{(1.265\times10^{-5} \text{ m}^2\text{/s})^2}(0.7383) = 3.869\times10^{10}$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/\text{Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(3.869\times10^{10})^{1/6}}{\left[1+\left(0.559/0.7383\right)^{9/16}\right]^{8/27}}\right\}^2 = 374.1$$

$$h = \frac{k}{D}Nu = \frac{0.02299 \text{ W/m.°C}}{1.5 \text{ m}}(374.1) = 5.733 \text{ W/m}^2\text{.°C}$$

$$A_s = \pi DL + 2\pi D^2/4 = \pi(1.5 \text{ m})(4 \text{ m}) + 2\pi(1.5 \text{ m})^2/4 = 22.38 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_\infty - T_s) = (5.733 \text{ W/m}^2\text{.°C})(22.38 \text{ m}^2)[(25-(-42)]\text{°C} = 8598 \text{ W}$$

The total mass and the rate of evaporation of propane are

$$m = \rho V = \rho\frac{\pi D^2}{4}L = (581 \text{ kg/m}^3)\frac{\pi(1.5 \text{ m})^2}{4}(4 \text{ m}) = 4107 \text{ kg}$$

$$\dot{m} = \frac{\dot{Q}}{h_{fg}} = \frac{8.598 \text{ kJ/s}}{425 \text{ kJ/kg}} = 0.02023 \text{ kg/s}$$

and it will take

$$\Delta t = \frac{m}{\dot{m}} = \frac{4107 \text{ kg}}{0.02023 \text{ kg/s}} = 202{,}996 \text{ s} = \mathbf{56.4 \text{ hours}}$$

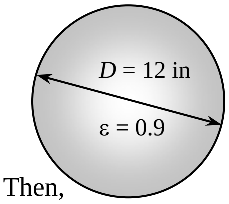
for the propane tank to empty.

9-43E The average surface temperature of a human head is to be determined when it is not covered.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** The head can be approximated as a 12-in.-diameter sphere.

Properties The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the surface temperature to be 120°F for the evaluation of the properties and *h*. We will check the accuracy of this guess later and repeat the calculations if necessary. The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2$ = (120+77)/2 = 98.5°F are (Table A-15E)

$$k = 0.01525\ \text{Btu/h.ft.°F}$$
$$\upsilon = 0.180\times10^{-3}\ \text{ft}^2/\text{s}$$
$$\text{Pr} = 0.7262$$
$$\beta = \frac{1}{T_f} = \frac{1}{(98.5+460)\text{R}} = 0.001791\ \text{R}^{-1}$$



Analysis The characteristic length for a spherical object is $L_c = D$ = 12/24 = 0.5 ft. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2}\text{Pr} = \frac{(32.2\ \text{ft/s}^2)(0.001791\ \text{R}^{-1})(95-77\ \text{R})(0.5\ \text{ft})^3}{(0.180\times10^{-3}\ \text{ft}^2/\text{s})^2}(0.7262) = 6.943\times10^6$$

$$Nu = 2 + \frac{0.589Ra^{1/4}}{\left[1+\left(\frac{0.469}{\text{Pr}}\right)^{9/16}\right]^{4/9}} = 2 + \frac{0.589(6.943\times10^6)^{1/4}}{\left[1+\left(\frac{0.469}{0.7262}\right)^{9/16}\right]^{4/9}} = 25.39$$

$$h = \frac{k}{D}Nu = \frac{0.01525\ \text{Btu/h.ft.°F}}{1\ \text{ft}}(25.39) = 0.7744\ \text{Btu/h.ft}^2\text{.°F}$$
$$A_s = \pi D^2 = \pi(0.5\ \text{ft})^2 = 0.7854\ \text{ft}^2$$

Considering both natural convection and radiation, the total rate of heat loss can be written as

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s^{\ 4} - T_{surr}^{\ 4})$$
$$(287/4\ \text{Btu/h}) = (0.7744\ \text{Btu/h.ft}^2\text{.°F})(0.7854\ \text{ft}^2)(T_s - 77)\text{°F}$$
$$+(0.9)(0.7854\ \text{m}^2)(0.1714\times10^{-8}\ \text{Btu/h.ft}^2\text{.R}^4)[(T_s+460\ \text{R})^4 - (77+460\ \text{R})^4]$$

Its solution is

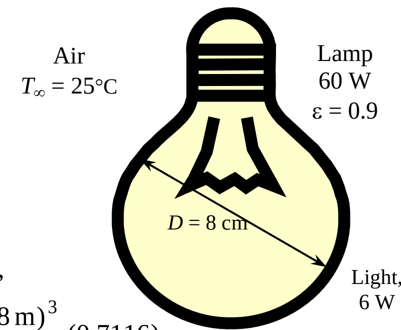
T_s = **125.9°F**

which is sufficiently close to the assumed value in the evaluation of the properties and *h*. Therefore, there is no need to repeat calculations.

9-44 The equilibrium temperature of a light glass bulb in a room is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** The light bulb is approximated as an 8-cm-diameter sphere.

Properties The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the surface temperature to be 170°C for the evaluation of the properties and *h*. We will check the accuracy of this guess later and repeat the calculations if necessary. The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2 = (170+25)/2 = 97.5$°C are (Table A-15)



$$k = 0.03077 \text{ W/m.°C}$$
$$\upsilon = 2.279\times10^{-5} \text{ m}^2/\text{s}$$
$$\text{Pr} = 0.7116$$
$$\beta = \frac{1}{T_f} = \frac{1}{(97.5+273)\text{K}} = 0.002699 \text{ K}^{-1}$$

Analysis The characteristic length in this case is $L_c = D = 0.08$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.002699 \text{ K}^{-1})(170-25 \text{ K})(0.08 \text{ m})^3}{(2.279\times10^{-5} \text{ m}^2/\text{s})^2}(0.7116)$$
$$= 2.694\times10^6$$

$$Nu = 2 + \frac{0.589Ra^{1/4}}{\left[1+\left(0.469/\text{Pr}\right)^{9/16}\right]^{4/9}} = 2 + \frac{0.589(2.694\times10^6)^{1/4}}{\left[1+\left(0.469/0.7116\right)^{9/16}\right]^{4/9}} = 20.42$$

Then

$$h = \frac{k}{D}Nu = \frac{0.03077 \text{ W/m.°C}}{0.08 \text{ m}}(20.42) = 7.854 \text{ W/m}^2\text{.°C}$$
$$A_s = \pi D^2 = \pi(0.08 \text{ m})^2 = 0.02011 \text{ m}^2$$

Considering both natural convection and radiation, the total rate of heat loss can be written as

$$\dot{Q} = hA_s(T_s - T_\infty) + \varepsilon A_s\sigma(T_s^{\,4} - T_{surr}^{\,4})$$
$$(0.90\times60) \text{ W} = (7.854 \text{ W/m}^2\text{.°C})(0.02011 \text{ m}^2)(T_s - 25)\text{°C}$$
$$+ (0.9)(0.02011 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2\text{.K}^4)[(T_s+273)^4 - (25+273 \text{ K})^4]$$

Its solution is

T_s = **169.4°C**

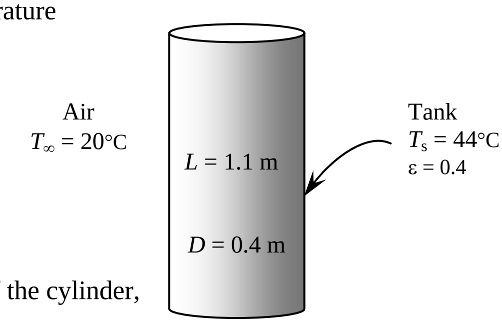
which is sufficiently close to the value assumed in the evaluation of properties and *h*. Therefore, there is no need to repeat calculations.

9-45 A vertically oriented cylindrical hot water tank is located in a bathroom. The rate of heat loss from the tank by natural convection and radiation is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** The temperature of the outer surface of the tank is constant.

Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2$ = (44+20)/2 = 32°C are (Table A-15)

$$k = 0.02603\ \text{W/m.°C}$$
$$\upsilon = 1.627\times10^{-5}\ \text{m}^2/\text{s}$$
$$\text{Pr} = 0.7276$$
$$\beta = \frac{1}{T_f} = \frac{1}{(32+273)\text{K}} = 0.003279\ \text{K}^{-1}$$



Analysis The characteristic length in this case is the height of the cylinder, $L_c = L = 1.1\,\text{m}$. Then,

$$\text{Gr} = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2} = \frac{(9.81\,\text{m/s}^2)(0.003279\,\text{K}^{-1})(44-20\,\text{K})(1.1\,\text{m})^3}{(1.627\times10^{-5}\ \text{m}^2/\text{s})^2} = 3.883\times10^9$$

A vertical cylinder can be treated as a vertical plate when

$$D(=0.4\ \text{m}) \ge \frac{35L}{\text{Gr}^{1/4}} = \frac{35(1.1\,\text{m})}{(3.883\times10^9)^{1/4}} = 0.1542\ \text{m}$$

which is satisfied. That is, the Nusselt number relation for a vertical plate can be used for the side surfaces. For the top and bottom surfaces we use the relevant Nusselt number relations. First, for the side surfaces,

$$\text{Ra} = \text{GrPr} = (3.883\times10^9)(0.7276) = 2.825\times10^9$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(2.825\times10^9)^{1/6}}{\left[1+\left(\frac{0.492}{0.7276}\right)^{9/16}\right]^{8/27}}\right\}^2 = 170.2$$

$$h = \frac{k}{L}Nu = \frac{0.02603\ \text{W/m.°C}}{1.1\,\text{m}}(170.2) = 4.027\ \text{W/m}^2\text{.°C}$$

$$A_s = \pi DL = \pi(0.4\ \text{m})(1.1\,\text{m}) = 1.382\ \text{m}^2$$

$$\dot{Q}_{\text{side}} = hA_s(T_s - T_\infty) = (4.027\ \text{W/m}^2\text{.°C})(1.382\ \text{m}^2)(44-20)\text{°C} = 133.6\ \text{W}$$

For the top surface,

$$L_c = \frac{A_s}{p} = \frac{\pi D^2/4}{\pi D} = \frac{D}{4} = \frac{0.4\ \text{m}}{4} = 0.1\,\text{m}$$

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)L_c^3}{\upsilon^2}\text{Pr} = \frac{(9.81\,\text{m/s}^2)(0.003279\,\text{K}^{-1})(44-20\,\text{K})(0.1\,\text{m})^3}{(1.627\times10^{-5}\ \text{m}^2/\text{s})^2}(0.7276) = 2.123\times10^6$$

$$\text{Nu} = 0.54\text{Ra}^{1/4} = 0.54(2.123\times10^6)^{1/4} = 20.61$$

$$h = \frac{k}{L_c}Nu = \frac{0.02603\ \text{W/m.°C}}{0.1\,\text{m}}(20.61) = 5.365\ \text{W/m}^2\text{.°C}$$

$$A_s = \pi D^2/4 = \pi(0.4\ \text{m})^2/4 = 0.1257\ \text{m}^2$$

$$\dot{Q}_{\text{top}} = hA_s(T_s - T_\infty) = (5.365\ \text{W/m}^2\text{.°C})(0.1257\ \text{m}^2)(44-20)\text{°C} = 16.2\ \text{W}$$

For the bottom surface,

$$\text{Nu} = 0.27\text{Ra}^{1/4} = 0.27(2.123\times10^6)^{1/4} = 10.31$$

$$h = \frac{k}{L_c} Nu = \frac{0.02603\,\text{W/m.°C}}{0.1\,\text{m}}(10.31) = 2.683\,\text{W/m}^2.\text{°C}$$

$$\dot{Q}_{\text{bottom}} = hA_s(T_s - T_\infty) = (2.683\,\text{W/m}^2.\text{°C})(0.1257\,\text{m}^2)(44-20)\text{°C} = 8.1\,\text{W}$$

The total heat loss by natural convection is

$$\dot{Q}_{\text{conv}} = \dot{Q}_{\text{side}} + \dot{Q}_{\text{top}} + \dot{Q}_{\text{bottom}} = 133.6 + 16.2 + 8.1 = \mathbf{157.9\ W}$$

The radiation heat loss from the tank is

$$\begin{aligned}\dot{Q}_{\text{rad}} &= \varepsilon A_s \sigma (T_s^{\,4} - T_{surr}^{\,4}) \\ &= (0.4)(1.382 + 0.1257 + 0.1257\,\text{m}^2)(5.67\times10^{-8}\ \text{W/m}^2.\text{K}^4)\left[(44+273\,\text{K})^4 - (20+273\,\text{K})^4\right] \\ &= \mathbf{101.1\ W}\end{aligned}$$

9-46 A rectangular container filled with cold water is gaining heat from its surroundings by natural convection and radiation. The water temperature in the container after a 3 hours and the average rate of heat transfer are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** The heat transfer coefficient at the top and bottom surfaces is the same as that on the side surfaces.

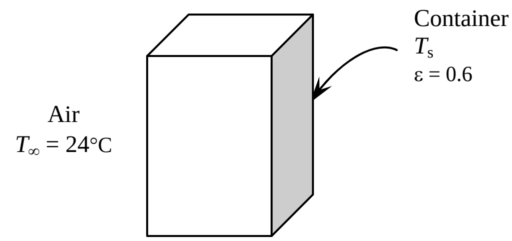
Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2 = (10+24)/2 = 17°C$ are (Table A-15)

$$k = 0.02491 \text{ W/m.°C}$$

$$\upsilon = 1.489\times10^{-5} \text{ m}^2\text{/s}$$

$$\text{Pr} = 0.7317$$

$$\beta = \frac{1}{T_f} = \frac{1}{(17+273)\text{K}} = 0.003448 \text{ K}^{-1}$$



The properties of water at 2°C are (Table A-7)

$$\rho = 1000 \text{ kg/m}^3 \text{ and } C_p = 4214 \text{ J/kg.°C}$$

Analysis We first evaluate the heat transfer coefficient on the side surfaces. The characteristic length in this case is the height of the container,

$L_c = L = 0.28$ m. Then,

$$\text{Ra} = \frac{g\beta(T_\infty - T_s)L^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003448 \text{ K}^{-1})(24-10 \text{ K})(0.28 \text{ m})^3}{(1.489\times10^{-5} \text{ m}^2\text{/s})^2}(0.7317) = 1.133\times10^7$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(1.133\times10^7)^{1/6}}{\left[1+\left(\frac{0.492}{0.7317}\right)^{9/16}\right]^{8/27}}\right\}^2 = 30.52$$

$$h = \frac{k}{L}Nu = \frac{0.02491 \text{ W/m.°C}}{0.28 \text{ m}}(30.52) = 4.224 \text{ W/m}^2\text{.°C}$$

$$A_s = 2(0.28\times0.18 + 0.28\times0.18 + 0.18\times0.18) = 0.2664 \text{ m}^2$$

The rate of heat transfer can be expressed as

$$\dot{Q} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = hA_s\left(T_\infty - \frac{T_1+T_2}{2}\right) + \varepsilon\sigma A_s\left[T_{\text{surr}}^4 - \left(\frac{T_1+T_2}{2}\right)^4\right]$$

$$= (4.224 \text{ W/m}^2\text{.°C})(0.2664 \text{ m}^2)\left[297 - \left(\frac{275+T_2}{2}\right)\right] \qquad \text{(Eq. 1)}$$

$$+ (0.6)(0.2664 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2\text{.K}^4)\left[297^4 - \left(\frac{275+T_2}{2}\right)^4\right]$$

where $(T_1 + T_2)/2$ is the average temperature of water (or the container surface). The mass of water in the container is

$$m = \rho V = (1000 \text{ kg/m}^3)(0.28\times0.18\times0.18)\text{m}^3 = 9.072 \text{ kg}$$

Then the amount of heat transfer to the water is

$$Q = mC_p(T_2 - T_1) = (9.072 \text{ kg})(4214 \text{ J/kg.°C})(T_2 - 275)\text{°C} = 38{,}229(T_2 - 275)$$

The average rate of heat transfer can be expressed as

$$\dot{Q} = \frac{Q}{\Delta t} = \frac{38{,}229(T_2 - 275)}{3\times3600 \text{ s}} = 3.53976(T_2 - 275) \qquad \text{(Eq. 2)}$$

Setting Eq. 1 and Eq. 2 equal to each other, we obtain the final water temperature.

$$T_2 = 284.7 \text{ K} = \mathbf{11.7°C}$$

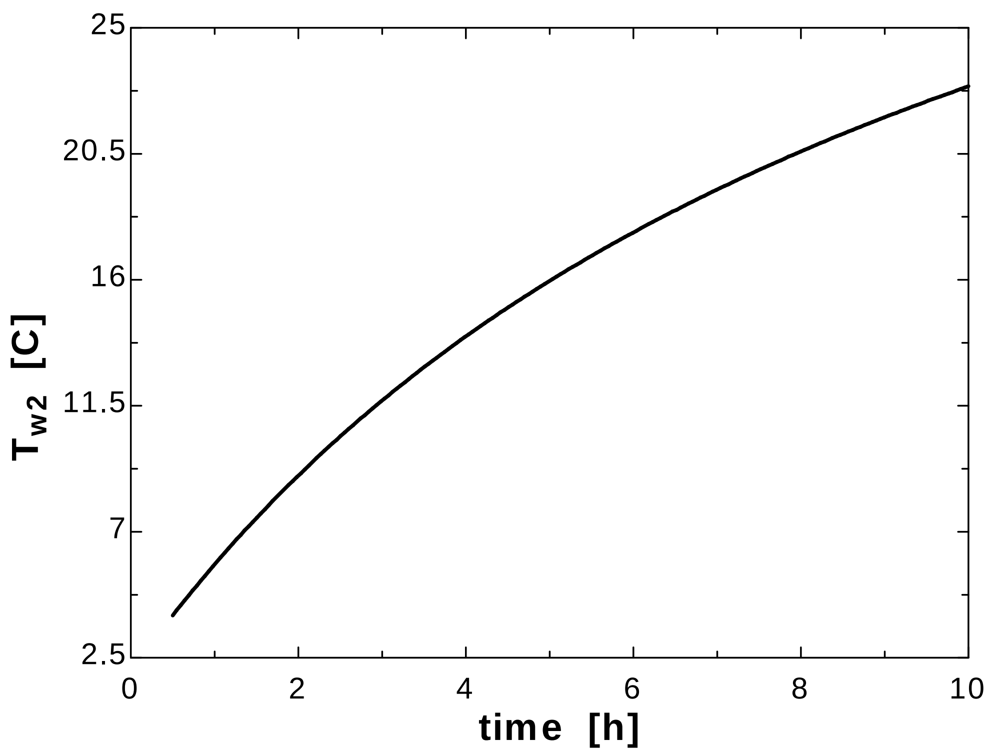
We could repeat the solution using air properties at the new film temperature using this value to increase the accuracy. However, this would only affect the heat transfer value somewhat, which would not have significant effect on the final water temperature. The average rate of heat transfer can be determined from Eq. 2

$$\dot{Q} = 3.53976(11.7 - 2) = \mathbf{34.3\ W}$$

9-47 "!PROBLEM 9-47"

```
"GIVEN"
height=0.28 "[m]"
L=0.18 "[m]"
w=0.18 "[m]"
T_infinity=24 "[C]"
T_w1=2 "[C]"
epsilon=0.6
T_surr=T_infinity
"time=3 [h], parameter to be varied"

"PROPERTIES"
Fluid$='air'
k=Conductivity(Fluid$, T=T_film)
Pr=Prandtl(Fluid$, T=T_film)
rho=Density(Fluid$, T=T_film, P=101.3)
mu=Viscosity(Fluid$, T=T_film)
nu=mu/rho
beta=1/(T_film+273)
T_film=1/2*(T_w_ave+T_infinity)
T_w_ave=1/2*(T_w1+T_w2)
rho_w=Density(water, T=T_w_ave, P=101.3)
C_p_w=CP(water, T=T_w_ave, P=101.3)*Convert(kJ/kg-C, J/kg-C)
sigma=5.67E-8 "[W/m^2-K^4], Stefan-Boltzmann constant"
g=9.807 "[m/s^2], gravitational acceleration"

"ANALYSIS"
delta=height
Ra=(g*beta*(T_infinity-T_w_ave)*delta^3)/nu^2*Pr
Nusselt=0.59*Ra^0.25
h=k/delta*Nusselt
A=2*(height*L+height*w+w*L)
Q_dot=h*A*(T_infinity-T_w_ave)+epsilon*A*sigma*((T_surr+273)^4-(T_w_ave+273)^4)

m_w=rho_w*V_w
V_w=height*L*w
Q=m_w*C_p_w*(T_w2-T_w1)
Q_dot=Q/(time*Convert(h, s))
```

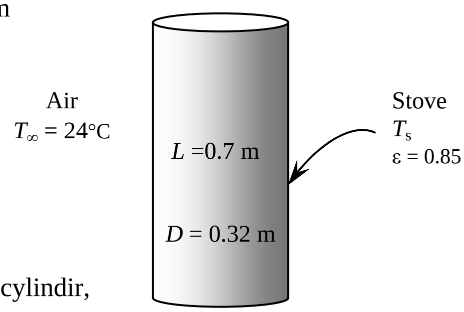
time [h]	T_{w2} [C]
0.5	4.013
1	5.837
1.5	7.496
2	9.013
2.5	10.41
3	11.69
3.5	12.88
4	13.98
4.5	15
5	15.96
5.5	16.85
6	17.69
6.5	18.48
7	19.22
7.5	19.92
8	20.59
8.5	21.21
9	21.81
9.5	22.37
10	22.91



9-48 A room is to be heated by a cylindrical coal-burning stove. The surface temperature of the stove and the amount of coal burned during a 30-day-period are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** The temperature of the outer surface of the stove is constant. **5** The heat transfer from the bottom surface is negligible. **6** The heat transfer coefficient at the top surface is the same as that on the side surface.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2$ = (130+24)/2 = 77°C are (Table A-1)



$$k = 0.02931\ \text{W/m.°C}$$
$$\upsilon = 2.066\times10^{-5}\ \text{m}^2\text{/s}$$
$$\text{Pr} = 0.7161$$
$$\beta = \frac{1}{T_f} = \frac{1}{(77+273)\text{K}} = 0.002857\ \text{K}^{-1}$$

Analysis The characteristic length in this case is the height of the cylindir, $L_c = L = 0.7$ m. Then,

$$\text{Gr} = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2} = \frac{(9.81\ \text{m/s}^2)(0.002857\ \text{K}^{-1})(130-24\ \text{K})(0.70\ \text{m})^3}{(2.066\times10^{-5}\ \text{m}^2\text{/s})^2} = 2.387\times10^9$$

A vertical cylinder can be treated as a vertical plate when

$$D(=0.32\ \text{m}) \geq \frac{35L}{\text{Gr}^{1/4}} = \frac{35(0.7\ \text{m})}{(2.387\times10^9)^{1/4}} = 0.1108\ \text{m}$$

which is satisfied. That is, the Nusselt number relation for a vertical plate can be used for side surfaces.

$$\text{Ra} = \text{GrPr} = (2.387\times10^9)(0.7161) = 1.709\times10^9$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(1.709\times10^9)^{1/6}}{\left[1+\left(\frac{0.492}{0.7161}\right)^{9/16}\right]^{8/27}}\right\}^2 = 145.2$$

$$h = \frac{k}{L}Nu = \frac{0.02931\ \text{W/m.°C}}{0.7\ \text{m}}(145.2) = 6.080\ \text{W/m}^2\text{.°C}$$

$$A_s = \pi DL + \pi D^2/4 = \pi(0.32\ \text{m})(0.7\ \text{m}) + \pi(0.32\ \text{m})^2/4 = 0.7841\ \text{m}^2$$

Then the surface temperature of the stove is determined from

$$\dot{Q} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = hA_s(T_s - T_\infty) + \varepsilon\sigma A_s(T_s^4 - T_{\text{surr}}^4)$$

$$1200\ \text{W} = (6.080\ \text{W/m}^2\text{.°C})(0.7841\ \text{m}^2)(T_s - 297) + (0.85)(0.7841\ \text{m}^2)(5.67\times10^{-8}\ \text{W/m}^2\text{.K}^4)(T_s^4 - 290^4)$$

$$\longrightarrow T_s = 400.6\ \text{K} = \mathbf{127.6°C}$$

The amount of coal used is determined from

$$Q = \dot{Q}\Delta t = (1.2\ \text{kJ/s})(14\ \text{h/day}\times3600\ \text{s/h}) = 60{,}480\ \text{kJ}$$

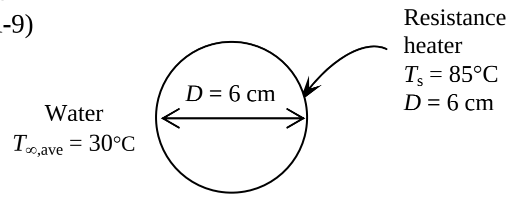
$$m_{\text{coal}} = \frac{Q/\eta}{HV} = \frac{(60{,}480\ \text{kJ})/0.65}{30{,}000\ \text{kJ/kg}} = \mathbf{3.102\ kg}$$

9-49 Water in a tank is to be heated by a spherical heater. The heating time is to be determined.

Assumptions **1** Steady operating conditions exist. **2** The temperature of the outer surface of the sphere is constant.

Properties Using the average temperature for water (15+45)/2=30 as the fluid temperature, the properties of water at the film temperature of $(T_s+T_\infty)/2$ = (85+30)/2 = 57.5°C are (Table A-9)

$$k = 0.6515 \text{ W/m.°C}$$
$$\upsilon = 0.474\times10^{-6} \text{ m}^2\text{/s}$$
$$\Pr = 3.12$$
$$\beta = 0.501\times10^{-3} \text{ K}^{-1}$$



Also, the properties of water at 30°C are (Table A-9)

$$\rho = 996 \text{ kg/m}^3 \text{ and } C_p = 4178 \text{ J/kg.°C}$$

Analysis The characteristic length in this case is $L_c = D$ = 0.06 m. Then,

$$\text{Ra} = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2}\Pr = \frac{(9.81 \text{ m/s}^2)(0.501\times10^{-3} \text{ K}^{-1})(85-30 \text{ K})(0.06 \text{ m})^3}{(0.474\times10^{-6} \text{ m}^2\text{/s})^2}(3.12) = 8.108\times10^8$$

$$Nu = 2 + \frac{0.589Ra^{1/4}}{\left[1+\left(0.469/\Pr\right)^{9/16}\right]^{4/9}} = 2 + \frac{0.589(8.108\times10^8)^{1/4}}{\left[1+\left(0.469/3.12\right)^{9/16}\right]^{4/9}} = 89.14$$

$$h = \frac{k}{D}Nu = \frac{0.6515 \text{ W/m.°C}}{0.06 \text{ m}}(89.14) = 967.9 \text{ W/m}^2\text{.°C}$$

$$A_s = \pi D^2 = \pi(0.06 \text{ m})^2 = 0.01131 \text{ m}^2$$

The rate of heat transfer by convection is

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) = (967.9 \text{ W/m}^2\text{.°C})(0.01131\text{m}^2)(85-30) = 602.1 \text{W}$$

The mass of water in the container is

$$m = \rho V = (996 \text{ kg/m}^3)(0.040 \text{ m}^3) = 39.84 \text{ kg}$$

The amount of heat transfer to the water is

$$Q = mC_p(T_2 - T_1) = (39.84 \text{ kg})(4178 \text{ J/kg.°C})(45\text{-}15)\text{°C} = 4.994\times10^6 \text{ J}$$

Then the time the heater should be on becomes

$$\Delta t = \frac{Q}{\dot{Q}} = \frac{4.994\times10^6 \text{ J}}{602.1 \text{ J/s}} = 8294 \text{ s} = \mathbf{2.304 \text{ hours}}$$

Combined Natural and Forced Convection

9-72C In combined natural and forced convection, the natural convection is negligible when $Gr/\mathrm{Re}^2 < 0.1$. Otherwise it is not.

9-73C In assisting or transverse flows, natural convection enhances forced convection heat transfer while in opposing flow it hurts forced convection.

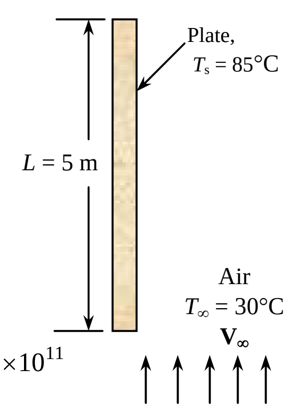
9-74C When neither natural nor forced convection is negligible, it is not correct to calculate each separately and to add them to determine the total convection heat transfer. Instead, the correlation

$$\mathrm{Nu}_{\text{combined}} = \left(\mathrm{Nu}_{\text{forced}}^{n} + \mathrm{Nu}_{\text{natural}}^{n}\right)^{1/n}$$

based on the experimental studies should be used.

9-75 A vertical plate in air is considered. The forced motion velocity above which natural convection heat transfer from the plate is negligible is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The atmospheric pressure at that location is 1 atm.



Properties The properties of air at 1 atm and 1 atm and the film temperature of $(T_s+T_\infty)/2$ = (85+30)/2 = 57.5°C are (Table A-15)

$$\upsilon = 1.871\times10^{-5}\ \mathrm{m^2/s}$$

$$\beta = \frac{1}{T_f} = \frac{1}{(57.5+273)\mathrm{K}} = 0.003026\ \mathrm{K^{-1}}$$

Analysis The characteristic length is the height of the plate, $L_c = L = 5$ m. The Grashof and Reynolds numbers are

$$Gr = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2} = \frac{(9.81\ \mathrm{m/s^2})(0.003026\ \mathrm{K^{-1}})(85-30\ \mathrm{K})(5\ \mathrm{m})^3}{(1.871\times10^{-5}\ \mathrm{m^2/s})^2} = 5.829\times10^{11}$$

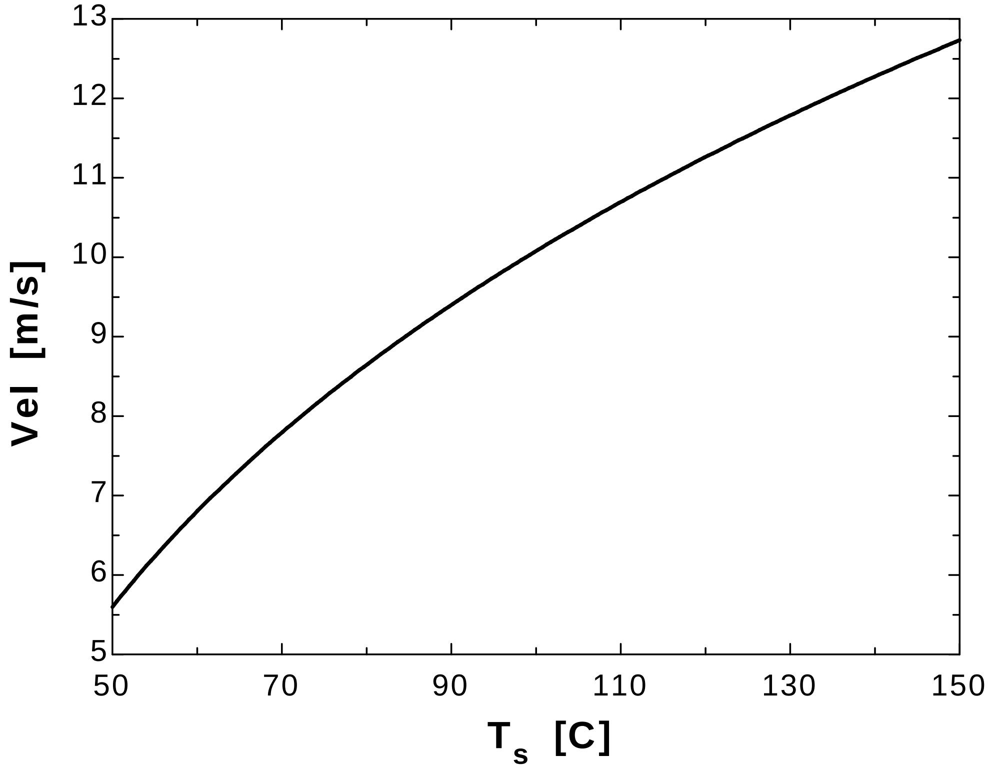
$$\mathrm{Re} = \frac{V_\infty L}{\upsilon} = \frac{V_\infty(5\ \mathrm{m})}{1.871\times10^{-5}\ \mathrm{m^2/s}} = 2.67\times10^5 V_\infty$$

and the forced motion velocity above which natural convection heat transfer from this plate is negligible is

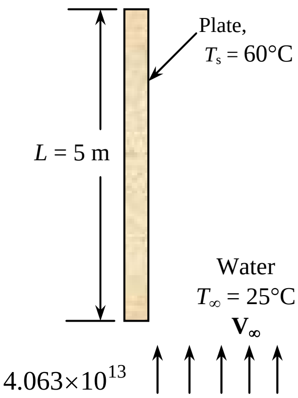
$$\frac{Gr}{\mathrm{Re}^2} = 0.1 \longrightarrow \frac{5.829\times10^{11}}{(2.67\times10^5 V_\infty)^2} = 0.1 \longrightarrow V_\infty = \mathbf{9.04\ m/s}$$

9-76
```
"!PROBLEM 9-76"

"GIVEN"
L=5 "[m]"
"T_s=85 [C], parameter to be varied"
T_infinity=30 "[C]"

"PROPERTIES"
Fluid$='air'
rho=Density(Fluid$, T=T_film, P=101.3)
mu=Viscosity(Fluid$, T=T_film)
nu=mu/rho
beta=1/(T_film+273)
T_film=1/2*(T_s+T_infinity)
g=9.807 "[m/s^2], gravitational acceleration"

"ANALYSIS"
Gr=(g*beta*(T_s-T_infinity)*L^3)/nu^2
Re=(Vel*L)/nu
Gr/Re^2=0.1
```

T_s [C]	Vel [m/s]
50	5.598
55	6.233
60	6.801
65	7.318
70	7.793
75	8.233
80	8.646
85	9.033
90	9.4
95	9.747
100	10.08
105	10.39
110	10.69
115	10.98
120	11.26
125	11.53
130	11.79
135	12.03
140	12.27
145	12.51
150	12.73


13
12
11
10
9
8
7
6
5
Vel [m/s]
50
70
90
110
130
150
Ts [C]

9-77 A vertical plate in water is considered. The forced motion velocity above which natural convection heat transfer from the plate is negligible is to be determined.



Assumptions **1** Steady operating conditions exist.

Properties The properties of water at the film temperature of $(T_s+T_\infty)/2$ = (60+25)/2 = 42.5°C are (Table A-15)

$$\upsilon = 0.65\times10^{-6}\ \text{m}^2/\text{s}$$

$$\beta = 0.00040\ \text{K}^{-1}$$

Analysis The characteristic length is the height of the plate $L_c = L = 5$ m. The Grashof and Reynolds numbers are

$$Gr = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2} = \frac{(9.81\,\text{m/s}^2)(0.0004\,\text{K}^{-1})(60-25\,\text{K})(5\,\text{m})^3}{(0.65\times10^{-6}\ \text{m}^2/\text{s})^2} = 4.063\times10^{13}$$

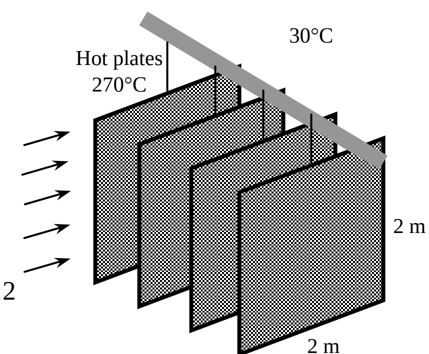
$$\text{Re} = \frac{V_\infty L}{\upsilon} = \frac{V_\infty(5\,\text{m})}{0.65\times10^{-6}\ \text{m}^2/\text{s}} = 4.6\times10^6 V_\infty$$

and the forced motion velocity above which natural convection heat transfer from this plate is negligible is

$$\frac{Gr}{\text{Re}^2} = 0.1 \longrightarrow \frac{4.063\times10^{13}}{(4.6\times10^6 V_\infty)^2} = 0.1 \longrightarrow V_\infty = \mathbf{2.62\,m/s}$$

9-78 Thin square plates coming out of the oven in a production facility are cooled by blowing ambient air horizontally parallel to their surfaces. The air velocity above which the natural convection effects on heat transfer are negligible is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The atmospheric pressure at that location is 1 atm.



Properties The properties of air at 1 atm and 1 atm and the film temperature of $(T_s+T_\infty)/2$ = (270+30)/2 = 150°C are (Table A-15)

$$\upsilon = 2.859\times10^{-5}\ \text{m}^2/\text{s}$$

$$\beta = \frac{1}{T_f} = \frac{1}{(150+273)\text{K}} = 0.002364\,\text{K}^{-1}$$

Analysis The characteristic length is the height of the plate $L_c = L = 2$ m. The Grashof and Reynolds numbers are

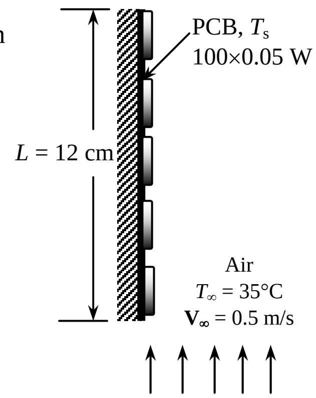
$$Gr = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2} = \frac{(9.81\,\text{m/s}^2)(0.002364\,\text{K}^{-1})(270-30\,\text{K})(2\,\text{m})^3}{(2.859\times10^{-5}\ \text{m}^2/\text{s})^2} = 5.447\times10^{10}$$

$$\text{Re} = \frac{V_\infty L}{\upsilon} = \frac{V_\infty(2\,\text{m})}{2.859\times10^{-5}\ \text{m}^2/\text{s}} = 6.995\times10^4 V_\infty$$

and the forced motion velocity above which natural convection heat transfer from this plate is negligible is

$$\frac{Gr}{\text{Re}^2} = 0.1 \longrightarrow \frac{5.447\times10^{10}}{(6.995\times10^4 V_\infty)^2} = 0.1 \longrightarrow V_\infty = \mathbf{10.6\ m/s}$$

9-79 A circuit board is cooled by a fan that blows air upwards. The average temperature on the surface of the circuit board is to be determined for two cases.



Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The atmospheric pressure at that location is 1 atm.

Properties The properties of air at 1 atm and 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2$ = (60+35)/2 = 47.5°C are (Table A-15)

$$k = 0.02717 \text{ W/m.°C}$$
$$\upsilon = 1.774\times10^{-5} \text{ m}^2\text{/s}$$
$$\Pr = 0.7235$$
$$\beta = \frac{1}{T_f} = \frac{1}{(47.5+273)\text{K}} = 0.00312 \text{ K}^{-1}$$

Analysis We assume the surface temperature to be 60°C. We will check this assumption later on and repeat calculations with a better assumption, if necessary. The characteristic length in this case is the length of the board in the flow (vertical) direction, $L_c = L$ = 0.12 m. Then the Reynolds number becomes

$$\text{Re} = \frac{V_\infty L}{\upsilon} = \frac{(0.5 \text{ m/s})(0.12 \text{ m})}{1.774\times10^{-5} \text{ m}^2\text{/s}} = 3383$$

which is less than critical Reynolds number (5×10^5). Therefore the flow is laminar and the forced convection Nusselt number and h are determined from

$$Nu = \frac{hL}{k} = 0.664 \text{Re}^{0.5} \Pr^{1/3} = 0.664(3383)^{0.5}(0.7235)^{1/3} = 34.67$$

$$h = \frac{k}{L} Nu = \frac{0.02717 \text{ W/m.°C}}{0.12 \text{ m}}(34.67) = 7.85 \text{ W/m}^2\text{.°C}$$

$$A_s = L\times W = (0.12 \text{ m})(0.2 \text{ m}) = 0.024 \text{ m}^2$$

Then,

$$\dot{Q} = hA_s(T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{Q}}{hA_s} = 35°\text{C} + \frac{(100)(0.05 \text{ W})}{(7.85 \text{ W/m}^2\text{.°C})(0.024 \text{ m}^2)} = \mathbf{61.5°C}$$

which is sufficiently close to the assumed value in the evaluation of properties. Therefore, there is no need to repeat calculations.

(b) The Rayleigh number is

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\Pr = \frac{(9.81 \text{ m/s}^2)(0.00312 \text{ K}^{-1})(60-35 \text{ K})(0.12 \text{ m})^3}{(1.774\times10^{-5} \text{ m}^2\text{/s})^2}(0.7235) = 3.041\times10^6$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(3.041\times10^6)^{1/6}}{\left[1+\left(\frac{0.492}{0.7235}\right)^{9/16}\right]^{8/27}}\right\}^2 = 22.42$$

This is an assisting flow and the combined Nusselt number is determined from

$$Nu_{combined} = (Nu_{forced}{}^n + Nu_{natural}{}^n)^{1/n} = (34.67^3 + 22.42^3)^{1/3} = 37.55$$

Then, $$h = \frac{k}{L} Nu_{combined} = \frac{0.02717 \text{ W/m.°C}}{0.12 \text{ m}}(37.55) = 8.502 \text{ W/m}^2\text{.°C}$$

and $$\dot{Q} = hA_s(T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{Q}}{hA_s} = 35°\text{C} + \frac{(100)(0.05\text{ W})}{(8.502\text{ W/m}^2.°\text{C})(0.024\text{ m}^2)} = \mathbf{59.5°C}$$

Therefore, natural convection lowers the surface temperature in this case by about 2°C.

Special Topic: Heat Transfer Through Windows

9-80C Windows are considered in three regions when analyzing heat transfer through them because the structure and properties of the frame are quite different than those of the glazing. As a result, heat transfer through the frame and the edge section of the glazing adjacent to the frame is two-dimensional. Even in the absence of solar radiation and air infiltration, heat transfer through the windows is more complicated than it appears to be. Therefore, it is customary to consider the windows in three regions when analyzing heat transfer through them: (1) the *center-of-glass*, (2) the *edge-of-glass*, and (3) the *frame* regions. When the heat transfer coefficient for all three regions are known, the overall U-value of the window is determined from

$$U_{\text{window}} = (U_{\text{center}}A_{\text{center}} + U_{\text{edge}}A_{\text{edge}} + U_{\text{frame}}A_{\text{frame}}) / A_{\text{window}}$$

where A_{window} is the window area, and A_{center}, A_{edge}, and A_{frame} are the areas of the center, edge, and frame sections of the window, respectively, and U_{center}, U_{edge}, and U_{frame} are the heat transfer coefficients for the center, edge, and frame sections of the window.

9-81C Of the three similar double pane windows with air gab widths of 5, 10, and 20 mm, the U-factor and thus the rate of heat transfer through the window will be a minimum for the window with 10-mm air gab, as can be seen from Fig. 9-44.

9-82C In an ordinary double pane window, about half of the heat transfer is by radiation. A practical way of reducing the radiation component of heat transfer is to reduce the emissivity of glass surfaces by coating them with low-emissivity (or "low-e") material.

9-83C When a thin polyester film is used to divide the 20-mm wide air of a double pane window space into two 10-mm wide layers, both (*a*) convection and (*b*) radiation heat transfer through the window will be reduced.

9-84C When a double pane window whose air space is flashed and filled with argon gas, (*a*) convection heat transfer will be reduced but (*b*) radiation heat transfer through the window will remain the same.

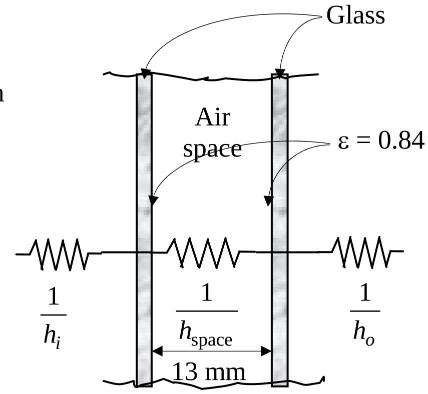
9-85C The heat transfer rate through the glazing of a double pane window is higher at the edge section than it is at the center section because of the two-dimensional effects due to heat transfer through the frame.

9-86C The *U*-factors of windows with aluminum frames will be highest because of the higher conductivity of aluminum. The U-factors of wood and vinyl frames are comparable in magnitude.

9-87 The U-factor for the center-of-glass section of a double pane window is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Heat transfer through the window is one-dimensional. **3** The thermal resistance of glass sheets is negligible.

Properties The emissivity of clear glass is given to be 0.84. The values of h_i and h_o for winter design conditions are h_i = 8.29 W/m^2.°C and h_o = 34.0 W/m^2.°C (from the text).



Analysis Disregarding the thermal resistance of glass sheets, which are small, the U-factor for the center region of a double pane window is determined from

$$\frac{1}{U_{\text{center}}} \cong \frac{1}{h_i} + \frac{1}{h_{\text{space}}} + \frac{1}{h_o}$$

where h_i, h_{space}, and h_o are the heat transfer coefficients at the inner surface of window, the air space between the glass layers, and the outer surface of the window, respectively. The effective emissivity of the air space of the double pane window is

$$\varepsilon_{\text{effective}} = \frac{1}{1/\varepsilon_1 + 1/\varepsilon_2 - 1} = \frac{1}{1/0.84 + 1/0.84 - 1} = 0.72$$

For this value of emissivity and an average air space temperature of 10°C with a temperature difference across the air space to be 15°C, we read h_{space} = 5.7 W/m^2.°C from Table 9-3 for 13-mm thick air space. Therefore,

$$\frac{1}{U_{\text{center}}} = \frac{1}{8.29} + \frac{1}{5.7} + \frac{1}{34.0} \rightarrow U_{\text{center}} = \mathbf{3.07\ W/m^2.°C}$$

Discussion The overall U-factor of the window will be higher because of the edge effects of the frame.

9-88 The rate of heat loss through an double-door wood framed window and the inner surface temperature are to be determined for the cases of single pane, double pane, and low-e triple pane windows.

Assumptions **1** Steady operating conditions exist. **2** Heat transfer through the window is one-dimensional. **3** Thermal properties of the windows and the heat transfer coefficients are constant. **4** Infiltration heat losses are not considered.

Properties The U-factors of the windows are given in Table 9-6.

Analysis The rate of heat transfer through the window can be determined from

$$\dot{Q}_{\text{window}} = U_{\text{overall}} A_{\text{window}} (T_i - T_o)$$

where T_i and T_o are the indoor and outdoor air temperatures, respectively, U_{overall} is the U-factor (the overall heat transfer coefficient) of the window, and A_{window} is the window area which is determined to be

$$A_{window} = \text{Height} \times \text{Width} = (1.2\text{ m})(1.8\text{ m}) = 2.16\text{ m}^2$$

The U-factors for the three cases can be determined directly from Table 9-6 to be 5.57, 2.86, and 1.46 $W/m^2.°C$, respectively. Also, the inner surface temperature of the window glass can be determined from Newton's law,

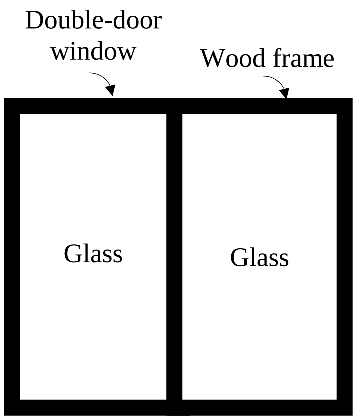
$$\dot{Q}_{\text{window}} = h_i A_{\text{window}} (T_i - T_{\text{glass}}) \quad \rightarrow \quad T_{\text{glass}} = T_i - \frac{\dot{Q}_{\text{window}}}{h_i A_{\text{window}}}$$

where h_i is the heat transfer coefficient on the inner surface of the window which is determined from Table 9-5 to be h_i = 8.3 $W/m^2.°C$. Then the rate of heat loss and the interior glass temperature for each case are determined as follows:

(*a*) Single glazing:

$$\dot{Q}_{\text{window}} = (5.57\text{ W/m}^2.°\text{C})(2.16\text{ m}^2)[20-(-8)]°\text{C} = \mathbf{337\ W}$$

$$T_{\text{glass}} = T_i - \frac{\dot{Q}_{\text{window}}}{h_i A_{\text{window}}} = 20°\text{C} - \frac{337\text{ W}}{(8.29\text{ W/m}^2.°\text{C})(2.16\text{ m}^2)} = \mathbf{1.2°C}$$



(*b*) Double glazing (13 mm air space):

$$\dot{Q}_{\text{window}} = (2.86\text{ W/m}^2.°\text{C})(2.16\text{ m}^2)[20-(-8)]°\text{C} = \mathbf{173\ W}$$

$$T_{\text{glass}} = T_i - \frac{\dot{Q}_{\text{window}}}{h_i A_{\text{window}}} = 20°\text{C} - \frac{173\text{ W}}{(8.29\text{ W/m}^2.°\text{C})(2.16\text{ m}^2)} = \mathbf{10.3°C}$$

(*c*) Triple glazing (13 mm air space, low-e coated):

$$\dot{Q}_{\text{window}} = (1.46\text{ W/m}^2.°\text{C})(2.16\text{ m}^2)[20-(-8)]°\text{C} = \mathbf{88.3\ W}$$

$$T_{\text{glass}} = T_i - \frac{\dot{Q}_{window}}{h_i A_{window}} = 20 - \frac{88.3\text{ W}}{(8.3\text{ W/m}^2.°\text{C})(2.16\text{ m}^2)} = \mathbf{15.1°C}$$

Discussion Note that heat loss through the window will be reduced by 49 percent in the case of double glazing and by 74 percent in the case of triple glazing relative to the single glazing case. Also, in the case of single glazing, the low inner glass surface temperature will cause considerable discomfort in the occupants because of the excessive heat loss from the body by radiation. It is raised from 1.2°C to 10.3°C in the case of double glazing and to 15.1°C in the case of triple glazing.

9-89 The overall U-factor for a double-door type window is to be determined, and the result is to be compared to the value listed in Table 9-6.

Assumptions **1** Steady operating conditions exist. **2** Heat transfer through the window is one-dimensional.

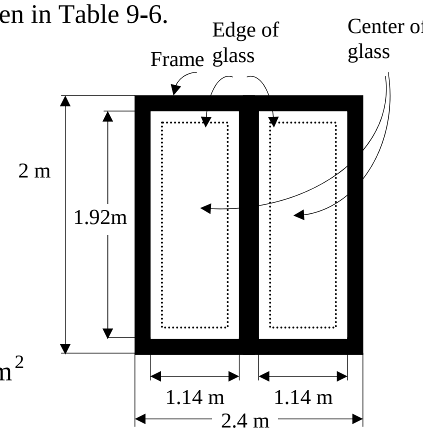
Properties The U-factors for the various sections of windows are given in Table 9-6.

Analysis The areas of the window, the glazing, and the frame are

$$A_{\text{window}} = \text{Height} \times \text{width} = (2\text{ m})(2.4\text{ m}) = 4.80\text{ m}^2$$

$$A_{\text{glazing}} = 2 \times (\text{Height} \times \text{width}) = 2(1.92\text{ m})(1.14\text{ m}) = 4.38\text{ m}^2$$

$$A_{\text{frame}} = A_{\text{window}} - A_{\text{glazing}} = 4.80 - 4.38 = 0.42\text{ m}^2$$



The edge-of-glass region consists of a 6.5-cm wide band around the perimeter of the glazings, and the areas of the center and edge sections of the glazing are determined to be

$$A_{\text{center}} = 2(\text{Height} \times \text{Width}) = 2(1.92 - 0.13\text{ m})(1.14 - 0.13\text{ m}) = 3.62\text{ m}^2$$

$$A_{\text{edge}} = A_{\text{glazing}} - A_{\text{center}} = 4.38 - 3.62 = 0.76\text{ m}^2$$

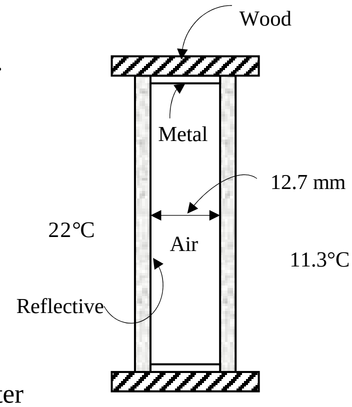
The U-factor for the frame section is determined from Table 9-4 to be U_{frame} = 2.8 W/m^2.°C. The U-factor for the center and edge sections are determined from Table 9-6 to be U_{center} = 2.78 W/m^2.°C and U_{edge} =3.40 W/m^2.°C. Then the overall U-factor of the entire window becomes

$$\begin{aligned} U_{\text{window}} &= (U_{\text{center}} A_{\text{center}} + U_{\text{edge}} A_{\text{edge}} + U_{\text{frame}} A_{\text{frame}}) / A_{\text{window}} \\ &= (2.78 \times 3.62 + 3.40 \times 0.76 + 2.8 \times 0.42) / 4.80 \\ &= \mathbf{2.88\ W/m^2.°C} \end{aligned}$$

Discussion The overall U-factor listed in Table 9-6 for the specified type of window is 2.86 W/m^2.°C, which is sufficiently close to the value obtained above.

9-90 The windows of a house in Atlanta are of double door type with wood frames and metal spacers. The average rate of heat loss through the windows in winter is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Heat transfer through the window is one-dimensional. **3** Thermal properties of the windows and the heat transfer coefficients are constant. **4** Infiltration heat losses are not considered.



Properties The U-factor of the window is given in Table 9-6 to be 2.13 W/m^2.°C.

Analysis The rate of heat transfer through the window can be determined from

$$\dot{Q}_{\text{window, ave}} = U_{\text{overall}} A_{\text{window}} (T_i - T_{o,\text{ave}})$$

where T_i and T_o are the indoor and outdoor air temperatures, respectively, U_{overall} is the U-factor (the overall heat transfer coefficient) of the window, and A_{window} is the window area. Substituting,

$$\dot{Q}_{\text{window, ave}} = (2.13\ \text{W/m}^2\cdot{}^\circ\text{C})(20\ \text{m}^2)(22-11.3)^\circ\text{C} = \mathbf{456\ W}$$

Discussion This is the "average" rate of heat transfer through the window in winter in the absence of any infiltration.

9-91E The *R*-value of the common double door windows that are double pane with 1/4-in of air space and have aluminum frames is to be compared to the *R*-value of *R*-13 wall. It is also to be determined if more heat is transferred through the windows or the walls.

Assumptions **1** Steady operating conditions exist. **2** Heat transfer through the window is one-dimensional. **3** Thermal properties of the windows and the heat transfer coefficients are constant. **4** Infiltration heat losses are not considered.

Properties The U-factor of the window is given in Table 9-6 to be 4.55×0.176 = 0.801 Btu/h.ft^2.°F.

Analysis The *R*-value of the windows is simply the inverse of its *U*-factor, and is determined to be

$$R_{\text{window}} = \frac{1}{U} = \frac{1}{0.801\,\text{Btu/h}\cdot\text{ft}^2\cdot{}^\circ\text{F}} = 1.25\,\text{h}\cdot\text{ft}^2\cdot{}^\circ\text{F/Btu}$$

which is less than 13. Therefore, the *R*-value of a double pane window is **much less** than the *R*-value of an *R*-13 wall.

Now consider a 1-ft^2 section of a wall. The solid wall and the window areas of this section are A_{wall} = 0.8 ft^2 and A_{window} = 0.2 ft^2. Then the rates of heat transfer through the two sections are determined to be

$$\dot{Q}_{\text{wall}} = U_{wall} A_{\text{wall}} (T_i - T_o) = A_{\text{wall}} \frac{T_i - T_o}{\text{R}-\text{value, wall}} = (0.8\,\text{ft}^2)\frac{\Delta T(^\circ\text{F})}{(13\,\text{h.ft}^2.^\circ\text{F/Btu}} = 0.0615\Delta T \text{ Btu/h}$$

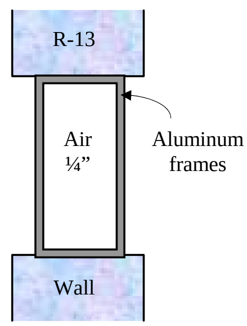
$$\dot{Q}_{\text{window}} = U_{\text{window}} A_{\text{window}} (T_i - T_o) = A_{\text{window}} \frac{T_i - T_o}{\text{R}-\text{value}} = (0.2\,\text{ft}^2)\frac{\Delta T(^\circ\text{F})}{(1.25\,\text{h.ft}^2.^\circ\text{F/Btu}} = 0.160\Delta T \text{ Btu/h}$$

Therefore, the rate of heat transfer through a double pane window is **much more** than the rate of heat transfer through an *R*-13 wall.

Discussion The ratio of heat transfer through the wall and through the window is

$$\frac{\dot{Q}_{\text{window}}}{\dot{Q}_{\text{wall}}} = \frac{0.160\,\text{Btu/h}}{0.0615\,\text{Btu/h}} = 2.60$$

Therefore, 2.6 times more heat is lost through the windows than through the walls although the windows occupy only 20% of the wall area.



9-92 The overall U-factor of a window is given to be U = 2.76 W/m^2.°C for 12 km/h winds outside. The new U-factor when the wind velocity outside is doubled is to be determined.

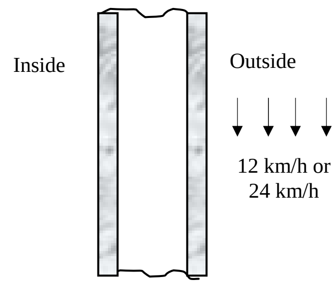
Assumptions Thermal properties of the windows and the heat transfer coefficients are constant.

Properties The heat transfer coefficients at the outer surface of the window are h_o = 22.7 W/m^2.°C for 12 km/h winds, and h_o = 34.0 W/m^2.°C for 24 km/h winds (from the text).

Analysis The corresponding convection resistances for the outer surfaces of the window are

$$R_{\text{o, 12 km/h}} = \frac{1}{h_{\text{o, 12 km/h}}} = \frac{1}{22.7\ \text{W/m}^2\cdot°\text{C}} = 0.044\ \text{m}^2\cdot°\text{C/W}$$

$$R_{\text{o, 24 km/h}} = \frac{1}{h_{\text{o, 24 km/h}}} = \frac{1}{34.0\ \text{W/m}^2\cdot°\text{C}} = 0.029\ \text{m}^2\cdot°\text{C/W}$$



Also, the R-value of the window at 12 km/h winds is

$$R_{\text{window, 12 km/h}} = \frac{1}{U_{\text{window, 12 km/h}}} = \frac{1}{2.76\ \text{W/m}^2\cdot°\text{C}} = 0.362\ \text{m}^2\cdot°\text{C/W}$$

Noting that all thermal resistances are in series, the thermal resistance of the window for 24 km/h winds is determined by replacing the convection resistance for 12 km/h winds by the one for 24 km/h:

$$R_{\text{window, 24 km/h}} = R_{\text{window, 12 km/h}} - R_{\text{o, 12 km/h}} + R_{\text{o, 24 km/h}} = 0.362 - 0.044 + 0.029 = 0.347\ \text{m}^2\cdot°\text{C/W}$$

Then the U-factor for the case of 24 km/h winds becomes

$$U_{\text{window, 24 km/h}} = \frac{1}{R_{\text{window, 24 km/h}}} = \frac{1}{0.347\ \text{m}^2\cdot°\text{C/W}} = \mathbf{2.88\ W/m^2\cdot°C}$$

Discussion Note that doubling of the wind velocity increases the U-factor only slightly (about 4%) from 2.76 to 2.88 W/m^2.°C.

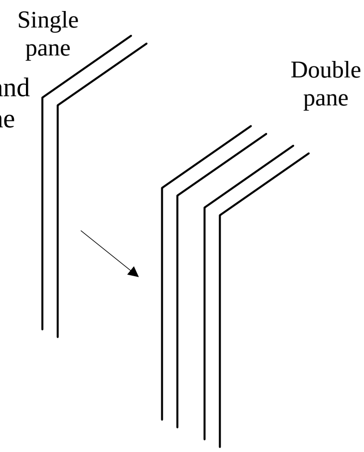
9-93 The existing wood framed single pane windows of an older house in Wichita are to be replaced by double-door type vinyl framed double pane windows with an air space of 6.4 mm. The amount of money the new windows will save the home owner per month is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Heat transfer through the window is one-dimensional. **3** Thermal properties of the windows and the heat transfer coefficients are constant. **4** Infiltration heat losses are not considered.

Properties The U-factors of the windows are 5.57 W/m^2.°C for the old single pane windows, and 3.20 W/m^2.°C. for the new double pane windows (Table 9-6).

Analysis The rate of heat transfer through the window can be determined from

$$Q_{\text{window}} = U_{\text{overall}} A_{\text{window}} (T_i - T_o)$$



where T_i and T_o are the indoor and outdoor air temperatures, respectively, U_{overall} is the U-factor (the overall heat transfer coefficient) of the window, and A_{window} is the window area. Noting that the heaters will turn on only when the outdoor temperature drops below 18°C, the rates of heat transfer due to electric heating for the old and new windows are determined to be

$$\dot{Q}_{\text{window, old}} = (5.57\ \text{W/m}^2.°\text{C})(12\ \text{m}^2)(18-7.1)°\text{C} = 729\ \text{W}$$

$$\dot{Q}_{\text{window, new}} = (3.20\ \text{W/m}^2.°\text{C})(12\ \text{m}^2)(18-7.1)°\text{C} = 419\ \text{W}$$

$$\dot{Q}_{\text{saved}} = \dot{Q}_{\text{window, old}} - \dot{Q}_{\text{window, new}} = 729 - 419 = 310\ \text{W}$$

Then the electrical energy and cost savings per month becomes

$$\text{Energy savings} = \dot{Q}_{\text{saved}} \Delta t = (0.310\ \text{kW})(30 \times 24\ \text{h/month}) = 223\ \text{kWh/month}$$

$$\text{Cost savings} = (\text{Energy savings})(\text{Unit cost of energy}) = (223\ \text{kWh/month})(\$0.07/\text{kWh}) = \mathbf{\$15.62/month}$$

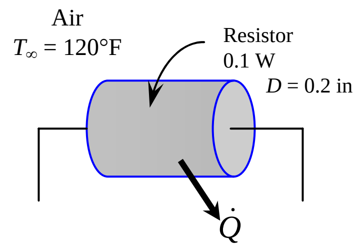
Discussion We would obtain the same result if we used the actual indoor temperature (probably 22°C) for T_i instead of the balance point temperature of 18°C.

Review Problems

9-94E A small cylindrical resistor mounted on the lower part of a vertical circuit board. The approximate surface temperature of the resistor is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** Radiation effects are negligible. **5** Heat transfer through the connecting wires is negligible.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2 = (220+120)/2 = 170$°F are (Table A-15E)



$$k = 0.01692 \text{ Btu/h.ft.°F}$$
$$\upsilon = 0.222\times10^{-3} \text{ ft}^2\text{/s}$$
$$\Pr = 0.7161$$
$$\beta = \frac{1}{T_f} = \frac{1}{(170+460)\text{R}} = 0.001587 \text{ R}^{-1}$$

Analysis The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the surface temperature to be 220°F for the evaluation of the properties and h. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the diameter of resistor, $L_c = D = 0.2$ in. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D^3}{\upsilon^2}\Pr = \frac{(32.2 \text{ ft/s}^2)(0.001587 \text{ R}^{-1})(220-120 \text{ R})(0.2/12 \text{ ft})^3}{(0.222\times10^{-3} \text{ ft}^2\text{/s})^2}(0.7161) = 343.8$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/\Pr\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(343.8)^{1/6}}{\left[1+\left(0.559/0.7161\right)^{9/16}\right]^{8/27}}\right\}^2 = 2.105$$

$$h = \frac{k}{D}Nu = \frac{0.01692 \text{ Btu/h.ft.°F}}{0.2/12 \text{ ft}}(2.105) = 2.138 \text{ Btu/h.ft}^2\text{.°F}$$
$$A_s = \pi DL + 2D^2/4 = \pi(0.2/12 \text{ ft})(0.3/12 \text{ ft}) + 2\pi(0.2/12 \text{ ft})^2/4 = 0.00175 \text{ ft}^2$$

and
$$\dot{Q} = hA_s(T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{Q}}{hA_s} = 120\text{°F} + \frac{(0.1\times3.412)\text{ Btu/h}}{(2.138 \text{ Btu/h.ft}^2\text{.°F})(0.00175 \text{ ft}^2)} = \mathbf{211.5°F}$$

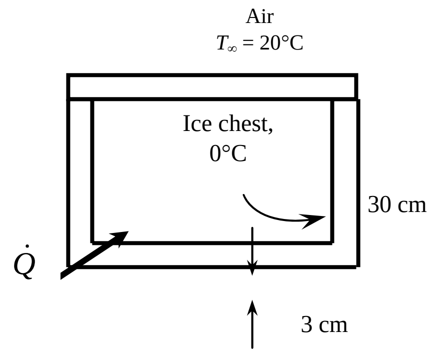
which is sufficiently close to the assumed temperature for the evaluation of properties. Therefore, there is no need to repeat calculations.

9-95 An ice chest filled with ice at 0°C is exposed to ambient air. The time it will take for the ice in the chest to melt completely is to be determined for natural and forced convection cases.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** Heat transfer from the base of the ice chest is disregarded. **4** Radiation effects are negligible. **5** Heat transfer coefficient is the same for all surfaces considered. **6** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2$ = (15+20)/2 = 17.5°C are (Table A-15)

$$k = 0.02495 \text{ W/m.°C}$$
$$\upsilon = 1.493\times10^{-5} \text{ m}^2\text{/s}$$
$$\text{Pr} = 0.7316$$
$$\beta = \frac{1}{T_f} = \frac{1}{(17.5+273)\text{K}} = 0.003442 \text{ K}^{-1}$$



Analysis The solution of this problem requires a trial-and-error approach since the determination of the Rayleigh number and thus the Nusselt number depends on the surface temperature which is unknown. We start the solution process by "guessing" the surface temperature to be 15°C for the evaluation of the properties and h. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length for the side surfaces is the height of the chest, $L_c = L$ = 0.3 m Then,

$$Ra = \frac{g\beta(T_\infty - T_s)L^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003442 \text{ K}^{-1})(20-15 \text{ K})(0.3 \text{ m})^3}{(1.493\times10^{-5} \text{ m}^2\text{/s})^2}(0.7316) = 1.495\times10^7$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(1.495\times10^7)^{1/6}}{\left[1+\left(\frac{0.492}{0.7316}\right)^{9/16}\right]^{8/27}}\right\}^2 = 35.15$$

$$h = \frac{k}{L}Nu = \frac{0.02495 \text{ W/m.°C}}{0.3 \text{ m}}(35.15) = 2.923 \text{ W/m}^2\text{.°C}$$

The heat transfer coefficient at the top surface can be determined similarly. However, the top surface constitutes only about one-fourth of the heat transfer area, and thus we can use the heat transfer coefficient for the side surfaces for the top surface also for simplicity. The heat transfer surface area is

$$A_s = 4(0.3 \text{ m})(0.4 \text{ m}) + (0.4 \text{ m})(0.4 \text{ m}) = 0.64 \text{ m}^2$$

Then the rate of heat transfer becomes

$$\dot{Q} = \frac{T_\infty - T_{s,i}}{R_{wall} + R_{conv,o}} = \frac{T_\infty - T_{s,i}}{\frac{L}{kA_s} + \frac{1}{hA_s}} = \frac{(20-0)\text{°C}}{\frac{0.03 \text{ m}}{(0.033 \text{ W/m.°C})(0.64 \text{ m}^2)} + \frac{1}{(2.923 \text{ W/m}^2\text{.°C})(0.64 \text{ m}^2)}} = 10.23 \text{ W}$$

The outer surface temperature of the ice chest is determined from Newton's law of cooling to be

$$\dot{Q} = hA_s(T_\infty - T_s) \rightarrow T_s = T_\infty - \frac{\dot{Q}}{hA_s} = 20\text{°C} - \frac{10.4 \text{ W}}{(2.923 \text{ W/m}^2\text{.C})(0.64 \text{ m}^2)} = 14.53\text{°C}$$

which is almost identical to the assumed value of 15°C used in the evaluation of properties and h. Therefore, there is no need to repeat the calculations. Then the rate at which the ice will melt becomes

$$\dot{Q} = \dot{m}h_{if} \rightarrow \dot{m} = \frac{\dot{Q}}{h_{if}} = \frac{10.23\times10^{-3}\ \text{kJ/s}}{333.7\ \text{kJ/kg}} = 3.066\times10^{-5}\ \text{kg/s}$$

Therefore, the melting of the ice in the chest completely will take

$$m = \dot{m}\Delta t \longrightarrow \Delta t = \frac{m}{\dot{m}} = \frac{30\ \text{kg}}{3.066\times10^{-5}\ \text{kg/s}} = 9.786\times10^5\ \text{s} = \mathbf{271.8\ h = 11.3\ days}$$

(b) The temperature drop across the styrofoam will be much greater in this case than that across thermal boundary layer on the surface. Thus we assume outer surface temperature of the styrofoam to be $19\,°\text{C}$. Radiation heat transfer will be neglected. The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (19+20)/2 = 19.5°\text{C}$ are (Table A-15)

$$k = 0.0251\ \text{W/m.°C}$$
$$\upsilon = 1.512\times10^{-5}\ \text{m}^2/\text{s}$$
$$\text{Pr} = 0.7311$$
$$\beta = \frac{1}{T_f} = \frac{1}{(19.5+273)\text{K}} = 0.00342\ \text{K}^{-1}$$

The characteristic length in this case is the width of the chest, $L_c = W = 0.4$ m. Then,

$$\text{Re} = \frac{V_\infty W}{\upsilon} = \frac{(50\times1000/3600\ \text{m/s})(0.4\ \text{m})}{1.512\times10^{-5}\ \text{m}^2/\text{s}} = 367{,}538$$

which is less than critical Reynolds number (5×10^5). Therefore the flow is laminar, and the Nusselt number is determined from

$$Nu = \frac{hW}{k} = 0.664\,\text{Re}^{0.5}\,\text{Pr}^{1/3} = 0.664(367{,}538)^{0.5}(0.7311)^{1/3} = 362.6$$

$$h = \frac{k}{W}Nu = \frac{0.0251\ \text{W/m.°C}}{0.4\ \text{m}}(362.6) = 22.76\ \text{W/m}^2.°\text{C}$$

Then the rate of heat transfer becomes

$$\dot{Q} = \frac{T_\infty - T_{s,i}}{R_{wall} + R_{conv,o}} = \frac{T_\infty - T_{s,i}}{\frac{L}{kA_s} + \frac{1}{hA_s}} = \frac{(20-0)°\text{C}}{\frac{0.03\ \text{m}}{(0.033\ \text{W/m.°C})(0.64\ \text{m}^2)} + \frac{1}{(22.76\ \text{W/m}^2.°\text{C})(0.64\ \text{m}^2)}} = 13.43\ \text{W}$$

The outer surface temperature of the ice chest is determined from Newton's law of cooling to be

$$\dot{Q} = hA_s(T_\infty - T_s) \rightarrow T_s = T_\infty - \frac{\dot{Q}}{hA_s} = 20°\text{C} - \frac{13.43\ \text{W}}{(22.76\ \text{W/m}^2.\text{C})(0.64\ \text{m}^2)} = 19.1°\text{C}$$

which is almost identical to the assumed value of 19°C used in the evaluation of properties and h. Therefore, there is no need to repeat the calculations. Then the rate at which the ice will melt becomes

$$\dot{Q} = \dot{m}h_{if} \rightarrow \dot{m} = \frac{\dot{Q}}{h_{if}} = \frac{13.43\times10^{-3}\ \text{kJ/s}}{333.7\ \text{kJ/kg}} = 4.025\times10^{-5}\ \text{kg/s}$$

Therefore, the melting of the ice in the chest completely will take

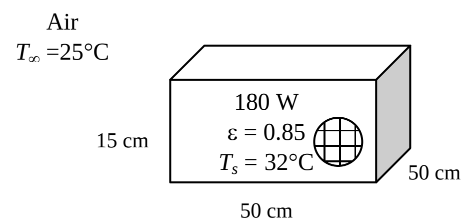
$$m = \dot{m}\Delta t \longrightarrow \Delta t = \frac{m}{\dot{m}} = \frac{30}{4.025\times10^{-5}} = 7.454\times10^5\ \text{s} = \mathbf{207.05\ h = 8.6\ days}$$

9-96 An electronic box is cooled internally by a fan blowing air into the enclosure. The fraction of the heat lost from the outer surfaces of the electronic box is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** Heat transfer from the base surface is disregarded. **4** The pressure of air inside the enclosure is 1 atm.

Properties The properties of air at 1 atm and the film temperature of
$(T_s+T_\infty)/2 = (32+15)/2 = 28.5°\text{C}$ are (Table A-15)

$$k = 0.02577 \text{ W/m.°C}$$
$$\upsilon = 1.594\times10^{-5} \text{ m}^2\text{/s}$$
$$\Pr = 0.7286$$
$$\beta = \frac{1}{T_f} = \frac{1}{(28.5+273)\text{K}} = 0.003317 \text{ K}^{-1}$$



Analysis Heat loss from the horizontal top surface:

The characteristic length in this case is $L_c = \frac{A_s}{p} = \frac{(0.5\text{ m})^2}{2[(0.5\text{ m})+(0.5\text{ m})]} = 0.125\text{ m}$. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\upsilon^2}\Pr = \frac{(9.81\text{ m/s}^2)(0.003317\text{ K}^{-1})(32-25\text{ K})(0.125\text{ m})^3}{(1.594\times10^{-5}\text{ m}^2\text{/s})^2}(0.7286) = 1.275\times10^6$$

$$Nu = 0.54Ra^{1/4} = 0.54(1.275\times10^6)^{1/4} = 18.15$$

$$h = \frac{k}{L_c}Nu = \frac{0.02577\text{ W/m.°C}}{0.125\text{ m}}(18.15) = 3.741\text{ W/m}^2\text{.°C}$$

$$A_{top} = (0.5\text{ m})^2 = 0.25\text{ m}^2$$

and $$\dot{Q}_{top} = hA_{top}(T_s - T_\infty) = (3.741\text{ W/m}^2\text{.°C})(0.25\text{ m}^2)(32-25)\text{°C} = 6.55\text{ W}$$

Heat loss from vertical side surfaces:

The characteristic length in this case is the height of the box $L_c = L$ =0.15 m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\Pr = \frac{(9.81\text{ m/s}^2)(0.003317\text{ K}^{-1})(32-25\text{ K})(0.15\text{ m})^3}{(1.594\times10^{-5}\text{ m}^2\text{/s})^2}(0.7286) = 2.204\times10^6$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(2.204\times10^7)^{1/6}}{\left[1+\left(\frac{0.492}{0.7286}\right)^{9/16}\right]^{8/27}}\right\}^2 = 20.55$$

$$h = \frac{k}{L}Nu = \frac{0.02577\text{ W/m.°C}}{0.15\text{ m}}(20.55) = 3.530\text{ W/m}^2\text{.°C}$$

$$A_{side} = 4(0.15\text{ m})(0.5\text{ m}) = 0.3\text{ m}^2$$

and

$$\dot{Q}_{side} = hA_{side}(T_s - T_\infty) = (3.530\text{ W/m}^2\text{.°C})(0.3\text{ m}^2)(32-25)\text{°C} = 7.41\text{ W}$$

The radiation heat loss is

$$\dot{Q}_{rad} = \varepsilon A_s\sigma(T_s^{\,4} - T_{surr}^{\,4})$$
$$= (0.85)(0.25+0.3\text{ m}^2)(5.67\times10^{-8}\text{ W/m}^2\text{.K}^4)[(32+273\text{ K})^4 - (25+273\text{ K})^4] = 20.34\text{ W}$$

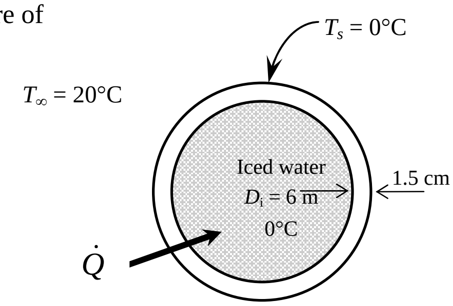
Then the fraction of the heat loss from the outer surfaces of the box is determined to be

$$f = \frac{(6.55+7.41+20.34)\text{W}}{180\text{ W}} = 0.1906 = \mathbf{19.1\%}$$

9-97 A spherical tank made of stainless steel is used to store iced water. The rate of heat transfer to the iced water and the amount of ice that melts during a 24-h period are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** Thermal resistance of the tank is negligible. **4** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (0+20)/2 = 10°C$ are (Table A-15)



$$k = 0.02439 \text{ W/m.°C}$$
$$\upsilon = 1.426\times10^{-5} \text{ m}^2\text{/s}$$
$$\text{Pr} = 0.7336$$
$$\beta = \frac{1}{T_f} = \frac{1}{(10+273)\text{K}} = 0.003534 \text{ K}^{-1}$$

Analysis (*a*) The characteristic length in this case is $L_c = D_o = 6.03$ m. Then,

$$Ra = \frac{g\beta(T_\infty - T_s)D_o^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003534 \text{ K}^{-1})(20-0 \text{ K})(6.03 \text{ m})^3}{(1.426\times10^{-5} \text{ m}^2\text{/s})^2}(0.7336) = 5.485\times10^{11}$$

$$Nu = 2 + \frac{0.589Ra^{1/4}}{\left[1+(0.469/\text{Pr})^{9/16}\right]^{4/9}} = 2 + \frac{0.589(5.485\times10^{11})^{1/4}}{\left[1+(0.469/0.7336)^{9/16}\right]^{4/9}} = 394.5$$

$$h = \frac{k}{D_o}Nu = \frac{0.02439 \text{ W/m.°C}}{6.03 \text{ m}}(394.5) = 1.596 \text{ W/m}^2\text{.°C}$$
$$A_s = \pi D_o^2 = \pi(6.03 \text{ m})^2 = 114.2 \text{ m}^2$$

and

$$\dot{Q} = hA_s(T_\infty - T_s) = (1.596 \text{ W/m}^2\text{.°C})(114.2 \text{ m}^2)(20-0)\text{°C} = 3646 \text{ W}$$

Heat transfer by radiation and the total rate of heat transfer are

$$\dot{Q}_{rad} = \varepsilon A_s\sigma({T_s}^4 - {T_{surr}}^4)$$
$$= (1)(114.2 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2\text{.K}^4)[(20+273 \text{ K})^4 - (0+273 \text{ K})^4] = 11{,}759 \text{ W}$$
$$\dot{Q}_{total} = 3646 + 11{,}759 = 15{,}404 \text{ W} \cong \mathbf{15.4 \text{ kW}}$$

(*b*) The total amount of heat transfer during a 24-hour period is

$$Q = \dot{Q}\Delta t = (15.4 \text{ kJ/s})(24 \text{ h/day}\times3600 \text{ s/h}) = 1.331\times10^6 \text{ kJ/day}$$

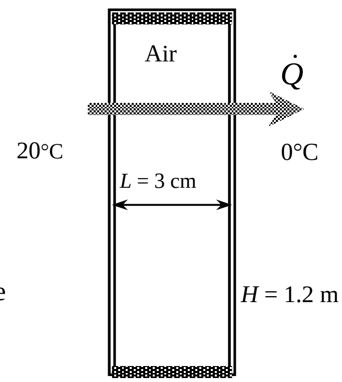
Then the amount of ice that melts during this period becomes

$$Q = mh_{if} \longrightarrow m = \frac{Q}{h_{if}} = \frac{1.331\times10^6 \text{ kJ}}{333.7 \text{ kJ/kg}} = \mathbf{3988 \text{ kg}}$$

9-98 A double-pane window consisting of two layers of glass separated by an air space is considered. The rate of heat transfer through the window and the temperature of its inner surface are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** Radiation effects are negligible. **4** The pressure of air inside the enclosure is 1 atm.

Properties We expect the average temperature of the air gap to be roughly the average of the indoor and outdoor temperatures, and evaluate The properties of air at 1 atm and the average temperature of $(T_{\infty 1}+T_{\infty 2})/2$ = (20 +0)/2 = 10°C are (Table A-15)



$$k = 0.02439 \text{ W/m.°C}$$
$$\upsilon = 1.426\times 10^{-5} \text{ m}^2/\text{s}$$
$$\text{Pr} = 0.7336$$
$$\beta = \frac{1}{T_f} = \frac{1}{(10+273)\text{K}} = 0.003534 \text{ K}^{-1}$$

Analysis We "guess" the temperature difference across the air gap to be 15°C = 15 K for use in the Ra relation. The characteristic length in this case is the air gap thickness, $L_c = L$ = 0.03 m. Then,

$$Ra = \frac{g\beta(T_1 - T_2)L^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003534 \text{ K}^{-1})(15 \text{ K})(0.03 \text{ m})^3}{(1.426\times 10^{-5} \text{ m}^2/\text{s})^2}(0.7336) = 5.065\times 10^4$$

Then the Nusselt number and the heat transfer coefficient are determined to be

$$Nu = 0.42Ra^{1/4}\text{ Pr}^{0.012}\left(\frac{H}{L}\right)^{-0.3} = 0.42(5.065\times 10^4)^{1/4}(0.7336)^{0.012}\left(\frac{1.2 \text{ m}}{0.03 \text{ m}}\right)^{-0.3} = 2.076$$

$$h_{air} = \frac{k}{L}Nu = \frac{0.02439 \text{ W/m.°C}}{0.03 \text{ m}}(2.076) = 1.688 \text{ W/m}^2.\text{°C}$$

Then the rate of heat transfer through this double pane window is determined to be

$$A_s = H\times W = (1.2 \text{ m})(2 \text{ m}) = 2.4 \text{ m}^2$$

$$\dot{Q} = \frac{T_{\infty,i} - T_{\infty,o}}{R_{conv,i} + R_{cond,glasses} + R_{conv,air} + R_{conv,o}} = \frac{T_\infty - T_{s,i}}{\frac{1}{h_i A_s} + \frac{2t_{\text{glass}}}{k_{glass}A_s} + \frac{1}{h_{air}A_s} + \frac{1}{h_o A_s}}$$

$$= \frac{20-0}{\frac{1}{(10)(2.4)} + \frac{2(0.003)}{(0.78)(2.4)} + \frac{1}{(1.688)(2.4)} + \frac{1}{(25)(2.4)}} = \mathbf{65\ W}$$

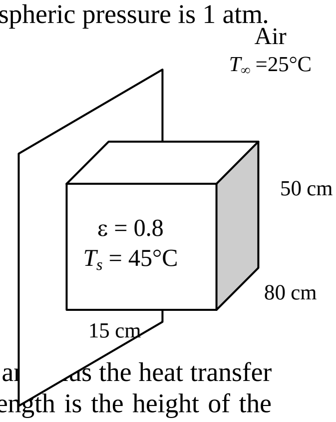
Check: The temperature drop across the air gap is determined from

$$\dot{Q} = hA_s\Delta T \ \rightarrow\ \Delta T = \frac{\dot{Q}}{hA_s} = \frac{65 \text{ W}}{(1.688 \text{ W/m}^2.\text{°C})(2.4 \text{ m}^2)} = 16.0\text{°C}$$

which is very close to the assumed value of 15°C used in the evaluation of the Ra number.

9-99 An electric resistance space heater filled with oil is placed against a wall. The power rating of the heater and the time it will take for the heater to reach steady operation when it is first turned on are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** Heat transfer from the back, bottom, and top surfaces are disregarded. **4** The local atmospheric pressure is 1 atm.



Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (45+25)/2 = 35$°C are (Table A-15)

$$k = 0.02625\ \text{W/m.°C}$$
$$\upsilon = 1.655\times10^{-5}\ \text{m}^2/\text{s}$$
$$\text{Pr} = 0.7268$$
$$\beta = \frac{1}{T_f} = \frac{1}{(35+273)\text{K}} = 0.003247\ \text{K}^{-1}$$

Analysis Heat transfer from the top and bottom surfaces are said to be negligible, and thus the heat transfer area in this case consists of the three exposed side surfaces. The characteristic length is the height of the box, $L_c = L = 0.5$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\text{Pr} = \frac{(9.81\,\text{m/s}^2)(0.003247\,\text{K}^{-1})(45-25\,\text{K})(0.5\,\text{m})^3}{(1.655\times10^{-5}\ \text{m}^2/\text{s})^2}(0.7268) = 2.114\times10^8$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(2.114\times10^8)^{1/6}}{\left[1+\left(\frac{0.492}{0.7268}\right)^{9/16}\right]^{8/27}}\right\}^2 = 76.68$$

$$h = \frac{k}{L}Nu = \frac{0.02625\ \text{W/m.°C}}{0.5\,\text{m}}(76.68) = 4.026\ \text{W/m}^2.\text{°C}$$
$$A_s = (0.5\,\text{m})(0.8\,\text{m}) + 2(0.15\,\text{m})(0.5\,\text{m}) = 0.55\,\text{m}^2$$

and $\dot{Q} = hA_s(T_s - T_\infty) = (4.026\ \text{W/m}^2.\text{°C})(0.55\,\text{m}^2)(45-25)\text{°C} = 44.3\,\text{W}$

The radiation heat loss is

$$\dot{Q}_{rad} = \varepsilon A_s\sigma(T_s{}^4 - T_{surr}{}^4) = (0.8)(0.55\,\text{m}^2)(5.67\times10^{-8}\ \text{W/m}^2.\text{K}^4)[(45+273\,\text{K})^4 - (25+273\,\text{K})^4] = 58.4\,\text{W}$$

Then the total rate of heat transfer, thus the power rating of the heater becomes

$$\dot{Q}_{total} = 44.3 + 58.4 = \mathbf{102.7\ W}$$

The specific heat of the oil at the average temperature of the oil is 1943 J/kg.°C. Then the amount of heat transfer needed to raise the temperature of the oil to the steady operating temperature and the time it takes become

$$Q = mC_p(T_2 - T_1) = (45\,\text{kg})(1943\ \text{J/kg.°C})(45-25)\text{°C} = 1.749\times10^6\ \text{J}$$

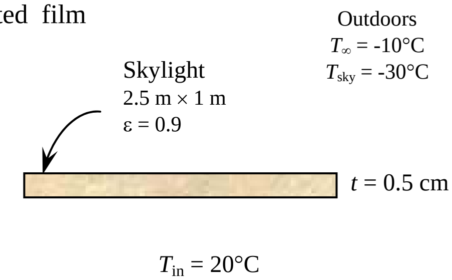
$$Q = \dot{Q}\Delta t \longrightarrow \Delta t = \frac{Q}{\dot{Q}} = \frac{1.749\times10^6\ \text{kJ}}{100\ \text{J/s}} = 17{,}034\,\text{s} = \mathbf{4.73\,h}$$

which is not practical. Therefore, the surface temperature of the heater must be allowed to be higher than 45°C.

9-100 A horizontal skylight made of a single layer of glass on the roof of a house is considered. The rate of heat loss through the skylight is to be determined for two cases.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2$ = (-4-10)/2 = -7°C are (Table A-15)



$$k = 0.0231\ \text{W/m.°C}$$
$$\upsilon = 1.278\times10^{-5}\ \text{m}^2/\text{s}$$
$$\text{Pr} = 0.738$$
$$\beta = \frac{1}{T_f} = \frac{1}{(-7+273)\text{K}} = 0.003759\ \text{K}^{-1}$$

Analysis We assume radiation heat transfer inside the house to be negligible. We start the calculations by "guessing" the glass temperature to be 4°C for the evaluation of the properties and h. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is determined from $L_c = \dfrac{A_s}{p} = \dfrac{(1\,\text{m})(2.5\,\text{m})}{2(1\,\text{m}+2.5\,\text{m})} = 0.357\ \text{m}$. Then,

$$Ra = \frac{g\beta(T_1-T_2)L_c^3}{\upsilon^2}\text{Pr} = \frac{(9.81\,\text{m/s}^2)(0.003759\,\text{K}^{-1})[-4-(-10)\,\text{K}](0.357\,\text{m})^3}{(1.278\times10^{-5}\ \text{m}^2/\text{s})^2}(0.738) = 4.553\times10^7$$

$$Nu = 0.15Ra^{1/3} = 0.15(4.553\times10^7)^{1/3} = 53.56$$

$$h_o = \frac{k}{L_c}Nu = \frac{0.0231\,\text{W/m.°C}}{0.357\,\text{m}}(53.56) = 3.465\ \text{W/m}^2.\text{°C}$$
$$A_s = (1\,\text{m})(2.5\,\text{m}) = 2.5\,\text{m}^2$$

Using the assumed value of glass temperature, the radiation heat transfer coefficient is determined to be

$$\begin{aligned} h_{rad} &= \varepsilon\sigma(T_s+T_{sky})(T_s{}^2+T_{sky}{}^2) \\ &= 0.9(5.67\times10^{-8}\ \text{W/m}^2.\text{K}^4)[(-4+273)+(-30+273)][(-4+273)^2+(-30+273)^2]\text{K}^3 \\ &= 3.433\,\text{W/m}^2.\text{K} \end{aligned}$$

Then the combined convection and radiation heat transfer coefficient outside becomes

$$h_{o,combined} = h_o + h_{rad} = 3.465+3.433 = 6.898\ \text{W/m}^2$$

Again we take the glass temperature to be -4°C for the evaluation of the properties and h for the inner surface of the skylight. The properties of air at 1 atm and the film temperature of T_f = (-4+20)/2 = 8°C are (Table A-15)

$$k = 0.02424\ \text{W/m.°C}$$
$$\upsilon = 1.409\times10^{-5}\ \text{m}^2/\text{s}$$
$$\text{Pr} = 0.7342$$
$$\beta = \frac{1}{T_f} = \frac{1}{(8+273)\text{K}} = 0.003559\ \text{K}^{-1}$$

The characteristic length in this case is also 0.357 m. Then,

$$Ra = \frac{g\beta(T_1 - T_2)L_c^3}{\upsilon^2}\Pr = \frac{(9.81\,\text{m/s}^2)(0.003559\,\text{K}^{-1})[20-(-4)\,\text{K})(0.357\,\text{m})^3}{(1.409\times10^{-5}\ \text{m}^2/\text{s})^2}(0.7342) = 1.412\times10^8$$

$$Nu = 0.27Ra^{1/4} = 0.27(1.412\times10^8)^{1/4} = 29.43$$

$$h_i = \frac{k}{L_c}Nu = \frac{0.02424\,\text{W/m.°C}}{0.357\,\text{m}}(29.43) = 1.998\,\text{W/m}^2\text{.°C}$$

Using the thermal resistance network, the rate of heat loss through the skylight is determined to be

$$\dot{Q}_{skylight} = \frac{T_{s,i} - T_{\infty,o}}{R_{conv,i} + R_{cond,glas} + R_{combined,o}}$$
$$= \frac{A_s(T_{room} - T_{out})}{\dfrac{1}{h_i} + \dfrac{t_{\text{glass}}}{k_{glass}} + \dfrac{1}{h}} = \frac{(2.5\,\text{m}^2)[20-(-10)]\text{°C}}{\dfrac{1}{1.998\,\text{W/m}^2\text{.°C}} + \dfrac{0.005\,\text{m}}{0.78\,\text{W/m.°C}} + \dfrac{1}{6.898\,\text{W/m}^2\text{.°C}}} = 115\,\text{W}$$

Using the same heat transfer coefficients for simplicity, the rate of heat loss through the roof in the case of R-5.34 construction is determined to be

$$\dot{Q}_{roof} = \frac{T_{s,i} - T_{\infty,o}}{R_{conv,i} + R_{cond} + R_{combined,o}}$$
$$= \frac{A_s(T_{room} - T_{out})}{\dfrac{1}{h_i} + R_{glass} + \dfrac{1}{h}} = \frac{(2.5\,\text{m}^2)[20-(-10)]\text{°C}}{\dfrac{1}{1.998\,\text{W/m}^2\text{.°C}} + 5.34\,\text{m}^2\text{.°C/W} + \dfrac{1}{6.898\,\text{W/m}^2\text{.°C}}} = 5.36\,\text{W}$$

Therefore, a house loses 115/5.36 ≅ 21 times more heat through the skylights than it does through an insulated wall of the same size.

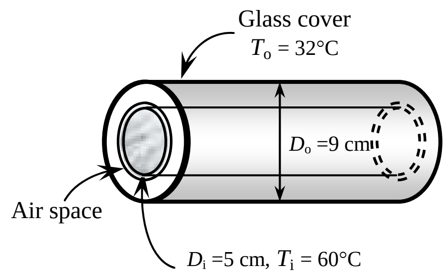
Using Newton’s law of cooling, the glass temperature corresponding to a heat transfer rate of 115 W is calculated to be –3.3°C, which is sufficiently close to the assumed value of -4°C. Therefore, there is no need to repeat the calculations.

9-101 A solar collector consists of a horizontal copper tube enclosed in a concentric thin glass tube. Water is heated in the tube, and the annular space between the copper and glass tube is filled with air. The rate of heat loss from the collector by natural convection is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** Radiation effects are negligible. **3** The pressure of air in the enclosure is 1 atm.

Properties The properties of air at 1 atm and the average temperature of $(T_i+T_o)/2 = (60+32)/2 = 46$°C are (Table A-15)

$$k = 0.02706 \text{ W/m.°C}$$
$$\upsilon = 1.759\times10^{-5} \text{ m}^2\text{/s}$$
$$\text{Pr} = 0.7239$$
$$\beta = \frac{1}{T_f} = \frac{1}{(46+273)\text{K}} = 0.003135 \text{ K}^{-1}$$



Analysis The characteristic length in this case is the distance between the two cylinders ,

$$L_c = \frac{D_o - D_i}{2} = \frac{(9-5)\text{ cm}}{2} = 2 \text{ cm}$$

and,

$$Ra = \frac{g\beta(T_i - T_o)L_c^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003135 \text{ K}^{-1})(60-32 \text{ K})(0.02 \text{ m})^3}{(1.759\times10^{-5} \text{ m}^2\text{/s})^2}(0.7239) = 16{,}106$$

The effective thermal conductivity is

$$F_{\text{cyl}} = \frac{\left[\ln\frac{D_o}{D_i}\right]^4}{L_c^3(D_i^{-3/5} + D_o^{-3/5})^5} = \frac{\left[\ln\frac{0.09 \text{ m}}{0.05 \text{ m}}\right]^4}{(0.02 \text{ m})^3\left[(0.05 \text{ m})^{-7/5} + (0.09 \text{ m})^{-7/5}\right]^5} = 0.1303$$

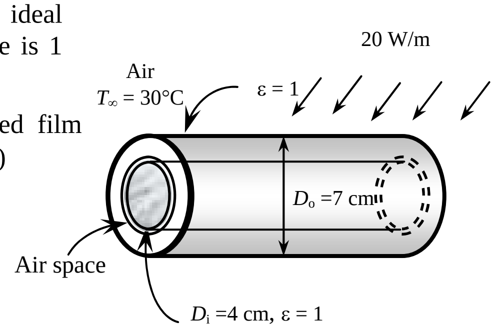
$$k_{\text{eff}} = 0.386k\left(\frac{\text{Pr}}{0.861+\text{Pr}}\right)^{1/4}(F_{\text{cyl}}\text{Ra})^{1/4}$$
$$= 0.386(0.02706 \text{ W/m.°C})\left(\frac{0.7239}{0.861+0.7239}\right)^{1/4}\left[(0.1303)(16{,}106)\right]^{1/4} = 0.05812 \text{ W/m.°C}$$

Then the heat loss from the collector per meter length of the tube becomes

$$\dot{Q} = \frac{2\pi k_{\text{eff}}}{\ln\left(\frac{D_o}{D_i}\right)}(T_i - T_o) = \frac{2\pi(0.05812 \text{ W/m.°C})}{\ln\left(\frac{0.09 \text{ m}}{0.05 \text{ m}}\right)}(60-32)\text{°C} = \mathbf{17.4\ W}$$

9-102 A solar collector consists of a horizontal tube enclosed in a concentric thin glass tube is considered. The pump circulating the water fails. The temperature of the aluminum tube when equilibrium is established is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.



Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2$ = (33+30)/2 = 31.5°C are (Table A-15)

$$k = 0.02599 \text{ W/m.°C}$$
$$\upsilon = 1.622\times10^{-5} \text{ m}^2\text{/s}$$
$$\Pr = 0.7278$$
$$\beta = \frac{1}{T_f} = \frac{1}{(31.5+273)\text{K}} = 0.003284 \text{ K}^{-1}$$

Analysis This problem involves heat transfer from the aluminum tube to the glass cover, and from the outer surface of the glass cover to the surrounding ambient air. When steady operation is reached, these two heat transfers will be equal to the rate of heat gain. That is,

$$\dot{Q}_{tube-glass} = \dot{Q}_{glass-ambient} = \dot{Q}_{solar\ gain} = 20 \text{ W} \quad \text{(per meter length)}$$

Now we assume the surface temperature of the glass cover to be 33°C. We will check this assumption later on, and repeat calculations with a better assumption, if necessary.

The characteristic length for the outer diameter of the glass cover $L_c = D_o$ =0.07 m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)D_o^3}{\upsilon^2}\Pr = \frac{(9.81 \text{ m/s}^2)(0.003284 \text{ K}^{-1})(33-30 \text{ K})(0.07 \text{ m})^3}{(1.622\times10^{-5} \text{ m}^2\text{/s})^2}(0.7278) = 91{,}679$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/\Pr\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(91{,}679)^{1/6}}{\left[1+\left(0.559/0.7278\right)^{9/16}\right]^{8/27}}\right\}^2 = 7.626$$

$$A_s = \pi D_o L = \pi(0.07 \text{ m})(1 \text{ m}) = 0.2199 \text{ m}^2$$

$$h = \frac{k}{D_o}Nu = \frac{0.02599 \text{ W/m.°C}}{0.07 \text{ m}}(7.626) = 2.832 \text{ W/m}^2\text{.°C}$$

and,

$$\dot{Q}_{conv} = hA_s(T_s - T_\infty) = (2.832 \text{ W/m}^2\text{.°C})(0.2199 \text{ m}^2)(T_{glass} - 30)\text{°C}$$

The radiation heat loss is

$$\dot{Q}_{rad} = \varepsilon A_s\sigma(T_s^{\,4} - T_{surr}^{\,4}) = (1)(0.2199 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2\text{.K}^4)\left[(T_{glass}+273 \text{ K})^4 - (20+273 \text{ K})^4\right]$$

The expression for the total rate of heat transfer is

$$\dot{Q}_{total} = \dot{Q}_{conv} + \dot{Q}_{rad}$$
$$20 \text{ W} = (2.832 \text{ W/m}^2\text{.°C})(0.2199 \text{ m}^2)(T_{glass} - 30)\text{°C}$$
$$+ (1)(0.2199 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2\text{.K}^4)\left[(T_{glass}+273 \text{ K})^4 - (20+273 \text{ K})^4\right]$$

Its solution is $T_{glass} = 33.34$°C , which is sufficiently close to the assumed value of 33°C.

Now we will calculate heat transfer through the air layer between aluminum tube and glass cover. We will assume the aluminum tube temperature to be 45°C and evaluate properties at the average temperature of

$(T_i+T_o)/2 = (45+33.34)/2 = 39.17$°C are (Table A-15)

$$k = 0.02656 \text{ W/m.°C}$$
$$\upsilon = 1.694\times10^{-5} \text{ m}^2/\text{s}$$
$$\text{Pr} = 0.7257$$
$$\beta = \frac{1}{T_f} = \frac{1}{(39.17+273)\text{K}} = 0.003203\,\text{K}^{-1}$$

The characteristic length in this case is the distance between the two cylinders,

$$L_c = (D_o - D_i)/2 = (7-4)/2 \text{ cm} = 1.5 \text{ cm}$$

Then,

$$Ra = \frac{g\beta(T_1-T_2)L_c^3}{\upsilon^2}\text{Pr} = \frac{(9.81\,\text{m/s}^2)(0.003203\,\text{K}^{-1})(45-33.34\,\text{K})(0.015\,\text{m})^3}{(1.694\times10^{-5}\,\text{m}^2/\text{s})^2}(0.7257) = 3125$$

The effective thermal conductivity is

$$F_{\text{cyl}} = \frac{\left[\ln\dfrac{D_o}{D_i}\right]^4}{L_c^3(D_i^{-3/5}+D_o^{-3/5})^5} = \frac{\left[\ln\dfrac{0.07\text{ m}}{0.04\text{ m}}\right]^4}{(0.015\text{ m})^3\left[(0.04\text{ m})^{-7/5}+(0.07\text{ m})^{-7/5}\right]^5} = 0.1254$$

$$k_{\text{eff}} = 0.386k\left(\frac{\text{Pr}}{0.861+\text{Pr}}\right)^{1/4}(F_{\text{cyl}}\text{Ra})^{1/4}$$
$$= 0.386(0.02656\text{ W/m.°C})\left(\frac{0.7257}{0.861+0.7257}\right)^{1/4}\left[(0.1254)(3125)\right]^{1/4} = 0.03751\text{ W/m.°C}$$

The heat transfer expression is

$$\dot{Q} = \frac{2\pi k_{\text{eff}}}{\ln\left(\dfrac{D_o}{D_i}\right)}(T_1-T_2) = \frac{2\pi(0.03751\text{ W/m.°C})}{\ln\left(\dfrac{0.07\text{ m}}{0.04\text{ m}}\right)}(T_{tube}-33.34)\text{°C}$$

The radiation heat loss is

$$\dot{Q}_{rad} = \varepsilon A_s\sigma(T_s{}^4 - T_{surr}{}^4) = (1)(0.1684\text{ m}^2)(5.67\times10^{-8}\text{ W/m}^2.\text{K}^4)\left[(T_{tube}+273\text{ K})^4-(33.34+273\text{ K})^4\right]$$

The expression for the total rate of heat transfer is

$$\dot{Q}_{total} = \dot{Q}_{conv} + \dot{Q}_{rad}$$
$$20\text{ W} = \frac{2\pi(0.03751\text{ W/m.°C})}{\ln\left(\dfrac{0.07\text{ m}}{0.04\text{ m}}\right)}(T_{tube}-33.34)\text{°C}$$
$$+(1)(0.1684\text{ m}^2)(5.67\times10^{-8}\text{ W/m}^2.\text{K}^4)\left[(T_{tube}+273\text{ K})^4-(33.34+273\text{ K})^4\right]$$

Its solution is $T_{tube} = \mathbf{45.9°\,C}$,

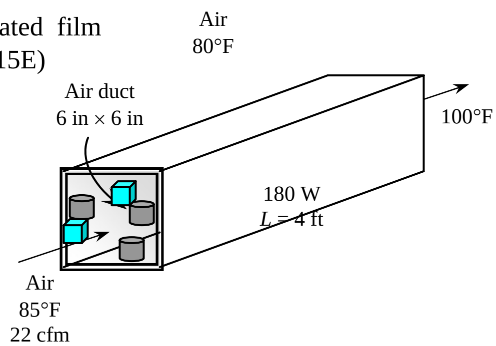
which is sufficiently close to the assumed value of 45°C. Therefore, there is no need to repeat the calculations.

9-103E The components of an electronic device located in a horizontal duct of rectangular cross section is cooled by forced air. The heat transfer from the outer surfaces of the duct by natural convection and the average temperature of the duct are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** Radiation effects are negligible. **5** The thermal resistance of the duct is negligible.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2$ = (120+80)/2 = 100°F are (Table A-15E)

$$k = 0.01529 \text{ Btu/h.ft.°F}$$
$$\upsilon = 0.1808\times10^{-3} \text{ ft}^2\text{/s}$$
$$\text{Pr} = 0.726$$
$$\beta = 1/T_f = 1/(100+460)\text{R} = 0.001786 \text{ R}^{-1}$$



Analysis (*a*) Using air properties at the average temperature of (85+100))/2 = 92.5°F and 1 atm for the forced air, the mass flow rate of air and the heat transfer rate by forced convection are determined to be

$$\dot{m} = \rho\dot{V} = (0.07186 \text{ lbm/ft}^3)(22 \text{ ft}^3\text{/min}) = 1.581 \text{ lbm/min}$$

$$\dot{Q}_{forced} = \dot{m}C_p(T_{out} - T_{in}) = (1.581\times 60 \text{ lbm/h})(0.2405 \text{ Btu/lbm.°F})(100-85)\text{°F} = 342.1 \text{ Btu/h}$$

Noting that radiation heat transfer is negligible, the rest of the 180 W heat generated must be dissipated by natural convection,

$$\dot{Q}_{natural} = \dot{Q}_{total} - \dot{Q}_{forced} = (180\times3.412) - 342.1 = \mathbf{272\ Btu/h}$$

(*b*) We start the calculations by "guessing" the surface temperature to be 120°F for the evaluation of the properties and *h*. We will check the accuracy of this guess later and repeat the calculations if necessary.

Horizontal top surface: The characteristic length is $L_c = \dfrac{A_s}{P} = \dfrac{(4 \text{ ft})(6/12 \text{ ft})}{2(4 \text{ ft} + 6/12 \text{ ft})} = 0.2222 \text{ ft}$. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\upsilon^2}\text{Pr} = \frac{(32.2 \text{ ft/s}^2)(0.001786 \text{ R}^{-1})(120-80 \text{ R})(0.2222 \text{ ft})^3}{(0.1808\times10^{-3} \text{ ft}^2\text{/s})^2}(0.726) = 5.604\times10^5$$

$$Nu = 0.54Ra^{1/4} = 0.54(5.604\times10^5)^{1/4} = 14.77$$

$$h_{top} = \frac{k}{L_c}Nu = \frac{0.01529 \text{ Btu/h.ft.°F}}{0.2222 \text{ ft}}(14.77) = 1.016 \text{ Btu/h.ft}^2\text{.°F}$$

$$A_{top} = (4 \text{ ft})(6/12 \text{ ft}) = 2 \text{ ft}^2 = A_{bottom}$$

Horizontal bottom surface: The Nusselt number for this geometry and orientation can be determined from

$$Nu = 0.27Ra^{1/4} = 0.27(5.604\times10^5)^{1/4} = 7.387$$

$$h_{bottom} = \frac{k}{L_c}Nu = \frac{0.01529 \text{ Btu/h.ft.°F}}{0.2222 \text{ ft}}(7.387) = 0.5082 \text{ Btu/h.ft}^2\text{.°F}$$

Vertical side surfaces: The characteristic length in this case is the height of the duct, $L_c = L$ = 6 in. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\text{Pr} = \frac{(32.2 \text{ ft/s}^2)(0.001786 \text{ R}^{-1})(120-80 \text{ R})(0.5 \text{ ft})^3}{(0.1808\times10^{-3} \text{ ft}^2\text{/s})^2}(0.726) = 6.383\times10^6$$

$$Nu = \left\{ 0.825 + \frac{0.387 \text{Ra}^{1/6}}{\left[1 + \left(\frac{0.492}{\Pr} \right)^{9/16} \right]^{8/27}} \right\}^2 = \left\{ 0.825 + \frac{0.387(6.383 \times 10^6)^{1/6}}{\left[1 + \left(\frac{0.492}{0.726} \right)^{9/16} \right]^{8/27}} \right\}^2 = 27.57$$

$$h_{side} = \frac{k}{L} Nu = \frac{0.01529 \text{ Btu/h.ft.°F}}{0.5 \text{ ft}} (27.57) = 0.843 \text{ Btu/h.ft}^2\text{.°F}$$

$$A_{side} = 2(4 \text{ ft})(0.5 \text{ ft}) = 4 \text{ ft}^2$$

Then the total heat loss from the duct can be expressed as

$$\dot{Q}_{total} = \dot{Q}_{top} + \dot{Q}_{bottom} + \dot{Q}_{side} = [(hA)_{top} + (hA)_{bottom} + (hA)_{side}](T_s - T_\infty)$$

Substituting and solving for the surface temperature,

$$272 \text{ Btu/h} = [(1.016 \times 2 + 0.5082 \times 2 + 0.843 \times 4) \text{ Btu/h.°F}](T_s - 80)\text{°F}$$

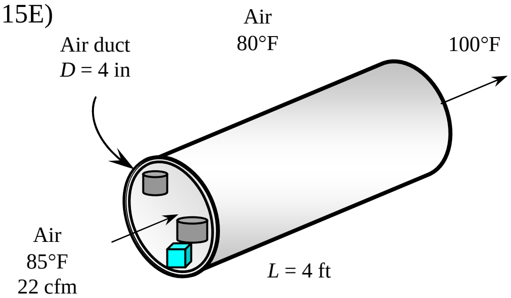
T_s = **122.4°F**

which is sufficiently close to the assumed value of 120°F used in the evaluation of properties and *h*. Therefore, there is no need to repeat the calculations.

9-104E The components of an electronic system located in a horizontal duct of circular cross section is cooled by forced air. The heat transfer from the outer surfaces of the duct by natural convection and the average temperature of the duct are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** Radiation effects are negligible. **5** The thermal resistance of the duct is negligible.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2$ = (150+80)/2 = 115°F are (Table A-15E)



$$k = 0.01564 \text{ Btu/h.ft.°F}$$
$$\upsilon = 0.1894\times10^{-3} \text{ ft}^2\text{/s}$$
$$\text{Pr} = 0.7268$$
$$\beta = \frac{1}{T_f} = \frac{1}{(115+460)\text{ R}} = 0.001739 \text{ R}^{-1}$$

Analysis (a) Using air properties at the average temperature of (85+100))/2 = 92.5°F and 1 atm for the forced air, the mass flow rate of air and the heat transfer rate by forced convection are determined to be

$$\dot{m} = \rho\dot{V} = (0.07186 \text{ lbm/ft}^3)(22 \text{ ft}^3\text{/min}) = 1.581 \text{ lbm/min}$$

$$\dot{Q}_{forced} = \dot{m}C_p(T_{out} - T_{in}) = (1.581\times60 \text{ lbm/h})(0.2405 \text{ Btu/lbm.°F})(100-85)\text{°F} = 342.1 \text{ Btu/h}$$

Noting that radiation heat transfer is negligible, the rest of the 180 W heat generated must be dissipated by natural convection,

$$\dot{Q}_{natural} = \dot{Q}_{total} - \dot{Q}_{forced} = (180\times3.412) - 342.1 = \mathbf{272 \text{ Btu/h}}$$

(b) We start the calculations by "guessing" the surface temperature to be 150°F for the evaluation of the properties and h. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the outer diameter of the duct, $L_c = D$ = 4 in. Then,

$$Ra = \frac{g\beta(T_1 - T_2)D^3}{\upsilon^2}\text{Pr} = \frac{(32.2 \text{ ft/s}^2)(0.001739 \text{ R}^{-1})(150-80 \text{ R})(4/12 \text{ ft})^3}{(0.1894\times10^{-3} \text{ ft}^2\text{/s})^2}(0.7268) = 2.930\times10^6$$

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/\text{Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(2.930\times10^6)^{1/6}}{\left[1+\left(0.559/0.7268\right)^{9/16}\right]^{8/27}}\right\}^2 = 19.79$$

$$h = \frac{k}{D}Nu = \frac{0.01564 \text{ Btu/h.ft.°F}}{4/12 \text{ ft}}(19.79) = 0.9287 \text{ Btu/h.ft}^2\text{.°F}$$
$$A_s = \pi DL = \pi(4/12 \text{ ft})(4 \text{ ft}) = 4.19 \text{ ft}^2$$

Then the surface temperature is determined to be

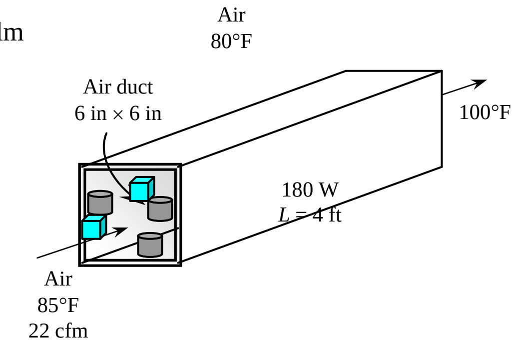
$$\dot{Q} = hA_s(T_s - T_\infty) \;\rightarrow\; T_s = T_\infty + \frac{\dot{Q}}{hA_s} = 80\text{°F} + \frac{272 \text{ Btu/h}}{(0.9287 \text{ Btu/h.ft}^2\text{.°F})(4.19 \text{ ft}^2)} = \mathbf{149.9°F}$$

which is practically equal to the assumed value of 150°F used in the evaluation of properties and h. Therefore, there is no need to repeat the calculations.

9-105E The components of an electronic system located in a horizontal duct of rectangular cross section is cooled by natural convection. The heat transfer from the outer surfaces of the duct by natural convection and the average temperature of the duct are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** Radiation effects are negligible. **5** The thermal resistance of the duct is negligible.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2$ = (160+80)/2 = 120°F are (Table A-15E)



$$k = 0.01576 \text{ Btu/h.ft.°F}$$
$$\upsilon = 0.1923\times10^{-3} \text{ ft}^2/\text{s}$$
$$\text{Pr} = 0.723$$
$$\beta = 1/T_f = 1/(120+460 \text{ R}) = 0.001724 \text{ R}^{-1}$$

Analysis (a) Noting that radiation heat transfer is negligible and no heat is removed by forced convection because of the failure of the fan, the entire 180 W heat generated must be dissipated by natural convection,

$$\dot{Q}_{natural} = \dot{Q}_{total} = \mathbf{180\ W}$$

(b) We start the calculations by "guessing" the surface temperature to be 160°F for the evaluation of the properties and h. We will check the accuracy of this guess later and repeat the calculations if necessary.

Horizontal top surface: The characteristic length is $L_c = \dfrac{A_s}{p} = \dfrac{(4\text{ ft})(6/12\text{ ft})}{2(4\text{ ft}+6/12\text{ ft})} = 0.2222\text{ ft}$. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L_c^3}{\upsilon^2}\text{Pr} = \frac{(32.2\text{ ft/s}^2)(0.001724\text{ R}^{-1})(160-80\text{ R})(0.2222\text{ ft})^3}{(0.1923\times10^{-3}\text{ ft}^2/\text{s})^2}(0.723) = 9.534\times10^5$$

$$Nu = 0.54Ra^{1/4} = 0.54(9.534\times10^5)^{1/4} = 16.87$$

$$h_{top} = \frac{k}{L_c}Nu = \frac{0.01576\text{ Btu/h.ft.°F}}{0.2222\text{ ft}}(16.87) = 1.197\text{ Btu/h.ft}^2\text{.°F}$$

$$A_{top} = (4\text{ ft})(6/12\text{ ft}) = 2\text{ ft}^2 = A_{bottom}$$

Horizontal bottom surface: The Nusselt number for this geometry and orientation can be determined from

$$Nu = 0.27Ra^{1/4} = 0.27(9.534\times10^5)^{1/4} = 8.437$$

$$h_{bottom} = \frac{k}{L_c}Nu = \frac{0.01576\text{ Btu/h.ft.°F}}{0.2222\text{ ft}}(8.437) = 0.5983\text{ Btu/h.ft}^2\text{.°F}$$

Vertical side surfaces: The characteristic length in this case is the height of the duct, $L_c = L = 6$ in. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\text{Pr} = \frac{(32.2\text{ ft/s}^2)(0.001724\text{ R}^{-1})(160-80\text{ R})(0.5\text{ ft})^3}{(0.1923\times10^{-3}\text{ ft}^2/\text{s})^2}(0.723) = 1.086\times10^7$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(1.086\times10^7)^{1/6}}{\left[1+\left(\frac{0.492}{0.723}\right)^{9/16}\right]^{8/27}}\right\}^2 = 32.03$$

$$h_{side} = \frac{k}{L}Nu = \frac{0.01576\text{ Btu/h.ft.°F}}{0.5\text{ ft}}(32.03) = 1.009\text{ Btu/h.ft}^2\text{.°F}$$

$$A_{side} = 2(4\text{ ft})(0.5\text{ ft}) = 4\text{ ft}^2$$

Then the total heat loss from the duct can be expressed as

$$\dot{Q}_{total} = \dot{Q}_{top} + \dot{Q}_{bottom} + \dot{Q}_{side} = [(hA)_{top} + (hA)_{bottom} + (hA)_{side}](T_s - T_\infty)$$

Substituting and solving for the surface temperature,

$$180\ \text{W}\left(\frac{3.41214\ \text{Btu/h}}{1\ \text{W}}\right) = [(1.197\times 2 + 0.5983\times 2 + 1.009\times 4)\ \text{Btu/h.°F}](T_s - 80)\text{°F}$$

T_s = **160.5°F**

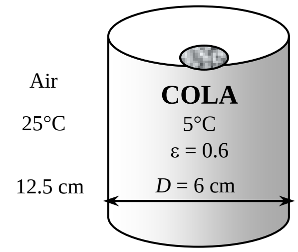
which is sufficiently close to the assumed value of 160°F used in the evaluation of properties and *h*. Therefore, there is no need to repeat the calculations.

9-106 A cold aluminum canned drink is exposed to ambient air. The time it will take for the average temperature to rise to a specified value is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** Any heat transfer from the bottom surface of the can is disregarded. **5** The thermal resistance of the can is negligible.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2$ = (6+25)/2 = 15.5°C are (Table A-15)

$$k = 0.0248 \text{ W/m.°C}$$
$$\upsilon = 1.475\times10^{-5} \text{ m}^2\text{/s}$$
$$\Pr = 0.7321$$
$$\beta = 1/T_f = 1/(15.5+273\text{ K}) = 0.003466 \text{ K}^{-1}$$



Analysis We assume the surface temperature of aluminum can to be equal to the temperature of the drink in the can since the can is made of a very thin layer of aluminum. Noting that the temperature of the drink rises from 5°C to 7°C, we take the average surface temperature to be 6°C. The characteristic length in this case is the height of the box $L_c = L$ = 0.125 m. Then,

$$Ra = \frac{g\beta(T_\infty - T_s)L^3}{\upsilon^2}\Pr = \frac{(9.81\text{ m/s}^2)(0.003466\text{ K}^{-1})(25-6\text{ K})(0.125\text{ m})^3}{(1.475\times10^{-5}\text{ m}^2\text{/s})^2}(0.7321) = 4.246\times10^6$$

At this point we should check if we can treat this aluminum can as a vertical plate. The criteria is

$$D \ge \frac{35L}{Gr^{1/4}} \longrightarrow \frac{35(12.5\text{ cm})}{(4.246\times10^6 / 0.7321)^{1/4}} = 9.92\text{ cm}$$

which is not smaller than the diameter of the can (6 cm), but close to it. Therefore, we can still use vertical plate relation approximately (besides, we do not have another relation available). Then the Nusselt number becomes from

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(4.246\times10^6)^{1/6}}{\left[1+\left(\frac{0.492}{0.7321}\right)^{9/16}\right]^{8/27}}\right\}^2 = 23.81$$

$$h = \frac{k}{L}Nu = \frac{0.0248\text{ W/m.°C}}{0.125\text{ m}}(23.81) = 4.887\text{ W/m}^2\text{.°C}$$

$$A_s = \pi DL + \frac{\pi D^2}{4} = \pi(0.06\text{ m})(0.125\text{ m}) + \frac{\pi(0.06\text{ m})^2}{4} = 0.02639\text{ m}^2$$

Note that we also include top surface area of the can to the total surface area, and assume the heat transfer coefficient for that area to be the same for simplicity (actually, it will be a little lower). Then heat transfer rate from outer surfaces of the can by natural convection becomes

$$\dot{Q} = hA_s(T_\infty - T_s) = (4.887\text{ W/m}^2\text{.°C})(0.02639\text{ m}^2)(25-6)\text{°C} = 2.45\text{ W}$$

The radiation heat loss is

$$\dot{Q}_{rad} = \varepsilon A_s\sigma(T_{surr}{}^4 - T_s{}^4)$$
$$= (0.6)(0.02639\text{ m}^2)(5.67\times10^{-8}\text{ W/m}^2\text{.K}^4)[(25+273\text{ K})^4 - (6+273\text{ K})^4] = 1.64\text{ W}$$

and $\dot{Q}_{total} = 2.41 + 1.64 = 4.09\text{ W}$

Using the properties of water for the cold drink at 6°C, the amount of heat transfer to the drink is determined from

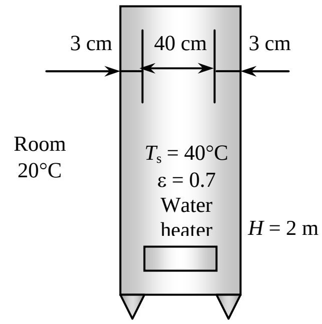
$$m = \rho V = \rho\frac{\pi D^2}{4}L = (1000\text{ kg/m}^3)\frac{\pi(0.06\text{ m})^2}{4}(0.125\text{ m}) = 0.3534\text{ kg}$$
$$Q = mC_p(T_2 - T_1) = (0.353\text{ kg})(4197\text{ J/kg.°C})(7-5)\text{°C} = 2967\text{ J}$$

Then the time required for the temperature of the cold drink to rise to 7°C becomes

$$Q = \dot{Q}\Delta t \longrightarrow \Delta t = \frac{Q}{\dot{Q}} = \frac{2967\ \text{J}}{4.09\ \text{J/s}} = 725\ \text{s} = \mathbf{12.1\ min}$$

9-107 An electric hot water heater is located in a small room. A hot water tank insulation kit is available for $30. The payback period of this insulation to pay for itself from the energy it saves is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** Any heat transfer from the top and bottom surfaces of the tank is disregarded. **5** The thermal resistance of the metal sheet is negligible.



Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (40+20)/2 = 30$°C are (Table A-15)

$$k = 0.02588 \text{ W/m.°C}$$
$$\upsilon = 1.608\times10^{-5} \text{ m}^2\text{/s}$$
$$\text{Pr} = 0.7282$$
$$\beta = \frac{1}{T_f} = \frac{1}{(30+273)\text{K}} = 0.0033 \text{ K}^{-1}$$

Analysis The characteristic length in this case is the height of the heater, $L_c = L = 2$ m. Then,

$$Ra = \frac{g\beta(T_\infty - T_s)L^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.0033 \text{ K}^{-1})(40-20 \text{ K})(2 \text{ m})^3}{(1.608\times10^{-5} \text{ m}^2\text{/s})^2}(0.7282) = 1.459\times10^{10}$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(1.459\times10^{10})^{1/6}}{\left[1+\left(\frac{0.492}{0.7282}\right)^{9/16}\right]^{8/27}}\right\}^2 = 285.4$$

$$h = \frac{k}{L}Nu = \frac{0.02588 \text{ W/m.°C}}{2 \text{ m}}(285.4) = 3.693 \text{ W/m}^2\text{.°C}$$
$$A_s = \pi DL = \pi(0.46 \text{ m})(2 \text{ m}) = 2.89 \text{ m}^2$$

and $\dot{Q} = hA_s(T_\infty - T_s) = (3.693 \text{ W/m}^2\text{.°C})(2.89 \text{ m}^2)(40-20)\text{°C} = 213.5 \text{ W}$

The radiation heat loss is

$$\dot{Q}_{rad} = \varepsilon A_s\sigma(T_{surr}{}^4 - T_s{}^4) = (0.7)(2.89 \text{ m}^2)(5.67\times10^{-8} \text{ W/m}^2\text{.K}^4)[(40+273 \text{ K})^4 - (20+273 \text{ K})^4] = 255.6 \text{ W}$$

and $\dot{Q}_{total} = 213.5 + 255.6 = 469 \text{ W}$

The reduction in heat loss after adding insulation is

$$\dot{Q} = (0.80)(469 \text{ W}) = 375.2 \text{ W}$$

The amount of heat and money saved per hour is

$$Q_{saved} = \dot{Q}_{saved}\Delta t = (0.3752 \text{ kW})(1 \text{ h}) = 0.3752 \text{ kWh}$$

$$\text{Money saved} = (0.3752 \text{ kWh})(\$0.08/\text{kWh}) = \$0.03002$$

Then it will take

$$\Delta t = \frac{\$30}{\$0.03002} = 999.4 \text{ h} = \mathbf{41.64\ days}$$

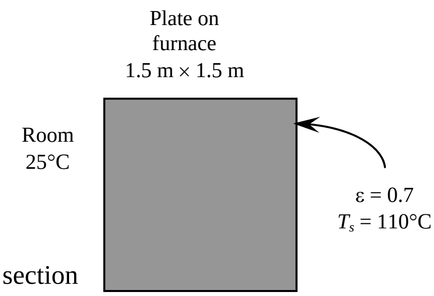
for the additional insulation to pay for itself from the energy it saves.

9-108 A hot part of the vertical front section of a natural gas furnace in a plant is considered. The rate of heat loss from this section and the annual cost of this heat loss are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** Any heat transfer from other surfaces of the tank is disregarded.

Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (110+25)/2 = 67.5°C$ are (Table A-15)

$$k = 0.02863\ \text{W/m.°C}$$
$$\upsilon = 1.97\times10^{-5}\ \text{m}^2/\text{s}$$
$$\text{Pr} = 0.7184$$
$$\beta = \frac{1}{T_f} = \frac{1}{(67.5+273)\text{K}} = 0.002937\ \text{K}^{-1}$$



Analysis The characteristic length in this case is the height of that section of furnace, $L_c = L = 15$ m. Then,

$$Ra = \frac{g\beta(T_s - T_\infty)L^3}{\upsilon^2}\text{Pr} = \frac{(9.81\ \text{m/s}^2)(0.002937\ \text{K}^{-1})(110-25\ \text{K})(1.5\ \text{m})^3}{(1.97\times10^{-5}\ \text{m}^2/\text{s})^2}(0.7184) = 1.530\times10^{10}$$

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\text{Pr}}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(1.530\times10^{10})^{1/6}}{\left[1+\left(\frac{0.492}{0.7184}\right)^{9/16}\right]^{8/27}}\right\}^2 = 289.1$$

$$h = \frac{k}{L}Nu = \frac{0.02863\ \text{W/m.°C}}{1.5\ \text{m}}(289.1) = 5.518\ \text{W/m}^2.°\text{C}$$
$$A_s = (1\ \text{m})(1.5\ \text{m}) = 1.5\ \text{m}^2$$

and

$$\dot{Q} = hA_s(T_s - T_\infty) = (5.518\ \text{W/m}^2.°\text{C})(1.5\ \text{m}^2)(110-25)°\text{C} = 703.5\ \text{W}$$

The radiation heat loss is

$$\begin{aligned}\dot{Q}_{rad} &= \varepsilon A_s\sigma(T_{surr}{}^4 - T_s{}^4) \\ &= (0.7)(1.5\ \text{m}^2)(5.67\times10^{-8}\ \text{W/m}^2.\text{K}^4)[(110+273\ \text{K})^4 - (25+273\ \text{K})^4] = 812\ \text{W}\end{aligned}$$

$$\dot{Q}_{total} = 703.5 + 812 = \mathbf{1515\ W}$$

The amount and cost of natural gas used to overcome this heat loss per year is

$$Q_{gas} = \dot{Q}_{gas}\Delta t = \frac{\dot{Q}_{total}}{0.79}\Delta t = \frac{(1.5151\ \text{kJ/s})}{0.79}(310\ \text{days/yr}\times10\ \text{hr/day}\times3600\ \text{s/hr}) = 2.14\times10^7\ \text{kJ}$$

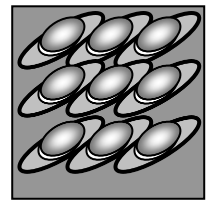
$$\text{Cost} = (2.14\times10^7 / 105{,}500\ \text{therm})(\$0.75/\text{therm}) = \mathbf{\$152.2}$$

9-109 A group of 25 transistors are cooled by attaching them to a square aluminum plate and mounting the plate on the wall of a room. The required size of the plate to limit the surface temperature to 50°C is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** Any heat transfer from the back side of the plate is negligible.

Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (50+30)/2 = 40$°C are (Table A-15)

$$k = 0.02662 \text{ W/m.°C}$$
$$\upsilon = 1.702\times10^{-5} \text{ m}^2\text{/s}$$
$$\Pr = 0.7255$$
$$\beta = \frac{1}{T_f} = \frac{1}{(40+273)\text{K}} = 0.003195 \text{ K}^{-1}$$



Analysis The Rayleigh number can be determined in terms of the characteristic length (length of the plate) to be

$$Ra = \frac{g\beta(T_\infty - T_s)L_c^3}{\upsilon^2}\Pr = \frac{(9.81 \text{ m/s}^2)(0.003195 \text{ K}^{-1})(50-30 \text{ K})(L)^3}{(1.702\times10^{-5} \text{ m}^2\text{/s})^2}(0.7255) = 1.571\times10^9 L^3$$

The Nusselt number relation is

$$Nu = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{\left[1+\left(\frac{0.492}{\Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.825 + \frac{0.387(1.571\times10^9 L^3)^{1/6}}{\left[1+\left(\frac{0.492}{0.7255}\right)^{9/16}\right]^{8/27}}\right\}^2$$

The heat transfer coefficient is

$$h = \frac{k}{L}Nu = \frac{0.02662 \text{ W/m.°C}}{L}Nu$$
$$A_s = L^2$$

Noting that both the surface and surrounding temperatures are known, the rate of convection and radiation heat transfer are expressed as

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) = \frac{0.02662 \text{ W/m.°C}}{L}NuL^2(50-30)\text{°C}$$
$$\dot{Q}_{\text{rad}} = \varepsilon A_s\sigma(T_s^4 - T_{sky}^4) = (0.9)L^2(5.67\times10^{-8} \text{ W/m}^2.\text{K}^4)[(50+273)^4 - (30+273)^4]\text{K}^4 = 125.3L^2$$

The rate of total heat transfer is expressed as

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}}$$
$$25\times(1.5 \text{ W}) = \frac{0.02662 \text{ W/m.°C}}{L}NuL^2(50-30)\text{°C} + 125.3L^2$$

Substituting Nusselt number expression above into this equation and solving for L, the length of the plate is determined to be

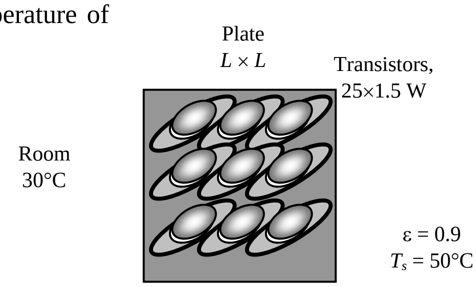
L = **0.426 m**

9-110 A group of 25 transistors are cooled by attaching them to a square aluminum plate and positioning the plate horizontally in a room. The required size of the plate to limit the surface temperature to 50°C is to be determined for two cases.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** Any heat transfer from the back side of the plate is negligible.

Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (50+30)/2 = 40$°C are (Table A-15)

$$k = 0.02662 \text{ W/m.°C}$$
$$\upsilon = 1.702\times10^{-5} \text{ m}^2\text{/s}$$
$$\text{Pr} = 0.7255$$
$$\beta = \frac{1}{T_f} = \frac{1}{(40+273)\text{K}} = 0.003195 \text{ K}^{-1}$$



Analysis The characteristic length and the Rayleigh number for the horizontal case are determined to be

$$L_c = \frac{A_s}{p} = \frac{L^2}{4L} = \frac{L}{4}$$

$$Ra = \frac{g\beta(T_\infty - T_s)L_c^3}{\upsilon^2}\text{Pr} = \frac{(9.81 \text{ m/s}^2)(0.003195 \text{ K}^{-1})(50-30 \text{ K})(L/4)^3}{(1.702\times10^{-5} \text{ m}^2\text{/s})^2}(0.7255) = 2.454\times10^7 L^3$$

Noting that both the surface and surrounding temperatures are known, the rate of radiation heat transfer is determined to be

$$\dot{Q}_{\text{rad}} = \varepsilon A_s\sigma(T_s{}^4 - T_{sky}{}^4) = (0.9)L^2(5.67\times10^{-8} \text{ W/m}^2.\text{K}^4)[(50+273)^4 - (30+273)^4]\text{K}^4 = 125.3L^2$$

(*a*) **Hot surface facing up:** We assume Ra < 10^7 and thus L <0.74 m so that we can determine the Nu number from Eq. 9-22. Then the Nusselt number and the convection heat transfer coefficient become

$$Nu = 0.54Ra^{1/4} = 0.54(2.454\times10^7 L^3)^{1/4} = 38.0L^{3/4}$$

Then,

$$h = \frac{k}{L}Nu = \frac{0.02662 \text{ W/m.°C}}{L/4}(38.0L^{3/4}) = 4.047L^{-1/4} \text{ W/m}^2.\text{°C}$$
$$A_s = L^2$$

The rate of convection heat transfer is

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) = (4.047L^{-1/4})L^2(50-30) = 80.94L^{7/8} \text{ W}$$

Then,

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}}$$
$$25\times(1.5 \text{ W}) = 80.94L^{7/4} + 125.3L^2 \text{ W}$$

Solving for L, the length of the plate is determined to be

L = **0.407 m**

Note that L < 0.75 m, and therefore the assumption of Ra < 10^7 is verified. That is,

(*b*) **Hot surface facing down:** The Nusselt number in this case is determined from

$$Nu = 0.27Ra^{1/4} = 0.27(2.454\times10^7 L^3)^{1/4} = 19.0L^{3/4}$$

Then,

$$h = \frac{k}{L_c}Nu = \frac{0.02662 \text{ W/m.°C}}{L/4}(19.0L^{3/4}) = 2.023L^{-1/4}$$

The rate of convection heat transfer is

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) = (2.023L^{-1/4})L^2(50-30) = 40.47L^{7/8} \quad \text{W}$$

Then,

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}}$$

$$25\times(1.5\text{ W}) = 40.47L^{7/8} + 125.3L^2 \text{ W}$$

Solving for *L*, the length of the plate is determined to be

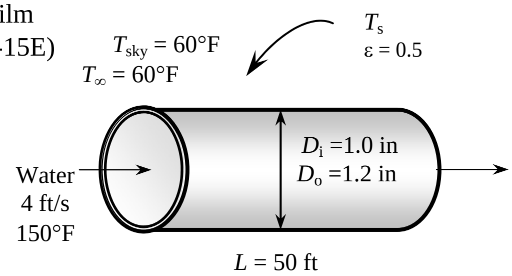
L = **0.464 m**

9-111E A hot water pipe passes through a basement. The temperature drop of water in the basement due to heat loss from the pipe is to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the anticipated film temperature of $(T_s+T_\infty)/2 = (150+60)/2 = 105°F$ are (Table A-15E)

$$k = 0.01541 \text{ Btu/h.ft.°F}$$
$$\upsilon = 0.1837\times10^{-3} \text{ ft}^2\text{/s}$$
$$\text{Pr} = 0.7253$$
$$\beta = \frac{1}{T_f} = \frac{1}{(105+460)\text{R}} = 0.00177 \text{ R}^{-1}$$



Analysis We expect the pipe temperature to be very close to the water temperature, and start the calculations by "guessing" the average outer surface temperature of the pipe to be 150°F for the evaluation of the properties and h. We will check the accuracy of this guess later and repeat the calculations if necessary. The characteristic length in this case is the outer diameter of the pipe, $L_c = D_o = 1.2$ in. Then,

$$Ra = \frac{g\beta(T_\infty - T_s)D_o^3}{\upsilon^2}\text{Pr} = \frac{(32.2 \text{ ft/s}^2)(0.00177 \text{ R}^{-1})(150-60 \text{ R})(1.2/12 \text{ ft})^3}{(0.1837\times10^{-3} \text{ ft}^2\text{/s})^2}(0.7253) = 1.102\times10^5$$

The natural convection Nusselt number can be determined from

$$Nu = \left\{0.6 + \frac{0.387Ra^{1/6}}{\left[1+\left(0.559/\text{Pr}\right)^{9/16}\right]^{8/27}}\right\}^2 = \left\{0.6 + \frac{0.387(1.1023\times10^5)^{1/6}}{\left[1+\left(0.559/0.7253\right)^{9/16}\right]^{8/27}}\right\}^2 = 7.999$$

$$h_o = \frac{k}{D_o}Nu = \frac{0.01541 \text{ W/m.°C}}{(1.2/12) \text{ ft}}(7.999) = 1.232 \text{ Btu/h.ft}^2\text{.°F}$$
$$A_i = \pi D_i L = \pi(1/12 \text{ ft})(50 \text{ ft}) = 13.09 \text{ ft}^2$$
$$A_o = \pi D_o L = \pi(1.2/12 \text{ ft})(50 \text{ ft}) = 15.708 \text{ ft}^2$$

Using the assumed value of glass temperature, the radiation heat transfer coefficient is determined to be

$$\begin{aligned} h_{rad} &= \varepsilon\sigma(T_s + T_{surr})(T_s^{\,2} + T_{surr}^{\,2}) \\ &= (0.5)(0.1714\times10^{-8} \text{ Btu/h.ft}^2\text{.R}^4)[(150+460)+(60+460)][(150+460)^2+(60+460)^2]R^3 \\ &= 0.6222 \text{ Btu/ft}^2\text{.R} \end{aligned}$$

Then the combined convection and radiation heat transfer coefficient outside becomes

$$h_{o,combined} = h_o + h_{rad} = 1.232 + 0.6222 = 1.854 \text{ Btu/ft}^2\text{.R}$$

and

$$\dot{Q} = \frac{T_{water} - T_\infty}{\dfrac{1}{h_i A_i} + \dfrac{\ln(D_o/D_i)}{4\pi kL} + \dfrac{1}{h_o A_o}} = \frac{150-60}{\dfrac{1}{(30)(13.09)} + \dfrac{\ln(1.2/1)}{4\pi(30)(50)} + \dfrac{1}{(1.854)(15.708)}} = 2440 \text{ Btu/h}$$

The mass flow rate of water

$$\dot{m} = \rho A_c V = (62.2 \text{ lbm/ft}^3)\left[\pi(1/12 \text{ ft})^2/4\right](4 \text{ ft/s}) = 1.357 \text{ lbm/s} = 4885 \text{ lbm/h}$$

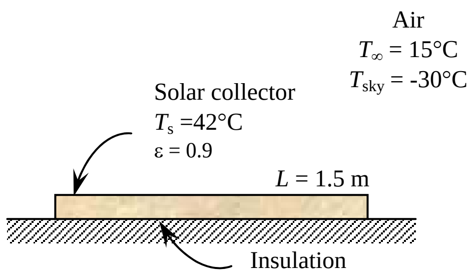
Then the temperature drop of water as it flows through the pipe becomes

$$\dot{Q} = \dot{m}C_p\Delta T \;\rightarrow\; \Delta T = \frac{\dot{Q}}{\dot{m}C_p} = \frac{2440 \text{ Btu/h}}{(4885 \text{ lbm/h})(1.0 \text{ Btu/lbm.°F})} = \mathbf{0.50°F}$$

9-112 A flat-plate solar collector placed horizontally on the flat roof of a house is exposed to the calm ambient air. The rate of heat loss from the collector by natural convection and radiation are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm.

Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2$ = (42+15)/2 = 28.5°C are (Table A-15)



$$k = 0.02577 \text{ W/m.°C}$$
$$\upsilon = 1.594\times10^{-5} \text{ m}^2\text{/s}$$
$$\Pr = 0.7286$$
$$\beta = \frac{1}{T_f} = \frac{1}{(28.5+273)\text{K}} = 0.003317 \text{ K}^{-1}$$

Analysis The characteristic length in this case is determined from

$$L_c = \frac{A_s}{p} = \frac{(1.5\text{ m})(6\text{ m})}{2(1.5\text{ m}+6\text{ m})} = 0.6 \text{ m}$$

Then,

$$Ra = \frac{g\beta(T_\infty - T_s)L_c^3}{\upsilon^2}\Pr = \frac{(9.81\text{ m/s}^2)(0.003317\text{ K}^{-1})(42-15\text{ K})(0.6\text{ m})^3}{(1.594\times10^{-5}\text{ m}^2\text{/s})^2}(0.7286) = 5.443\times10^8$$

$$Nu = 0.15Ra^{1/3} = 0.15(5.443\times10^8)^{1/3} = 122.5$$

$$h = \frac{k}{L_c}Nu = \frac{0.02577\text{ W/m.°C}}{0.6\text{ m}}(122.5) = 5.26 \text{ W/m}^2\text{.°C}$$
$$A_s = (1.5\text{ m})(6\text{ m}) = 9\text{ m}^2$$

and

$$\dot{Q}_{conv} = hA_s(T_s - T_\infty) = (5.26\text{ W/m}^2\text{.°C})(9\text{ m}^2)(42-15)\text{°C} = \mathbf{1278\ W}$$

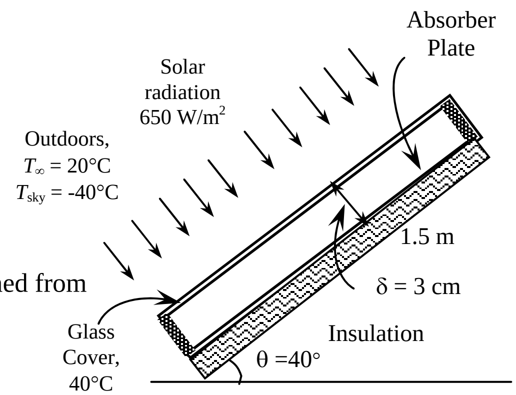
Heat transfer rate by radiation is

$$\begin{aligned}\dot{Q}_{rad} &= \varepsilon A_s\sigma(T_{surr}{}^4 - T_s{}^4)\\ &= (0.9)(9\text{ m}^2)(5.67\times10^{-8}\text{ W/m}^2\text{.K}^4)[(42+273\text{ K})^4 - (-30+273\text{ K})^4] = \mathbf{2920\ W}\end{aligned}$$

9-113 A flat-plate solar collector tilted 40°C from the horizontal is exposed to the calm ambient air. The total rate of heat loss from the collector, the collector efficiency, and the temperature rise of water in the collector are to be determined.

Assumptions **1** Steady operating conditions exist. **2** Air is an ideal gas with constant properties. **3** The local atmospheric pressure is 1 atm. **4** There is no heat loss from the back surface of the absorber plate.

Properties The properties of air at 1 atm and the film temperature of $(T_s+T_\infty)/2 = (40+20)/2 = 30$°C are (Table A-15)



$$k = 0.02588\ \text{W/m.°C}$$
$$\upsilon = 1.608\times10^{-5}\ \text{m}^2/\text{s}$$
$$\text{Pr} = 0.7282$$
$$\beta = \frac{1}{T_f} = \frac{1}{(30+273)\text{K}} = 0.0033\ \text{K}^{-1}$$

Analysis (*a*) The characteristic length in this case is determined from

$$L_c = \frac{A_s}{p} = \frac{(1.5\text{ m})(2\text{ m})}{2(1.5\text{ m}+2\text{ m})} = 0.429\text{ m}^2$$

Then,

$$Ra = \frac{g\beta(T_\infty - T_s)L_c^3}{\upsilon^2}\text{Pr} = \frac{(9.81\text{ m/s}^2)(\cos 40°)(0.003311\text{ K}^{-1})(40-20\text{ K})(0.429\text{ m})^3}{(1.608\times10^{-5}\text{ m}^2/\text{s})^2}(0.7282) = 1.100\times10^8$$

$$Nu = 0.15Ra^{1/3} = 0.15(1.100\times10^8)^{1/3} = 71.87$$
$$h = \frac{k}{L_s}Nu = \frac{0.02588\text{ W/m.°C}}{0.429\text{ m}}(71.87) = 4.340\text{ W/m}^2.\text{°C}$$
$$A_s = (1.5\text{ m})(2\text{ m}) = 3\text{ m}^2$$

and

$$\dot{Q}_{conv} = hA_s(T_s - T_\infty) = (4.340\text{ W/m}^2.\text{°C})(3\text{ m}^2)(40-20)\text{°C} = 260.4\text{ W}$$

Heat transfer rate by radiation is

$$\dot{Q}_{rad} = \varepsilon A_s\sigma(T_{surr}{}^4 - T_s{}^4)$$
$$= (0.9)(3\text{ m}^2)(5.67\times10^{-8}\text{ W/m}^2.\text{K}^4)[(40+273\text{ K})^4 - (-40+273\text{ K})^4] = 1018\text{ W}$$

and

$$\dot{Q}_{total} = 260.4 + 1018 = \mathbf{1279\ W}$$

(*b*) The solar energy incident on the collector is

$$\dot{Q}_{incident} = \alpha\dot{q}A_s = (0.88)(650\text{ W/m}^2)(3\text{ m}^2) = 1716\text{ W}$$

Then the collector efficiency becomes

$$efficiency = \frac{\dot{Q}_{incident} - \dot{Q}_{lost}}{\dot{Q}_{incident}} = \frac{1716-1279}{1716} = 0.255 = \mathbf{25.5\%}$$

(*c*) The temperature rise of the water as it passes through the collector is

$$\dot{Q} = \dot{m}C_p\Delta T \rightarrow \Delta T = \frac{\dot{Q}}{\dot{m}C_p} = \frac{(1716-1279)\text{W}}{(1/60\text{ kg/s})(4180\text{ J/kg.°C})} = \mathbf{6.3\ °C}$$

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